

Debiased Inference for Dynamic Panels with Low-Rank Coefficient Heterogeneity

Abstract

We develop estimation and inference for dynamic panel models in which the coefficients on lagged outcomes and observed regressors vary across units and time, while unobserved interactive effects enter as a nuisance component. Each unit-by-time coefficient matrix is low rank, so high-dimensional heterogeneity is generated by a small number of unit exposures and latent time paths. The main targets are entries, group and full cross-sectional means, between-group contrasts, and cumulative persistence, which is a linear combination of autoregressive coefficients. Dynamic generated regressors create an own-error leakage problem: a held-out idiosyncratic error can enter the estimated nuisance functions and Riesz weights through future outcomes or through contemporaneously dependent neighboring units. We address this with joint rank-constrained estimation and a single space–time-buffered cross-fitting construction. Under the stated data-generating, identification, procedure, and target assumptions—including weak exogeneity, stable lag dynamics, finite $(8 + \eta)$ th moments, low-rank identification, and conditional spatial mixing—the one-step estimators admit a cell-predictable full-covariance influence representation and are asymptotically normal. Under explicit bandwidth, covariance-tail, and local fourth-moment conditions stated with the studentization theorem, a spatial-kernel covariance estimator is ratio-consistent; conditional cross-sectional independence is nested as a special case. The statistical theory is stated for the ideal joint minimizer; alternating ridge and Riemannian least squares retain their role as operational implementations, with a separate objective-gap implementation requirement only when a finite nonconvex iterate is substituted for the ideal estimator.

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Keywords: dynamic panel data; heterogeneous coefficients; low-rank estimation; interactive fixed effects; common factors; debiased inference; cross-sectional dependence.

1 Introduction

How persistent is a macroeconomic series once common business-cycle movements are removed, and does that persistence differ between volatile and stable series? Does house-price momentum differ between cheap and expensive market segments, and has it changed over time? Questions of this kind concern *dynamic* responses that are *heterogeneous*—they differ across units and across time—and they ask for averages and comparisons of those responses, with confidence intervals. This paper develops an estimation and inference method for exactly these objects.

We observe a panel of outcomes y_{it} for units $i = 1, \dots, N$ and periods $t = 1, \dots, T$, generated by a dynamic equation in which every coefficient may differ across both units and time:

$$y_{it} = \sum_{\ell=1}^P a_{0,it,\ell} y_{i,t-\ell} + \sum_{k=1}^K x_{it,k} \beta_{0,it,k} + h_{0,it} + u_{it}. \quad (1.1)$$

Here $y_{i,t-1}, \dots, y_{i,t-P}$ are lagged outcomes, $x_{it,1}, \dots, x_{it,K}$ are observed regressors, u_{it} is an idiosyncratic error, and $h_{0,it}$ is an unobserved interactive effect—the familiar factor-times-loading component of large-panel econometrics—which we treat as a nuisance. The coefficient $a_{0,it,\ell}$ is unit i 's loading on its own ℓ th lag at time t , and $\beta_{0,it,k}$ is unit i 's slope on regressor k at time t . Collecting a given coefficient over all time periods and units produces a *coefficient matrix*: a matrix with one row per unit and one column per period. The model therefore has P dynamic coefficient matrices, K slope matrices, and one interactive-effect component. The number of lags P and regressors K are fixed and known; P is arbitrary, and because applied work most often uses one or two lags, Section 3.2 works the entire procedure out in the leading case of one lag and one regressor before any general notation is introduced.

The parameters of interest are linear summaries of the heterogeneous coefficient matrices. We refer to any such scalar parameter as a *target*. The main targets are the standard objects of applied panel work: an individual coefficient $a_{0,it,\ell}$ or $\beta_{0,it,k}$ for a given unit and time period; the average of a coefficient over an economically meaningful group of units (an industry, a region, a price tier) in a given period; the full cross-sectional average; and the difference between two group averages, which we call a between-group contrast. Because each unit carries a full lag polynomial, we also report cumulative persistence, but only as a linear combination of lag-coefficient targets already covered by the main theory. Nonlinear fixed-coefficient impulse responses and long-run multipliers are not part of the main target class. Every linear-target claim is conditional on the target-specific informativeness, variance, and local-influence-dispersion clauses in [Assumption 6](#). A time-local entry or mean is therefore covered when its Riesz influence is sufficiently dispersed across cells and contemporaneous dependence neighborhoods, even if a nonnegligible fraction of its variance is concentrated in one period; when that condition fails, the corresponding normal approximation is not asserted.

Three difficulties, and how the paper resolves them

Estimating (1.1) without restrictions is not identified: there are $(P + K + 1)NT$ coefficients and only NT observations. Any restriction strong enough to help will distort naive standard errors, and in a dynamic panel it does so in a way with no static precedent. Three difficulties organize the paper.

(1) *More coefficients than observations.* We restrict each coefficient matrix to be *low-rank*: the $N \times T$ array of, say, lag-one loadings is a few latent time paths times latent unit exposures—a factor model on the *coefficients*. Low rank provides a continuous latent-factor restriction that accommodates many leading empirical specifications (constant coefficients, additive effects, latent groups, trend-times-exposure structures) without requiring known groups, smoothness in time, or any ordering of units, and permits transparent rank screening from the spectrum of a preliminary fit; Section 3.3 compares it with the alternatives.

(2) *The lagged outcome is internally generated.* Unlike a static regressor, $y_{i,t-1}$ is produced by the model from past idiosyncratic errors of the same unit. The maintained exogeneity concept is therefore *weak exogeneity for predetermined regressors*: each idiosyncratic error has conditional mean zero given its own current predetermined regressor and the past. We keep the model in its original form, treating lagged outcomes as *design* and the idiosyncratic error as the error term, so the estimator’s leading term stays a weighted sum of idiosyncratic errors and standard-normal inference remains possible. No strict-exogeneity restriction on future covariates, and no vector conditional-mean restriction given the full current covariate panel, is imposed.

(3) *Regularization bias and a dynamic leakage problem.* Any low-rank estimator is regularized and hence biased in the shrunk directions; the debiased-machine-learning remedy is a one-step correction that re-weights the first-stage residuals by target-specific *debiasing weights* (the Riesz representer of the target in the local least-squares geometry). In a dynamic panel this creates a new problem: the idiosyncratic error u_{it} propagates into $y_{i,t+1}, y_{i,t+2}, \dots$, so a weight estimated from a sample containing the unit’s *future* is correlated with the very error it multiplies, and debiasing collapses. We call this *own-error leakage*, and it is the distinctive obstacle the paper resolves. The solution is a cross-fitting scheme built for dynamics: held-out folds are *scattered* cells—a interleaved assignment over the time–unit grid, not blocks of periods—and each fold’s training sample additionally excludes a local space–time neighborhood of every held-out cell. Scattering keeps every time period and unit represented, the temporal radius severs the dynamic leakage channel, and the spatial radius separates contemporaneously dependent neighbors. Stability, local cross-unit propagation, and spatial mixing make the remaining tails negligible. The buffer radii are selected from prespecified diagnostic grids subject to the theorem’s rate restrictions; the resulting sensitivity analysis makes the associated bias–variance trade-off transparent.

Organization of the maintained framework. Section 4 states the assumptions in compact mathematical form. Assumptions 1–6 support the main linear-target theory. The analyst-chosen cross-fitting rates, numerical accuracy requirement, and studentization bandwidth restrictions are

stated in the substantive sections where they are used rather than embedded in the assumptions. Appendix A explains the role, restrictiveness, and verification of every assumption and gives the full first-stage, rowwise, fold-transport, decoupling, and covariance-estimation arguments. The statistical estimator is the ideal joint rank-constrained minimizer. Alternating ridge and Riemannian least squares are operational algorithms; their diagnostics do not constitute a proof of global optimality.

What the paper delivers

At its core this is the solution to a *dynamic generated-regressor leakage problem*: how to attach honest standard errors to functionals of heterogeneous dynamic coefficients when the regularized weights and the errors they multiply are linked through each unit’s own future. The low-rank interactive-effects component is where the problem is most severe, but the distinctive ideas are the scattered cross-fitting with a space–time exclusion buffer and a weak-mixing central limit theory conditional on the relevant information set. The contribution is fourfold: (1) a dynamic-panel framework for low-rank coefficient heterogeneity in which local identification is a no-multicollinearity condition on observed quantities, with the dynamic twist that identification must come from the idiosyncratic variation in lagged outcomes that the factors do not span; (2) debiased inference that respects the dynamics—subsample-specific weights solved on space–time-buffered training samples and evaluated on held-out cells, yielding a one-step estimator first-order equivalent to an infeasible benchmark whose leading term is a weighted weak-mixing array of idiosyncratic errors, hence asymptotically standard-normal t -statistics for targets satisfying the stated informativeness, balance, and rate clauses; (3) an operational implementation—a cross-fitted criterion that jointly selects all ranks, explicit tuning rules, and transparent numerical diagnostics, while separating the ideal statistical estimator from nonconvex computation; and (4) inference for the objects applied users most directly report—coefficients, group and full averages, linear dynamic summaries such as cumulative persistence, and between-group contrasts, with spatial-mixing-robust covariance estimation, a reproducible Monte Carlo study, and two applications (regional unemployment persistence; house-price momentum).

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 introduces the model, works the complete procedure in the one-lag, one-regressor case, defends the low-rank restriction, and develops the general framework. Section 4 states the assumptions, and Section 7 presents the main asymptotic results. Section 8 reports the Monte Carlo simulation results, and Section 9 presents the empirical application. Finally, Section 10 concludes. All proofs of the main theorems and auxiliary lemmas are provided in Appendix A. Additional simulation results and empirical findings are collected in Appendices B and C, respectively. The replication package for the simulations and applications is available at <https://github.com/navidbahadoran/dlrhcs>.¹

Notation. Throughout, matrices are uppercase bold, vectors lowercase bold, and calligraphic letters denote linear maps between matrix spaces. Econometric subscripts are written unit first and

¹The repository contains the Monte Carlo scripts, empirical data-construction scripts, configuration files, and table/figure builders for the results reported in this paper.

time second: y_{it} , x_{it} , u_{it} , $a_{0,it,\ell}$, $\beta_{0,it,k}$, and $h_{0,it}$. All panel matrices are stored in the same order. Thus a generic panel matrix $\mathbf{M} \in \mathbb{R}^{N \times T}$ has (i, t) entry $[\mathbf{M}]_{it}$. In particular,

$$[\mathbf{Y}]_{it} = y_{it}, \quad [\mathbf{U}]_{it} = u_{it}, \quad [\mathbf{A}_0^{(\ell)}]_{it} = a_{0,it,\ell}, \quad [\mathbf{B}_0^{(k)}]_{it} = \beta_{0,it,k}, \quad [\mathbf{H}_0]_{it} = h_{0,it}.$$

Limits are taken as $\min\{N, T\} \rightarrow \infty$, with the number of lags, observed covariates, and all ranks fixed.

2 Related Literature

The paper sits at the intersection of five literatures: interactive-effects panels, dynamic panels, heterogeneous-coefficient panels, low-rank matrix inference, and debiased machine learning. We isolate the key difference in each case; Table 1 summarizes the comparison with the closest recent work.

Interactive-effects and dynamic panels. Large-panel factor analysis (Bai and Ng, 2002; Bai, 2003) and interactive-effects regressions (Pesaran, 2006; Moon and Weidner, 2015) absorb unobserved heterogeneity in the outcome *level* with common slopes as the objects of interest; we retain their additive component $h_{0,it}$ but only as a nuisance, and make the *coefficient matrices* the targets. The classical dynamic panel literature (Anderson and Hsiao, 1981, 1982; Arellano and Bond, 1991) treats short panels with common slopes and instruments the lagged-outcome/unit-effect correlation; we are in the opposite corner, with T, N both large, an entire unit-by-time matrix of autoregressive coefficients, and a least-squares estimator. The price of least squares is the maintained weak-exogeneity restriction: if the errors are serially correlated so that the lagged outcome is endogenous, the least-squares first stage is no longer valid and the appropriate remedy is an instrumented or GMM first-stage criterion rather than merely a robust standard error.

Heterogeneous-coefficient panels. Random-coefficient and mean-group approaches (Swamy, 1970; Pesaran and Smith, 1995), Bayesian shrinkage (Hsiao et al., 1999), and grouped-pattern estimation (Bonhomme and Manresa, 2015) restrict coefficient heterogeneity toward exchangeability, a common mean, or finitely many groups. Low rank is a distinct continuous latent-factor restriction: it permits latent heterogeneity in both dimensions, contains many finite-group and additive specifications as special cases, and needs no prespecified ordering or clustering of units. The targets—entries, subgroup and population averages, and contrasts—remain directly comparable to mean-group output.

Low-rank matrix inference and matrix-completion causal panels. The closest low-rank slope paper, Chernozhukov et al. (2023), permits slopes to vary across individuals and time, uses low-rank regularization with sample splitting and orthogonalization, and develops inference for individual–time effects and cross-sectional averages. Related work studies linear functionals and

completion of low-rank matrices (Choi et al., 2024, 2025) and matrix-completion approaches to causal panels (Athey et al., 2021; Bai and Ng, 2021; Agarwal et al., 2023). Our departure is not low-rank heterogeneity by itself. It is the coupled system of several coefficient matrices behind generated lagged outcomes, together with an unobserved interactive-effect component and a space–time buffer that restores predictability of target-specific Riesz weights. The design operator is therefore neither the identity nor a missing-entry mask, and the fitted-value change induced by a low-rank perturbation need not be low rank. Technically, the entrywise first-stage analysis draws on the leave-one-out and entrywise-eigenvector tools of Abbe et al. (2020); Chen et al. (2019, 2020, 2021), and the rank-constrained least-squares objective is the factorized form of nuclear-norm regularization (Candès and Recht, 2009; Recht et al., 2010; Koltchinskii et al., 2011).

Debiased and orthogonalized inference. The one-step correction follows the influence-function logic of Newey (1990, 1994), Bickel et al. (1993), and van der Vaart (1998), with a sample-splitting construction in the spirit of double/debiased machine learning (Belloni et al., 2014; Chernozhukov et al., 2018; Semenova and Chernozhukov, 2021). Two features distinguish it. The weights are not an i.i.d. nuisance-regression score: they solve a weighted least-squares problem in the *local geometry of the low-rank model*, adapting to the latent structure target by target. And the cross-fitting is redesigned for dynamic panels—held-out cells scattered over the grid, training samples excluding a local space–time neighborhood—which restores the predictability of each weight for the idiosyncratic error it multiplies. This construction is related to the leave-out-the-neighbors cross-fitting of Semenova et al. (2018), but here the deleted set must cover both dynamic descendants and contemporaneously dependent metric neighborhoods inside a coupled low-rank tangent-space problem. The identification argument is rotation-invariant: it constrains observed fitted-value changes, not normalizations of latent factors, so rotations are never identifying restrictions for the targets.

In one sentence: relative to interactive-effects and dynamic panels we move the heterogeneity from the intercept into every coefficient; relative to heterogeneous-coefficient panels we replace group or shrinkage structure with a low-rank time-unit matrix; relative to low-rank matrix inference we couple several coefficient matrices behind generated regressors; and relative to debiased machine learning we redesign the orthogonalization and the sample splitting for dynamic panel data. The paper consistently selects the exact rank because the tangent space entering the Riesz equation is rank-specific. By contrast, Choi et al. (2025) show that over-ranking can be harmless for a different single-matrix procedure. Robustness of the present coupled dynamic Riesz construction to deliberate over-ranking is a separate question and is not claimed here.

Table 1: High-level comparison with the most closely related work.

Paper	Coef. heterogeneity	Lagged y	Nuisance IFE	Targets with inference
Arellano and Bond (1991)	none (common slopes)	yes	no	common slope vector
Pesaran and Smith (1995)	unit-specific, time-constant	yes	no	mean of unit slopes
Bonhomme and Manresa (2015)	latent groups, time-varying within group	no	no	group-specific paths
Moon and Weidner (2015)	none (common slopes)	yes	yes	common slope vector
Chernozhukov et al. (2023)	low-rank slope heterogeneity in unit and time	no	no	individual-time effects and cross-sectional averages
Athey et al. (2021); Agarwal et al. (2023)	low-rank outcome matrix	no	—	missing entries / treatment effects
This paper	low-rank time-unit matrices, one per lag and regressor	yes	yes	entries, group/full means, contrasts, cumulative persistence

Notes: “Coef. heterogeneity” indicates whether slope or lag coefficients vary across units and time. “Lagged y ” indicates whether the model includes lagged outcomes with valid inference. “Nuisance IFE” indicates whether interactive effects are absorbed as nuisance components. “Targets” lists the inferential objects.

3 Model, Targets, and the Estimation Idea

3.1 The model and the objects of interest

The data are a panel of usable outcomes y_{it} , $i = 1, \dots, N$, $t = 1, \dots, T$, together with the required initial observations. The usable sample has already been reindexed after the P lags, and every coefficient may vary across both units and time,

$$y_{it} = \sum_{\ell=1}^P a_{0,it,\ell} y_{i,t-\ell} + \sum_{k=1}^K x_{it,k} \beta_{0,it,k} + h_{0,it} + u_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T, \quad (3.1)$$

with lagged outcomes $y_{i,t-1}, \dots, y_{i,t-P}$, observed regressors $x_{it,1}, \dots, x_{it,K}$, an unobserved interactive-effect component $h_{0,it}$, and an idiosyncratic error u_{it} , as introduced in Section 1. Thus

$$\mathcal{I}_{NT} := \{1, \dots, N\} \times \{1, \dots, T\}$$

is the usable sample of NT cells $a = (i, t)$; we write $y_a = y_{it}$ and $u_a = u_{it}$ when the cell index is more convenient. The initial observations $(y_{i,0}, \dots, y_{i,1-P})$ are treated as observed.

Three families of unknown objects appear in (3.1). For each lag $\ell \leq P$, the *dynamic coefficient matrix* is the $\mathbf{A}_0^{(\ell)} = (a_{0,it,\ell})_{i \leq N, t \leq T}$; for each regressor $k \leq K$, the *slope matrix* is $\mathbf{B}_0^{(k)} = (\beta_{0,it,k})_{i \leq N, t \leq T}$; and the *unobserved interactive-effect component* is $\mathbf{H}_0 = (h_{0,it})_{i \leq N, t \leq T}$. The maintained restriction, formalized in Section 4, is that every one of these coefficient matrices has low rank.

It is convenient to collect unit i ’s autoregressive coefficients at time t into the vector $\mathbf{A}_{0,it} = (a_{0,it,1}, \dots, a_{0,it,P})'$. As in vector autoregressions, the dynamics of the model are summarized by the

companion form: for a lag-coefficient vector $\mathbf{a} = (a_1, \dots, a_P)'$, the companion matrix is

$$\mathbf{C}(\mathbf{a}) = \begin{pmatrix} a_1 & a_2 & \cdots & a_{P-1} & a_P \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}, \quad (3.2)$$

so that unit i 's recursion at time t has state-transition matrix $\mathbf{C}_{0,it} := \mathbf{C}(\mathbf{A}_{0,it})$. Stability of the heterogeneous autoregression will be expressed through products of these companion matrices, exactly as stationarity of an $\text{AR}(P)$ is expressed through powers of its companion matrix.

The targets of inference are linear summaries of the coefficient matrices. For a group of units $G_N \subset \{1, \dots, N\}$ with $g_N = |G_N|$ members, the leading examples are the entry, group-mean, and full-mean targets

$$a_{0,it,\ell}, \quad \theta_{0,G,t,\ell}^A = g_N^{-1} \sum_{i \in G_N} a_{0,it,\ell}, \quad \theta_{0,t,\ell}^A = N^{-1} \sum_{i=1}^N a_{0,it,\ell},$$

their analogues for the slope matrices, and between-group contrasts obtained by differencing two group means. Because each unit carries a full lag polynomial, we also report *cumulative persistence*,

$$\theta_{0,it}^\Sigma = \sum_{\ell=1}^P a_{0,it,\ell}, \quad \theta_{0,G,t}^\Sigma = g_N^{-1} \sum_{i \in G_N} \sum_{\ell=1}^P a_{0,it,\ell},$$

which is a *linear* combination of autoregression-coefficient targets and is therefore covered verbatim by the inference theory of Section 7. Nonlinear fixed-coefficient response paths and long-run multipliers are smooth functions of the lag polynomial. They require additional stability, differentiability, and bootstrap or Delta-method regularity and are therefore not part of the formal submission theory.

Remark 1 (The lag order is general). Nothing in the theory is special to one lag. The maintained stability condition is a bound on products of companion matrices (the stability display in Assumption 1 below), of which the familiar “coefficients sum to less than one in absolute value” condition is a sufficient special case, and the geometric decay rate ρ_* that controls the dynamic arguments is the companion-power modulus of Assumption 1 (equal to the ℓ_1 bound only when $P = 1$). The one condition that genuinely tightens as P grows is the no-multicollinearity requirement of Assumption 5: for highly persistent series, consecutive lags $y_{i,t-1}, \dots, y_{i,t-P}$ become nearly collinear, so the identifying variation degrades with P . This—not the theory—is the practical limit on the lag order, and it is the same limit familiar from lag-length selection in time-series regression.

3.2 The leading case: one lag and one regressor

This subsection walks through the *entire* procedure in the simplest dynamic case, $P = K = 1$, with notation local to the subsection; every step has an exact general counterpart in Sections 3.4–3.5.

The model. With one lag and one regressor, (3.1) reads

$$y_{it} = a_{0,it} y_{i,t-1} + \beta_{0,it} x_{it} + h_{0,it} + u_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T. \quad (3.3)$$

unit i 's persistence at time t is $a_{0,it}$; its slope on x is $\beta_{0,it}$. The three $N \times T$ matrices are $\mathbf{A}_0 = (a_{0,it})$, $\mathbf{B}_0 = (\beta_{0,it})$, and $\mathbf{H}_0 = (h_{0,it})$. Low rank means, say for the persistence matrix with rank one,

$$a_{0,it} = f_t \lambda_i :$$

a single latent time path f_t (“how tight are credit conditions this quarter”) scaled by a unit-specific exposure λ_i (“how sensitive is this market’s persistence to credit conditions”). Rank r allows r such components. The model has roughly $3NT$ free coefficients without the restriction and roughly $3r(N + T)$ with it—fewer than the NT observations once T and N are moderately large, which is the entire point.

Each cell contributes one equation and three unknowns, so unrestricted estimation is infeasible at any panel size; the low-rank restriction couples cells by making the unknowns the shared latent factors and exposures.

The three difficulties, concretely. (i) The parameters enter *multiplied by regressors*: the data are informative about $a_{0,it}$ only through the product $a_{0,it} y_{i,t-1}$, so wherever $y_{i,t-1}$ is small the cell carries little information about persistence, and a change in the persistence matrix can be confounded with a change in the interactive-effect component if the two produce the same product pattern. (ii) The regressor $y_{i,t-1}$ is generated by the model: substituting the equation into itself shows

$$y_{i,t-1} = a_{0,i,t-1} y_{i,t-2} + \beta_{0,i,t-1} x_{i,t-1} + h_{0,i,t-1} + u_{i,t-1},$$

so $y_{i,t-1}$ carries the unit’s past idiosyncratic errors and inherits the factor structure; least-squares validity rests on the maintained weak-exogeneity moment restriction, not on independence of errors and regressors. (iii) Any low-rank estimator is regularized and hence biased, and—the dynamic twist—the bias-removal step must not let u_{it} re-enter through $y_{i,t+1}, y_{i,t+2}, \dots$, which all contain it.

Step 1: first-stage fit. At a candidate rank vector, estimate the three matrices jointly by rank-constrained least squares on the space–time–buffered training cells. In the one-lag, one-regressor case this means minimizing

$$\sum_{(i,t) \in \mathcal{I}_{-j}^{\text{buf}}} (y_{it} - y_{i,t-1} A_{it} - x_{it} B_{it} - H_{it})^2$$

over matrices $(\mathbf{A}, \mathbf{B}, \mathbf{H})$ having the candidate ranks and entries in a fixed compact envelope containing the truth. The rank constraint is the regularizer; it couples cells and resolves the fact that one cell supplies only one equation for several coefficient blocks. The statistical estimator is a global (or objective-gap equivalent) minimizer of this criterion. In computation we use multiple-start alternating ridge, seeded by blockwise spectral pilots, and retain a solution only after the objective and rank-stability diagnostics in Appendix A.2.7. Importantly, we do *not* claim that cellwise ambient ridge followed by truncated SVD recovers the separate coefficient matrices: without the rank restriction that ambient problem identifies only the fitted scalar at each cell, not its decomposition across coefficient blocks.

Step 0 (done first in practice): how many components? Fix generous rank caps before estimation, fit the model once at those caps, and use the singular values of each fitted matrix to order the candidates inside the fixed box. Then choose among the complete candidate set by comparing *out-of-sample* prediction error on held-out cells, penalized by model dimension. Section 5 makes this precise and proves the selection is consistent.

Step 2: why the first stage alone is not enough. Ridge shrinkage and the rank projection create bias of the same order as the noise in any single entry $\hat{a}_{it}$, so “ $\hat{a}_{it} \pm 1.96 \hat{\text{se}}$ ” is mis-centered. The remedy is a one-step correction: add back a weighted average of the first-stage residuals $\hat{u}_{it} = y_{it} - \hat{a}_{it}y_{i,t-1} - \hat{\beta}_{it}x_{it} - \hat{h}_{it}$, with weights chosen to remove the first-order bias.

Step 3: the debiasing weights. Suppose the target is one persistence coefficient a_{0,i_0t_0} . The correct weights Ψ_{it} are defined by a weighted least-squares requirement: for *every* local perturbation of the coefficient matrices that the low-rank model permits—move $\mathbf{A}$ by Δ^A , $\mathbf{B}$ by Δ^B , $\mathbf{H}$ by Δ^H , each within its factor structure—the weighted sum of the induced fitted-value changes must reproduce the induced change in the target:

$$\sum_{i,t} \Psi_{it} (y_{i,t-1} \Delta_{it}^A + x_{it} \Delta_{it}^B + \Delta_{it}^H) = \Delta_{i_0t_0}^A \quad \text{for all admissible } (\Delta^A, \Delta^B, \Delta^H). \quad (3.4)$$

This is the first-order condition of a least-squares projection: in a regression $y = \mathbf{X}\beta + u$ the analogous weights are $\mathbf{e}'_j(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, and (3.4) replaces $\mathbf{X}$ by the map from coefficient perturbations to fitted values (the Riesz representer of the target). Solving it requires the local design to be nonsingular—no admissible perturbation produces a zero fitted-value pattern, the analogue of “no perfect multicollinearity” (Section 3.5).

Step 4: cross-fitting with a space–time exclusion buffer. The weights must be estimated, and two dependence channels must be broken. A held-out idiosyncratic error enters the unit’s future outcomes through the dynamic recursion, and it may be statistically linked to nearby units in the same period under Assumption 3. We split cells into scattered folds and, for every validation cell (i, t) , remove from its training complement all cells (ℓ, s) with $0 \leq s - t \leq q_T$ and $d_N(\ell, i) \leq r_N$.

Stability makes the remaining temporal influence of order ρ_*^{qT} , the local propagation envelope makes the surviving cross-unit causal influence of order $\Omega_x(r_N)$, and spatial mixing makes the remaining contemporaneous dependence of order $\bar{\alpha}(r_N)$. Scattering keeps every time period and unit represented, whereas the growing but asymptotically sparse buffer supplies the decoupling used in the appendix proofs.

Step 5: the estimator and its confidence interval. The estimator adds the plug-in and held-out correction, averaged over folds:

$$\check{a}_{i_0 t_0} = \sum_{j=1}^J \frac{|\mathcal{I}_j|}{NT} \left[\hat{a}_{i_0 t_0}^{(-j)} + \frac{NT}{|\mathcal{I}_j|} \sum_{(i,t) \in \mathcal{I}_j} \hat{\Psi}_{it}^{(-j)} \hat{u}_{it}^{(-j)} \right].$$

Each cell uses weights and residuals estimated without its own dependence neighborhood. The main standard error is the spatial-kernel quadratic form

$$\hat{s}_{\text{sp}}^2 \sum_t \sum_{i,j} K\{d_N(i,j)/h_N\} \hat{\Psi}_{it} \hat{\Psi}_{jt} \hat{u}_{it} \hat{u}_{jt},$$

which reduces to the diagonal sandwich form under conditional cross-sectional independence. Section 7 proves the studentized normal limit under the maintained conditions and the bandwidth conditions stated with the studentization theorem.

The same construction for every other linear target. For a group mean only the right-hand side of (3.4) changes (to the group average of the perturbations); the weights re-solve and the interval is typically far narrower because averaging pools information across units. Full means and between-group contrasts are handled by changing the target direction in the same Riesz equation. With multiple lags, cumulative persistence is also linear because it is the sum of lag-coefficient targets. Nonlinear impulse-response paths require additional response-function analysis and are therefore kept out of the main target list.

3.3 Why low rank, rather than unrestricted or nonparametric heterogeneity?

Some restriction is unavoidable: (3.1) has $(P + K + 1)TN$ coefficients and TN observations, and unit-by-unit time-series regression identifies only time-constant coefficients, at the slow $\sqrt{T}$ rate and with no pooling. The question is how to restrict. Nonparametric smoothing restricts in a direction applications often do not support: smoothness in t rules out the abrupt coefficient shifts that motivate time variation, smoothness in i presumes a metric on units that rarely exists, and where a covariate metric does exist the curse of dimensionality returns averages at nonparametric rates. Low rank needs neither smoothness nor any ordering of units—a rank- r coefficient matrix may jump arbitrarily in both dimensions provided the variation is spanned by r latent components—and it nests the specifications applied work already uses: constant coefficients, additive unit and time effects,

finitely many latent groups (Bonhomme and Manresa, 2015), and smooth-trend-times-exposure structures are all low rank, while the parameter count stays at order $(N + T)$ per coefficient matrix, small enough for the $\sqrt{TN}$ -class rates established below. Finally, the restriction is *empirically assessable*: the singular-value profile of a preliminary fit reveals whether a few components dominate, the cross-fitted criterion of Section 5 selects the rank, and the stability diagnostic of Definition 1 flags cases where the low-rank description is too weak to trust. Under approximate low rank the framework still applies with the rank as a smoothing parameter.

3.4 Operator form, local geometry, and debiasing weights

The general procedure is the one-lag walkthrough written in operator notation; this subsection fixes only the objects the theorems below require, deferring the full development to Appendix A.3.5. Stack the coefficient matrices into $\Theta_0 = (\Gamma_0^{(1)}, \dots, \Gamma_0^{(M)}, H_0)$ in the product space $\mathbb{H}_{NT}$, and let the *design map* $\mathcal{A}$ send a tuple of coefficient matrices to its fitted-value matrix, $[\mathcal{A}(\Theta)]_{it} = \sum_m Z_{it}^{(m)} \Gamma_{it}^{(m)} + H_{it}$, where $Z^{(m)}$ is the regressor multiplying coefficient matrix m (a lagged outcome or an observed covariate); the model is $\mathbf{Y} = \mathcal{A}(\Theta_0) + \mathbf{U}$.

Targets as directions. Every target—an entry, a group or full mean, a contrast, cumulative persistence—is a linear functional $\phi_\nu(\Theta_0) = \langle \mathbf{D}_\nu, \Theta_0 \rangle_{\mathbb{H}}$ encoded by a *direction* $\mathbf{D}_\nu$ (e.g. $\mathcal{E}_\ell(\pi_G \mathbf{e}'_t)$ for a group- G mean of the ℓ th autoregressive coefficient, with group weight $\pi_G = g_N^{-1}(\mathbf{1}\{i \in G\})_i$; full means use $\pi_N = N^{-1}\mathbf{1}_N$, contrasts $\pi_\Delta = \pi_{G_1} - \pi_{G_2}$); see Appendix A.3.5 for the full list.

Local geometry. Because Θ_0 is low-rank, the admissible local perturbations form the product tangent space $\mathcal{T}_0$ (at a rank- r point $\mathbf{U}\Sigma\mathbf{V}'$, $\mathcal{T} = \{\mathbf{U}\mathbf{A}' + \mathbf{C}\mathbf{V}'\}$ with projection $\mathcal{P}_{\mathcal{T}}(\mathbf{A}) = \mathbf{P}_U\mathbf{A} + \mathbf{A}\mathbf{P}_V - \mathbf{P}_U\mathbf{A}\mathbf{P}_V$). The local least-squares geometry is summarized by the *information map* $\mathcal{G}_0 = \mathcal{P}_{\mathcal{T}_0}\mathcal{A}^*\mathcal{A}\mathcal{P}_{\mathcal{T}_0}$, the coefficient-matrix analogue of the regression Gram $\mathbf{X}'\mathbf{X}$ restricted to $\mathcal{T}_0$: for $\Delta \in \mathcal{T}_0$, $\langle \Delta, \mathcal{G}_0\Delta \rangle_{\mathbb{H}} = \|\mathcal{A}(\Delta)\|_F^2$.

Debiasing weights. At a realized design, the local Riesz direction for target $\mathbf{D}_\nu$ is characterized by $\mathbf{W}_\nu = \mathcal{P}_{\mathcal{T}_0}\mathbf{D}_\nu$, $\mathbf{q}_\nu^{\text{full}} = \mathcal{G}_0^{-1}\mathbf{W}_\nu$, and $\Psi_\nu^{\text{full}} = \mathcal{A}\mathbf{q}_\nu^{\text{full}}$, exactly as in (3.4). This full-design object is not called a population weight: in the dynamic model it may depend on every unit’s future and therefore need not be unpredictable for the idiosyncratic error it multiplies. The estimator consequently computes fold-specific Riesz directions on space–time-buffered training samples, while the central-limit benchmark uses the predictable population operator introduced in Section 7.

For later conditional statements, let

$$\begin{aligned} \mathcal{G}_0 &:= \sigma(\Theta_0, \text{initial conditions}, d_N, \mathcal{R}_{\text{fold}}, \text{time-invariant attributes}), \\ \mathcal{F}_t &:= \mathcal{G}_0 \vee \sigma(\mathbf{X}_s, \mathbf{u}_s : s \leq t), \quad \mathcal{P}_{it} := \mathcal{F}_{t-1} \vee \sigma(x_{it}). \end{aligned} \tag{3.5}$$

The current regressor x_{it} is observed before u_{it} , whereas future covariates are not included in $\mathcal{P}_{it}$. The covariate path is therefore stochastic rather than conditioned away. Conditional probability and expectation given $\mathcal{G}_0$ are denoted by $\mathbb{P}_0$ and $\mathbb{E}_0$. The fold randomization $\mathcal{R}_{\text{fold}}$ is generated independently of the panel. Assumption 2 imposes the original unit-level weak exogeneity restriction

$\mathbb{E}_0[u_{it} \mid \mathcal{P}_{it}] = 0$; it does not impose strict exogeneity or the stronger vector condition $\mathbb{E}_0[\mathbf{u}_t \mid \mathbf{X}_t, \mathcal{F}_{t-1}] = 0$.

Scattered cross-fitting with a space–time exclusion buffer. The auxiliary randomization $\mathcal{R}_{\text{fold}}$ assigns cells to scattered folds independently of the panel conditional on $\mathcal{G}_0$. the cross-fitting design in [Section 6.1](#) gives the canonical buffered training set (6.1). Its temporal radius removes the descendants through which a held-out idiosyncratic error propagates, while its spatial radius removes the contemporaneous mixing neighborhood. This buffer construction, the stability and mixing restrictions in Assumptions 1 and 3, and the rate regime (5.2) make the omitted tails negligible while preserving a nonvanishing training share. Scattering ensures that every time period and unit retains training support. The benchmark weights $\Psi_{\nu,j}^{\text{or}}$ solve the Riesz equation on the space–time-buffered training geometry and are evaluated on the held-out cells. They are not assumed to be exactly orthogonal merely because a cell is held out. Lemma 3 instead compares them with leave-one-slice predictable weights, while Proposition 7 controls the remaining geometric tail. Algebraically, every cell contributes once to the held-out score. That fact alone does not make a fixed fold count asymptotically valid: the fold-specific weights, retained training geometry, and leakage remainder still depend on $(q_T + 1)v_N(r_N)/J_{NT}$. The theory therefore lets the number of folds dominate the complete space–time buffer volume. A moderate fixed J is treated only as a finite-sample approximation and must be accompanied by retention, transport, and window sensitivity diagnostics (Remark 2 and Appendix A.2.7).

3.5 What identifies the model

Local identification is nonsingularity of the information map $\mathcal{G}_0$ on $\mathcal{T}_0$ —the requirement that no admissible perturbation of the coefficient matrices is invisible in the outcome. To see what this means in observable terms, define for each coefficient block the *fitted-value change* of its admissible perturbations,

$$\mathcal{L}_{\Gamma,m} := \{\mathbf{Z}^{(m)} \odot \Delta^{(m)} : \Delta^{(m)} \in \mathcal{T}_{\Gamma,m}\}, \quad (3.6)$$

the set of fitted-value patterns a local change in coefficient matrix m can produce, and $\mathcal{L}_H := \mathcal{T}_H$ for the nuisance block. The map $\mathcal{G}_0$ is singular exactly when some nonzero combination of such patterns cancels:

$$\sum_{m=1}^M \mathbf{Z}^{(m)} \odot \Delta^{(m)} + \Delta^H = \mathbf{0} \quad (3.7)$$

for a nonzero admissible Δ . Local identification is therefore a *no-multicollinearity condition among fitted-value changes*—the exact analogue, for coefficient matrices, of the rank condition on the regressor matrix in least squares. It permits arbitrarily strong factor structure in the regressors themselves; it rules out only the case in which a dynamic coefficient perturbation, a slope perturbation, and an interactive-effect perturbation are locally observationally equivalent.

Local collinearity. If a nonzero tangent perturbation has zero fitted-value change, the tangent information map is singular and the corresponding target is not locally identified. The local-collinearity diagnostic in Appendix A.3.1 records the corresponding finite-dimensional normal-equation failure.

Why lagged outcomes are mechanically factor-laden, and where identification comes from. Iterating (3.1) backward shows that $y_{i,t-m}$ loads on the same latent time paths and exposures that generate the coefficient matrices, *plus* a moving average of the unit’s past idiosyncratic errors. The regressor matrix $\mathbf{Z}^{(m)}$ therefore carries a pervasive common-factor component overlapping the nuisance space; decompose

$$\mathbf{Z}_F^{(m)} = \mathbf{\Lambda}_{z,m} \mathbf{F}'_{z,m}, \quad \mathbf{Z}_F^{(m)} = \mathbf{F}_{z,m} \mathbf{\Lambda}'_{z,m}, \quad (3.8)$$

into the common part $\mathbf{Z}_F^{(m)}$ (which may share spaces with $\mathbf{H}_0$ and the other coefficient matrices) and a residual $\mathbf{E}_Z^{(m)}$ —for a lagged outcome, the accumulated idiosyncratic-error history projected off the common factors, which under stability has nondegenerate energy. The maintained theory states the population conclusion directly in Assumption 5: the expected population design map must be uniformly nonsingular on all differences between candidate low-rank matrices and locally informative in row and column charts. Appendix A.3.1 derives the corresponding sample and fold-restricted statements. This formulation permits the common component of a lagged outcome to overlap the nuisance factor space; identification can then be supplied by residual identifying variation. Appendix A.3.1 gives two interpretable *sufficient* sufficient conditions—fitted-value no-cancellation and residual identifying variation. They are sufficient conditions for checking Assumption 5, not additional maintained assumptions.

3.6 How informative is a target? Leverage

The projected target direction $\mathbf{W}_\nu = \mathcal{P}_{\mathcal{T}_0} \mathbf{D}_\nu$ measures how much of the target survives in the identified local low-rank geometry. Assumption 6 retains only the irreducible target requirements: relevance, variance noncancellation, and population Riesz-influence dispersion. Entry, mean, and contrast leverage formulas are derived in Appendices A.4.2; the appendix also gives transparent sufficient design restrictions under which the influence-dispersion clause can be checked.

4 Assumptions

Throughout, T denotes the usable post-lag time dimension, $n_{NT} = NT$, $q = 8 + \eta$ for some $\eta > 0$, and $\min\{T, N\} \rightarrow \infty$ while P , K , and the true ranks remain fixed. For an arbitrarily slowly diverging L_{NT} , set

$$b_{NT} = (NT)^{1/q} L_{NT}.$$

This section states the assumptions used in the paper without extended interpretation. Assumptions 1–6 are maintained for the main linear-target theory. Their econometric role, restrictiveness, degree of primitiveness, and sufficient verification routes are discussed in Appendix A.2.

Assumption 1 (Stable dynamics and stochastic predetermined regressors). *(i) For the companion matrices $\mathbf{C}_{0,it}$ associated with the lag coefficients, there exist constants $C_* < \infty$ and $\rho_* \in (0, 1)$ such that*

$$\sup_{i,t,h \geq 1} \|\mathbf{C}_{0,t+h,i} \cdots \mathbf{C}_{0,t+1,i}\|_{\text{op}} \leq C_* \rho_*^h.$$

(ii) Each predetermined regressor $x_{it,k}$ has a recursive representation

$$x_{it,k} = G_{it,k}(\varepsilon_t^x, \varepsilon_{t-1}^x, \dots, \mathbf{u}_{t-1}, \mathbf{u}_{t-2}, \dots),$$

where $\mathbf{u}_t = (u_{1t}, \dots, u_{Nt})'$ and ε_t^x denotes the full time- t primitive innovation vector driving the covariate system. In particular, ε_t^x may be partitioned as

$$\varepsilon_t^x = \left((\varepsilon_{1t}^{x,\text{id}})', \dots, (\varepsilon_{Nt}^{x,\text{id}})', (\varepsilon_t^{x,\text{agg}})', (\eta_t^{x,\text{fac}})' \right)',$$

where $\varepsilon_{it}^{x,\text{id}}$ contains unit-specific covariate innovations, $\varepsilon_t^{x,\text{agg}}$ contains aggregate covariate innovations, and $\eta_t^{x,\text{fac}}$ contains primitive innovations that drive any latent common-factor process affecting the covariates. Thus, when a latent covariate factor follows a recursive law, the innovation driving that factor is included in ε_t^x , while the factor level itself is generated through the map $G_{it,k}$. Aggregate or common covariate components are admissible only insofar as the induced process satisfies Assumptions 2 and 3; in particular, this clause does not allow an unrestricted pervasive contemporaneous factor that violates the conditional spatial-mixing condition.

Let

$$e_t = ((\varepsilon_t^x)', \mathbf{u}_t')'.$$

For $h \geq 1$, let $x_{it,k}^{(h)}$ denote the coupled version of $x_{it,k}$ obtained by replacing e_{t-h} by an independent copy under the conditional law given $\mathcal{G}_0$, while holding all other primitive time slices fixed. Then

$$\sup_{t,i,k,h \geq 1} \|x_{it,k} - x_{it,k}^{(h)}\|_{q,0} \leq C_* \rho_*^h.$$

In addition, define the unit-specific primitive block

$$\zeta_{jt} = \left((\varepsilon_{jt}^{x,\text{id}})', u_{jt}' \right)'.$$

For a cell (s, j) , let $x_{\ell, s+h, k}^{(s,j)}$ be the coupled covariate obtained by replacing ζ_{js} by an independent copy under the conditional law given $\mathcal{G}_0$, while holding the aggregate/common innovation blocks and all other primitive components fixed. There is a nonincreasing spatial-propagation envelope $\omega_x(r) \downarrow 0$

such that, for the same metric-growth exponent d_s as in Assumption 3,

$$\sum_{r=0}^{\infty} (1+r)^{3d_s-1} \omega_x(r) < \infty,$$

and

$$\sup_{s,j,\ell,k,h \geq 0} \left\| x_{\ell,s+h,k} - x_{\ell,s+h,k}^{\langle s,j \rangle} \right\|_{q,0} \leq C_* \rho_*^h \omega_x(d_N(\ell, j)).$$

(iii) The initial conditions for the outcome recursion and any initial states entering the predetermined-covariate recursion have uniformly bounded q th moments.

Assumption 1(i) is a uniform stability condition for the heterogeneous dynamic panel. It allows the autoregressive coefficients to vary across units and time, but it requires the resulting autoregressive feedback to be uniformly stable. In other words, the effect of a shock today on future outcomes must decay geometrically. With one lag, this is implied by $\sup_{t,i} |a_{0,it,1}| < 1$; with several lags, the analogous requirement is stability of products of the time-varying companion matrices. Part (ii) imposes geometric forgetting on stochastic predetermined regressors. The innovation ε_t^x is a vector-valued primitive time slice that may include unit-specific, aggregate, and latent common-factor innovations affecting the covariate process, subject to the paper's weak-exogeneity and spatial-mixing restrictions. The time-slice coupling controls temporal memory, while the unit-specific coupling controls the propagation of a held-out unit shock across both time and space. Thus future covariates may depend on current idiosyncratic errors, but the effect of such shocks on later and spatially separated covariates must decay at the stated geometric and spatial rates.

Assumption 2 (Weak exogeneity and temporal error structure). *For every (i, t) ,*

$$\mathbb{E}_0[u_{it} \mid x_{it}, \mathcal{F}_{t-1}] = 0. \quad (4.1)$$

Conditional on $\mathcal{G}_0$, the joint innovations $e_t = ((\varepsilon_t^x)', \mathbf{u}_t')'$ are independent across t .

Assumption 2 is the standard predetermined-regressor condition for dynamic panels. It allows current covariates and lagged outcomes to depend on past idiosyncratic shocks, but it rules out a serially predictable component in the remaining idiosyncratic innovation once the current predetermined regressor and the past are observed. The independence of primitive time slices is imposed on the innovation representation, not on the persistent outcome or covariate levels generated by the recursion.

Assumption 3 (Moments and weak cross-sectional dependence). *For $q = 8 + \eta$ and constants $0 < c_\sigma < C_\sigma < \infty$,*

$$c_\sigma \leq \mathbb{E}_0[u_{it}^2 \mid x_{it}, \mathcal{F}_{t-1}] \leq C_\sigma, \quad \mathbb{E}_0[|u_{it}|^q \mid x_{it}, \mathcal{F}_{t-1}] \leq C$$

almost surely, uniformly in (i, t) , and

$$\sup_{t,i,k} \mathbb{E}_0 |x_{it,k}|^q \leq C, \quad \sup_i \mathbb{E}_0 \|(y_{i,1-P}, \dots, y_{i0})\|_2^q \leq C.$$

Conditional on $\mathcal{F}_{t-1}$, the within-period field $\{(u_{it}, x_{it}) : i \leq N\}$ is strongly mixing over $(\{1, \dots, N\}, d_N)$ with deterministic envelope $\alpha(r) \downarrow 0$. This condition is imposed on the process induced by Assumption 1; aggregate or common covariate shocks are therefore admissible only when the resulting conditional mixing envelope remains valid. Metric balls satisfy

$$v_N(r) := \max_i |\{j : d_N(i, j) \leq r\}| \leq C(1+r)^{d_s},$$

and, with $\chi_\eta = \eta/(8+\eta)$,

$$\sum_{r=0}^{\infty} (1+r)^{3d_s-1} \alpha(r)^{\chi_\eta} < \infty.$$

Tail envelopes. For later rate statements and shell summations, define

$$\begin{aligned} \Omega_x(r) &:= \sum_{\ell \geq r} (1+\ell)^{3d_s-1} \omega_x(\ell), \\ \bar{\alpha}(r) &:= \sum_{\ell \geq r} (1+\ell)^{3d_s-1} \alpha(\ell)^{\chi_\eta}. \end{aligned} \tag{4.2}$$

The summability clauses in Assumptions 1 and 3 imply $\Omega_x(r) \rightarrow 0$ and $\bar{\alpha}(r) \rightarrow 0$. The paper uses these tail envelopes, rather than the point values $\omega_x(r)$ and $\alpha(r)^{\chi_\eta}$, whenever a proof sums over all spatial shells beyond radius r .

Assumption 3 imposes finite $8+\eta$ moments and weak within-period dependence. The metric d_N describes which units are close in the cross section and is chosen from the economic structure of the application, such as geographic, input-output, or network distance. Conditional cross-sectional independence is the special case in which $\alpha(r)$ is zero beyond distance zero. When no credible metric is imposed, the spatial-mixing theory should not be interpreted as allowing arbitrary within-period dependence; Appendix A.2.4 gives a known-cluster alternative that allows arbitrary within-cluster dependence and conditional independence across clusters after conditioning on aggregate/common information. Under that route, the proof changes only in the decoupling, fold-buffer volume, and studentization steps: the rank selector, first-stage estimator, tangent-space construction, Riesz equation, and one-step correction are unchanged.

Assumption 4 (Pervasive, well-spread low-rank signal). *For every coefficient and nuisance block, $\mathbf{\Gamma}_0^{(m)} = \mathbf{U}_{0,m} \mathbf{\Sigma}_{0,m} \mathbf{V}_{0,m}'$ has fixed rank $r_{0,m}$,*

$$\sigma_{r_{0,m}}(\mathbf{\Gamma}_0^{(m)}) \asymp \sigma_1(\mathbf{\Gamma}_0^{(m)}) \asymp \sqrt{NT},$$

$$\max_i \|e_i' \mathbf{U}_{0,m}\|_2^2 \leq \mu_0 r_{0,m}/N, \quad \max_t \|e_t' \mathbf{V}_{0,m}\|_2^2 \leq \mu_0 r_{0,m}/T.$$

All true coefficient and interactive-effect components have uniformly bounded entries. Writing $\mathbf{r}_0 = (r_{0,1}, \dots, r_{0,M}, r_{0,H})$, the fixed search caps $\bar{\mathbf{r}} = (\bar{r}_1, \dots, \bar{r}_M, \bar{r}_H)$ satisfy $\mathbf{r}_0 \leq \bar{\mathbf{r}}$ componentwise.

Define the deterministic candidate box

$$\mathcal{R} = \prod_{m=1}^M \{0, 1, \dots, \bar{r}_m\} \times \{0, 1, \dots, \bar{r}_H\}, \quad (4.3)$$

let $\mathcal{S}_{\mathcal{R}}$ be its union of between-differences between candidate low-rank matrices, and let $\mathcal{T}_0$ denote the product tangent space at the truth. To make the curvature notation explicit, write a normalized factorization as $\vartheta = (\{\mathbf{z}_t\}_{t \leq T}, \{\mathbf{w}_i\}_{i \leq N})$, where $\mathbf{z}_t$ stacks the time factors and $\mathbf{w}_i$ stacks the unit loadings after the usual rotational normalization. Let

$$Q_0(\vartheta) = \frac{1}{2} \mathbb{E}_0 \|\mathcal{A}\{\boldsymbol{\Theta}(\vartheta) - \boldsymbol{\Theta}_0\}\|_F^2$$

be the population prediction-risk contrast on this normalized chart. The row and column Hessians are

$$H_t^F(\vartheta) = \nabla_{\mathbf{z}_t \mathbf{z}_t'}^2 Q_0(\vartheta), \quad H_i^\Lambda(\vartheta) = \nabla_{\mathbf{w}_i \mathbf{w}_i'}^2 Q_0(\vartheta),$$

and $H(\vartheta) = \nabla_{\vartheta \vartheta'}^2 Q_0(\vartheta)$ is the joint chart Hessian. Its block diagonal is

$$D(\vartheta) = \text{diag}\{H_1^F(\vartheta), \dots, H_T^F(\vartheta), H_1^\Lambda(\vartheta), \dots, H_N^\Lambda(\vartheta)\}.$$

Assumption 5 (Population identification and local curvature). *For constants $0 < c_S < C_S < \infty$,*

$$c_S \|\boldsymbol{\Delta}\|_{\mathbb{H}}^2 \leq \mathbb{E}_0 \|\mathcal{A}(\boldsymbol{\Delta})\|_F^2 \leq C_S \|\boldsymbol{\Delta}\|_{\mathbb{H}}^2, \quad \boldsymbol{\Delta} \in \mathcal{S}_{\mathcal{R}}. \quad (4.4)$$

Uniformly over a fixed neighborhood of the truth,

$$c_H \mathbf{I} \preceq H_t^F(\vartheta), \quad H_i^\Lambda(\vartheta) \preceq C_H \mathbf{I},$$

$$\|D(\vartheta)^{-1/2} \{H(\vartheta) - D(\vartheta)\} D(\vartheta)^{-1/2}\|_{\text{op}} \leq \rho_H < 1,$$

and the normalized Hessian is uniformly Lipschitz on that neighborhood.

Assumption 5(i) is a population identification condition and is the low-rank analog of the full-column-rank (or no perfect collinearity) condition in ordinary least squares. It says that two different low-rank coefficient specifications cannot produce the same fitted values. Thus, coefficient changes and interactive-effect changes cannot cancel each other observationally. The condition is imposed on $\mathcal{S}_{\mathcal{R}}$, rather than only on the tangent space at the truth, because rank selection compares different rank specifications. Assumption 5(ii) requires the local least-squares problem to be well-conditioned when the low-rank matrices are written in factor form. This condition is used only to upgrade global low-rank consistency to rowwise and entrywise control of the estimated singular spaces. It is analogous to local strong convexity and bounded leverage conditions used in

entrywise factor and matrix-completion theory. Specifically, the row and column Hessian bounds require enough information to estimate the factors and loadings separately. The condition involving $D(\vartheta)$ requires the cross terms between factors and loadings to be smaller than their own direct information. The local Hessian bounds and block-dominance restriction make row and column factor updates locally invertible and stable. While Assumption 5 is high level, Appendix A.3.1 gives sufficient conditions based on fitted-value no-cancellation and residual identifying variation after common factor components are removed.

The next assumption states target-specific regularity conditions for asymptotic inference on a fixed linear functional of the low-rank coefficient matrices. The main targets considered in the paper—individual coefficients, group and full cross-sectional means, between-group contrasts, and cumulative persistence—satisfy these conditions under primitive sufficient conditions verified in Appendices A.4 and A.4.2.

For a scalar target $\phi_\nu(\Theta_0)$, let $\mathbf{D}_\nu$ denote its linear target direction and define

$$\mathbf{W}_\nu = \mathcal{P}_{\mathcal{T}_0} \mathbf{D}_\nu, \quad \bar{\mathcal{G}}_0 = \mathbb{E}_0[\mathcal{P}_{\mathcal{T}_0} \mathcal{A}^* \mathcal{A} \mathcal{P}_{\mathcal{T}_0}], \quad \bar{\mathbf{q}}_\nu = \bar{\mathcal{G}}_0^{-1} \mathbf{W}_\nu, \quad \tilde{\Psi}_\nu = \mathcal{A} \bar{\mathbf{q}}_\nu.$$

Thus $\mathbf{W}_\nu$ is the identified component of the target direction, $\bar{\mathbf{q}}_\nu$ is its population Riesz representer, and $\tilde{\Psi}_\nu$ is the corresponding population debiasing weight. Define

$$\xi_{\nu,t} = \sum_{i=1}^N \tilde{\Psi}_{\nu,it} u_{it}, \quad V_{\nu,NT} = \sum_{t=1}^T \mathbb{E}_0(\xi_{\nu,t}^2 \mid \mathcal{F}_{t-1}), \quad s_{\nu,NT}^2 = \mathbb{E}_0 V_{\nu,NT}.$$

Assumption 6 (Target regularity). *The scalar target $\phi_\nu(\Theta_0)$ is regular if the following conditions hold.*

(i) (Relevance). *The target has nonnegligible projection onto the identified tangent space:*

$$\|\mathbf{W}_\nu\|_{\mathbb{H}}^2 \geq m_{\nu,NT} b_{NT}^2 (NT)^{-2}, \quad m_{\nu,NT} \rightarrow \infty.$$

(ii) (Nondegenerate variance). *The variance of the leading influence term is of the same order as the squared norm of its population debiasing weight:*

$$c_\nu \mathbb{E}_0 \|\tilde{\Psi}_\nu\|_F^2 \leq s_{\nu,NT}^2 \leq C_\nu \mathbb{E}_0 \|\tilde{\Psi}_\nu\|_F^2.$$

(iii) (Stable predictable variance).

$$V_{\nu,NT} / s_{\nu,NT}^2 \xrightarrow{p} 1.$$

(iv) (Influence dispersion). *No single cell and no contemporaneous dependence block carries a nonvanishing share of the influence variance. Let $\mathcal{N}_{NT}(k)$ denote the within-period dependence block of unit k . Under the maintained spatial-mixing framework, $\mathcal{N}_{NT}(k) = B_N(k, r_N) = \{i : d_N(i, k) \leq r_N\}$. Under the known-cluster alternative in Appendix A.2.4, $\mathcal{N}_{NT}(k) = \mathcal{C}_N(k)$,*

the known cluster containing k . The target must satisfy,

$$\frac{\|\tilde{\Psi}_\nu\|_{\max}^2}{\mathbb{E}_0\|\tilde{\Psi}_\nu\|_F^2} + \max_{t,k} \frac{\mathbb{E}_0 \left[\left(\sum_{i \in \mathcal{N}_{NT}(k)} \tilde{\Psi}_{\nu,it} u_{it} \right)^2 \right]}{s_{\nu,NT}^2} \rightarrow 0. \quad (4.5)$$

Under spatial mixing this condition is a no-dominant-local-neighborhood condition. Under known clusters it is the corresponding no-dominant-cluster score condition and allows arbitrary within-cluster covariance.

For a fixed finite target vector, these displays hold uniformly over variance-normalized linear combinations, and the corresponding correlation matrix converges to a positive-definite limit.

Assumption 6 is target-specific rather than a primitive restriction on the data-generating process. Regularity depends on how the chosen linear functional aligns with the identified tangent space and on how its debiasing weight is distributed over the panel. Condition (i) rules out weak identification of the target at the maintained rank. It fails when the target direction is nearly orthogonal to the identified tangent space. Conditions (ii)–(iii) ensure a nondegenerate and stable limiting variance, and condition (iv) is the no-dominant-leverage requirement needed for the stated central limit theorem. Failure of the assumption therefore indicates limited information for that specific target, typically because of weak identification or concentrated influence.

5 Choosing the Ranks

The rank criterion uses the same fold design as the estimator. Cells are assigned to validation folds $\mathcal{I}_j$, Π_j denotes projection onto fold j , and $\mathcal{I}_{-j}^{\text{buf}}$ denotes the training set obtained by removing the validation fold and its forward space–time buffer. The number of folds is J_{NT} , the forward exclusion length is q_T , and the spatial radius is r_N . The formal definition is given in Section 6.1.

Let $d_{NT} = N + T$, $a_{NT} = N^{-1} + T^{-1}$, and $\ell_{NT} = \{\log(NTJ_{NT})\}^{1/2}$. With the finite-moment envelope. b_{NT} defined in Section 4, set

$$\lambda_{NT} = b_{NT} \ell_{NT} \sqrt{a_{NT}}, \quad \zeta_{NT} = \lambda_{NT}^2, \quad \delta_{NT} = \lambda_{NT}. \quad (5.1)$$

These are derived stochastic rates, not additional tail assumptions; Appendix A.2 obtains them by truncation and Fuk–Nagaev/Rosenthal inequalities. The candidate box is (4.3), so $\mathbf{r}_0 \in \mathcal{R}$ by Assumption 4.

Define the one-neighborhood and overlap volumes

$$B_{NT} := (q_T + 1)v_N(r_N), \quad D_{NT} := (2q_T + 1)v_N(2r_N).$$

The theorem statements below impose the following common tuning-rate regime for the scattered

space–time-buffered procedure defined in [Section 6.1](#):

$$\sqrt{NT}\{\rho_*^{q_T} + \Omega_x(r_N) + \bar{\alpha}(r_N)\} \rightarrow 0, \quad \frac{B_{NT}}{J_{NT}} \rightarrow 0, \quad b_{NT}^4 \frac{(J_{NT} + D_{NT})(N + T) \log(NTJ_{NT})}{NT} \rightarrow 0. \quad (5.2)$$

The last restriction implies $\min\{T, N\}/\{J_{NT} \log(NTJ_{NT})\} \rightarrow \infty$ and $NT/J_{NT} \rightarrow \infty$. These are tuning and joint-asymptotic restrictions, not maintained data-generating assumptions. Fold balance, row/column coverage, retained-training share, leakage control, and information transport are proved in [Lemma 1](#), [Proposition 4](#), and [Lemma 3](#); none is maintained separately. For implementation, the growing-fold requirement is used through a finite-sample floor rule rather than through a literal fixed- J asymptotic sequence. Let

$$n_{\text{eff}} = \frac{NT}{N + T}, \quad L_{NT}^J = \max\{1, \log \log n_{\text{eff}}\}.$$

A theorem-admissible implementation may take

$$J_{NT} = \max\left\{J_{\min}, \lceil c_J B_{NT} L_{NT}^J \rceil\right\}, \quad (5.3)$$

where $J_{\min}$ is a conventional finite-sample floor and $c_J > 0$ is fixed. The second term grows slowly and enforces $B_{NT}/J_{NT} \rightarrow 0$, while the floor prevents the finite-sample validation folds from becoming too coarse. Thus a moderate fold count in simulations or applications should be interpreted as the finite-sample floor of the growing sequence in [\(5.3\)](#), not as a separate fixed- J asymptotic theorem. In the known-cluster alternative of [Appendix A.2.4](#), B_{NT} in [\(5.3\)](#) is replaced by B_{NT}^{cl} .

The ranks $(r_1, \dots, r_M, r_H)$ play the role that the number of factors plays in interactive-effects models: they index how rich the heterogeneity is allowed to be. This section gives the cross-fitted prediction criterion; implementation details are collected in [Appendix A.2.7](#). The candidate box is fixed in [\(4.3\)](#). The population underfitting gap is derived in the underfitting paragraph of [Appendix A.3.4](#) from [Assumptions 4](#) and [5](#), while the penalty separation used against overfitting follows algebraically from the explicit penalty definition below and the rate regime [\(5.2\)](#). [Section A.2.6](#) exhibits a panel regime in which the finite-moment rates, penalty, and buffer restrictions are jointly feasible.

5.1 A cross-fitted rank criterion

For a candidate rank vector $\mathbf{r} = (r_1, \dots, r_M, r_H) \in \mathcal{R}$, with $\mathcal{R}$ fixed by [\(4.3\)](#), define the corresponding model class

$$\mathcal{M}(\mathbf{r}) = \{\boldsymbol{\Theta} : \text{rank}(\mathbf{\Gamma}^{(m)}) \leq r_m, m \leq M, \text{rank}(\mathbf{H}) \leq r_H\}.$$

The population prediction risk of a candidate is the mean-squared error of the best fit available within its class,

$$L_0(\mathbf{r}) = \inf_{\boldsymbol{\Theta} \in \mathcal{M}_B(\mathbf{r})} \frac{1}{TN} \mathbb{E}_0 \left[\|\mathcal{A}(\boldsymbol{\Theta} - \boldsymbol{\Theta}_0)\|_F^2 \right], \quad (5.4)$$

where $\mathcal{A}$ is the design operator of [Section 3.4](#). Let $\Theta_{\mathbf{r}}^*$ denote a minimizer (the pseudo-true parameter at rank $\mathbf{r}$). The sample counterpart is an out-of-fold prediction loss: for each fold j and each candidate $\mathbf{r}$, compute the regularized estimator $\hat{\Theta}_{-j}^0(\mathbf{r}) \in \mathcal{M}(\mathbf{r})$ of [Section 6](#) using only the buffered training set $\mathcal{I}_{-j}^{\text{buf}}$, and evaluate its fit on the held-out cells:

$$\hat{L}(\mathbf{r}) = \frac{1}{TN} \sum_{j=1}^{J_{NT}} \left\| \Pi_j \{ \mathbf{Y} - \mathcal{A}(\hat{\Theta}_{-j}^0(\mathbf{r})) \} \right\|_F^2. \quad (5.5)$$

Because training excludes the first q_T dynamic descendants and the radius- r_N neighborhood of every held-out cell ([Appendix A.3.5](#)), the feedback remaining in (5.5) is of geometric-tail order. Hence the criterion is an *asymptotically leakage-controlled* forecast loss under $\{\rho_*^{q_T} + \Omega_x(r_N)\} \sqrt{NT} \rightarrow 0$, not an exactly independent validation loss in finite samples.

Richer candidates always fit better in sample, so the criterion charges each candidate for its effective number of free parameters,

$$d(\mathbf{r}) = \sum_{m=1}^M r_m(N + T - r_m) + r_H(N + T - r_H). \quad (5.6)$$

Because the cross-fitting rate regime (5.2) implies $\zeta_{NT} = a_{NT} b_{NT}^2 \ell_{NT}^2 \rightarrow 0$, fix any constant $c_\kappa \in (0, \infty)$ and define

$$\kappa_{NT} = c_\kappa b_{NT}^2 \ell_{NT}^2 \zeta_{NT}^{-1/2}. \quad (5.7)$$

[Lemma 2](#) proves that this choice dominates the overfitting-risk fluctuation while remaining negligible relative to the constant underfitting gap.

This is the dimension of the set of rank- r matrices, $r(N + T - r)$, not the raw count of factor coordinates; the subtraction of r^2 removes the rotational redundancy of factor models, so the penalty does not depend on the normalization used in estimation. The selected rank vector is

$$\hat{\mathbf{r}} = \arg \min_{\mathbf{r} \in \mathcal{R}} \left\{ \hat{L}(\mathbf{r}) + \kappa_{NT} \frac{d(\mathbf{r})}{TN} \right\}, \quad (5.8)$$

with ties broken in favor of the vector with the smaller effective dimension $d(\mathbf{r})$, and any remaining tie broken lexicographically, so the selected vector is unique. The explicit penalty in (5.7) satisfies

$$\frac{\kappa_{NT}}{b_{NT}^2 \ell_{NT}^2} = c_\kappa \zeta_{NT}^{-1/2} \rightarrow \infty, \quad a_{NT} \kappa_{NT} = c_\kappa \zeta_{NT}^{1/2} \rightarrow 0.$$

A consistent residual-scale multiplier may be used in computation without changing these orders; [Appendix A.2.7](#), Step 4, gives that operational version.

The conditional expectation in (5.4) makes every pseudo-true projection $\mathcal{G}_0$ -measurable and therefore predictable relative to a validation error. [Appendix A.3](#) proves the uniform out-of-fold risk expansion and rank-selection consistency. No standalone rank-margin or penalty assumption is imposed: candidate inclusion is built into (4.3), the underfitting gap is derived from [Assumptions 4](#)

and 5, and penalty separation follows from (5.7).

Two further questions—when distinct rank configurations are statistically distinguishable, and when underfitting is detectable out of fold—reduce to the population restricted-identification condition [Assumption 5](#); the corresponding underfitting margin is summarized in [Appendix A.3.4](#).

5.2 A practical roadmap to implementation

The criterion (5.8) uses the fixed candidate box (4.3) and requires choosing the number of folds J , the space–time exclusion buffer q_T , and the penalty scale. [Appendix A.2.7](#) gives an operational recipe based on a preliminary stability diagnostic, a one-sided persistence bound, spectral ordering of the fixed candidates, and a residual-scale calibration of the explicit penalty. Its *asymptotic* version lets J_{NT} dominate the complete buffer volume and is covered by the cross-fitting design in [Section 6.1](#), the rate regime (5.2), and the fixed candidate box and penalty in (4.3)–(5.7). In finite samples we implement the growing-fold requirement through the floor rule (5.3). The floor $J_{\min}$ is a conventional cross-fitting choice, such as five or ten folds, and inference is reported with retained-training-share and nearby- $J_{\min}$ robustness diagnostics. No separate fixed- J asymptotic theorem is claimed.

6 Estimation

Estimation is easiest to read as a sequence of computational steps, followed by mathematical definitions. The objects that enter this sequence have already been introduced. The operator $\mathcal{A}$ is the design operator in [Section 3.4](#); $\Theta = (\Gamma^{(1)}, \dots, \Gamma^{(M)}, \mathbf{H})$ collects the low-rank lag-coefficient, covariate-slope, and interactive-effect matrices; and linear targets have the form

$$\phi_\nu(\Theta_0) = \langle \mathbf{D}_\nu, \Theta_0 \rangle_{\mathbb{H}},$$

as in [Appendix A.3.5](#). The rank vector $\hat{\mathbf{r}}$ is selected by (5.8). For a chosen target direction $\mathbf{D}_\nu$, the researcher then implements the following steps.

Estimation roadmap.

Step 1. Create folds and buffers. Choose J using the finite-sample floor version of the growing-fold rule (5.3), and split cells into scattered validation folds. For each fold, remove from the training sample the forward dynamic descendants and the relevant contemporaneous dependence block of the held-out cells: a spatial neighborhood under the main spatial-mixing framework, or a known cluster under [Appendix A.2.4](#). This step defines the fold sets $\mathcal{I}_j$, the coordinate projections Π_j , the buffered training sets $\mathcal{I}_{-j}^{\text{buf}}$, the buffered projections Π_{-j}^{buf} , the fold shares $p_{j,NT}$, and the training-rescaling constants α_j . These objects are defined formally in [Section 6.1](#).

Step 2. Fit the low-rank coefficient matrices. On each buffered training set, solve the joint

rank-constrained least-squares problem at the selected rank $\hat{r}$. This produces fold-specific plug-in coefficient matrices

$$\widehat{\mathbf{A}}_{-j}^{(\ell)}, \quad \widehat{\mathbf{B}}_{-j}^{(k)}, \quad \widehat{\mathbf{H}}_{-j}.$$

Their entries are the fitted autoregressive coefficients, slope coefficients, and interactive components used for coefficient paths, maps, and numerical diagnostics.

Step 3. Construct the local tangent space. Compute the singular spaces of the fitted low-rank matrices. These singular spaces define the estimated tangent space, which is the local set of first-order coefficient perturbations allowed by the selected low-rank model.

Step 4. Compute debiasing weights. For the target direction $\mathbf{D}_\nu$, solve the Riesz normal equation on the estimated tangent space. The solution gives observation-space weights $\widehat{\Psi}_{\nu,-j}$ for the target.

Step 5. Form the one-step estimate and standard error. Add the held-out residual correction to the plug-in target and average across folds. The resulting estimator is the cross-fitted one-step estimate $\check{\phi}_\nu$. Standard errors are computed from the corresponding cross-fitted weights and residuals.

The plug-in coefficient matrices and the debiased target estimates serve different roles. The fitted coefficient matrices are descriptive estimates of heterogeneous dynamics and slopes. The confidence intervals in the paper are based on the one-step estimates $\check{\phi}_\nu$, not on raw plug-in values. The one-step correction removes the leading regularization and generated-regressor bias for the chosen linear target. The formal theory is stated for the ideal measurable rank-constrained minimizer. Finite alternating-ridge or Riemannian iterates are covered only when the objective-gap condition in (6.5) is verified; multistart agreement, rank-stability checks, and first-order residuals are numerical diagnostics, not certificates of global optimality. The subsections below give the mathematical definition of each step.

6.1 Step 1: Folds and space–time buffers

The first stage, rank criterion, and Riesz weights use a procedural fold design, not an additional data-generating assumption. Let $J_{NT} \geq 2$, $q_T \in \{0, \dots, T-1\}$, and $r_N \geq 0$ be analyst-chosen tuning sequences. Auxiliary labels $\{\sigma_a : a \in \mathcal{I}_{NT}\}$ are generated by randomness independent of the panel and included in $\mathcal{G}_0$, with

$$\sigma_a \stackrel{\text{iid}}{\sim} \text{Unif}\{1, \dots, J_{NT}\}.$$

They define the validation folds $\mathcal{I}_j = \{a \in \mathcal{I}_{NT} : \sigma_a = j\}$ and the buffered training sets

$$\mathcal{I}_{-j}^{\text{buf}} = \{(\ell, s) \notin \mathcal{I}_j : \nexists (i, t) \in \mathcal{I}_j \text{ such that } 0 \leq s - t \leq q_T, d_N(\ell, i) \leq r_N\}. \quad (6.1)$$

Let Π_j be the coordinate projection onto $\mathcal{I}_j$, let Π_{-j}^{buf} be the coordinate projection onto $\mathcal{I}_{-j}^{\text{buf}}$, write $p_{j,NT} = |\mathcal{I}_j|/(NT)$, and define the training-information rescaling $n_{-j}^{\text{buf}} = |\mathcal{I}_{-j}^{\text{buf}}|$ and $\alpha_j = NT/n_{-j}^{\text{buf}}$.

These quantities are procedural objects, not additional data-generating assumptions. The temporal radius removes dynamic descendants of each held-out cell, the spatial radius removes its local contemporaneous dependence neighborhood, and scattered labels keep time periods and units represented in every training problem. The tuning restrictions imposed in the theorem statements are collected in (5.2). $J_{NT}(N+T)\log(NTJ_{NT})/(NT)$ in those restrictions accounts for the fold-rescaling factor $p_{j,NT}^{-1} \asymp J_{NT}$. Appendix A.3.1 proves fold-information transport by conditioning on the realized panel and applying matrix concentration for the auxiliary fold randomization; buffer deletion and the temporal and spatial coupling tails are then controlled separately.

6.2 Step 2: Fitting the low-rank coefficient matrices

For a candidate rank vector $\mathbf{r}$, let $\mathcal{M}_B(\mathbf{r})$ be the rank class $\mathcal{M}(\mathbf{r})$ intersected with a fixed entrywise envelope $\|\Theta\|_{\max} \leq B$, where B is larger than the bound in Assumption 4. Define

$$\hat{\Theta}_{-j}^0(\mathbf{r}) \in \arg \min_{\Theta \in \mathcal{M}_B(\mathbf{r})} \frac{1}{2} \left\| \Pi_{-j}^{\text{buf}} \{ \mathbf{Y} - \mathcal{A}(\Theta) \} \right\|_F^2. \quad (6.2)$$

The compact envelope guarantees existence; any measurable tie-breaking rule may be used. An objective-gap equivalent estimator, with gap $o_{p,0}((TN)\zeta_{NT})$ uniformly over folds and candidates, has the same statistical conclusions. This definition is deliberately joint across blocks. A cellwise ambient ridge solution is not a valid coefficient-matrix estimator because y_{it} identifies only the scalar inner product of that cell's regressor vector and coefficient vector; the low-rank coupling across cells is what identifies the decomposition.

Computation. The criterion is nonconvex only because of the rank constraint. We compute it with multiple-start alternating ridge or Riemannian least squares, using blockwise spectral pilots and a vanishing ridge solely for numerical conditioning. To record the implementation, factor $\mathbf{\Gamma}^{(m)} = \mathbf{\Lambda}_m \mathbf{F}_m'$ and $\mathbf{H} = \mathbf{\Lambda}_H \mathbf{F}_H'$. Conditional on the loadings, stack $\mathbf{z}_t = (\mathbf{f}_{t,1}', \dots, \mathbf{f}_{t,M}', \mathbf{f}_{t,H}')'$ and $\mathbf{d}_{it} = (Z_{it}^{(1)} \boldsymbol{\lambda}'_{i,1}, \dots, Z_{it}^{(M)} \boldsymbol{\lambda}'_{i,M}, \boldsymbol{\lambda}'_{i,H})'$; one numerical time-factor update is

$$\hat{\mathbf{z}}_t = \left(\sum_{i:(i,t) \in \mathcal{I}_{-j}^{\text{buf}}} \mathbf{d}_{it} \mathbf{d}_{it}' + \mathbf{R}_F \right)^{-1} \left(\sum_{i:(i,t) \in \mathcal{I}_{-j}^{\text{buf}}} \mathbf{d}_{it} y_{it} \right). \quad (6.3)$$

Conditional on the factors, stack $\mathbf{w}_i = (\boldsymbol{\lambda}'_{i,1}, \dots, \boldsymbol{\lambda}'_{i,M}, \boldsymbol{\lambda}'_{i,H})'$ and $\mathbf{c}_{it} = (Z_{it}^{(1)} \mathbf{f}_{t,1}', \dots, Z_{it}^{(M)} \mathbf{f}_{t,M}', \mathbf{f}_{t,H}')'$; the companion unit-loading update is

$$\hat{\mathbf{w}}_i = \left(\sum_{t:(i,t) \in \mathcal{I}_{-j}^{\text{buf}}} \mathbf{c}_{it} \mathbf{c}_{it}' + \mathbf{R}_\Lambda \right)^{-1} \left(\sum_{t:(i,t) \in \mathcal{I}_{-j}^{\text{buf}}} \mathbf{c}_{it} y_{it} \right). \quad (6.4)$$

The numerical ridge matrices are positive semidefinite and vanish relative to the row and column information scales. The statistical theory concerns the exact measurable minimizer in (6.2), not a claim that every alternating iteration finds it. Appendix A.2.7 therefore reports the attained objective, first-order residual, multistart agreement, and rank-stability diagnostics. These diagnostics are reassuring numerical checks, but they do not certify global optimality or the objective-gap condition for a nonconvex problem. The main theorems therefore concern the ideal measurable global minimizer; a finite numerical iterate is covered only when an independent global lower bound or a separate optimization theorem verifies the required gap. Optional sufficient conditions for the row-wise tangent-rate envelope are collected in Appendix A.3.5; they are not additional maintained assumptions.

Implementation requirement for a finite nonconvex iterate. The statistical results concern the exact measurable minimizer of (6.2). If a computed estimator is instead a finite alternating-ridge or Riemannian iterate, uniformly over folds and candidate ranks we require

$$Q_{-j}(\hat{\Theta}_{-j}^{\text{alg}}) - \inf_{\Theta \in \mathcal{M}_B(r)} Q_{-j}(\Theta) = o_{p,0}(NT\zeta_{NT}), \quad (6.5)$$

and local stability of the selected numerical rank. This requirement is vacuous for the exact minimizer. Appendix A.3 gives local basin and contraction conditions sufficient for it, but no DGP assumption is claimed to certify global optimization of a nonconvex criterion.

Why the global and rowwise rates follow. The basic inequality for (6.2), the fixed-rank secant entropy, Assumptions 3 and 5, and the localization sequence b_{NT} give prediction and parameter risk $O_{p,0}(\zeta_{NT})$. Pervasiveness converts coefficient-matrix error into the global singular-space rate $O_{p,0}(\delta_{NT})$. Define the derived slice-deletion and coupling envelope

$$d_{\text{slice},NT} = \rho_*^{qT} + \Omega_x(r_N) + \bar{\alpha}(r_N) + b_{NT}^2 \left\{ \frac{qT + 1}{T} + \frac{v_N(r_N)}{N} \right\},$$

and the rowwise envelope

$$\delta_{\max,NT} = b_{NT}\ell_{NT} \left(N^{-1/2} + T^{-1/2} \right) + \delta_{NT}^2 + d_{\text{slice},NT}.$$

Proposition 6 uses the local row/column information, Davydov-type covariance and blocking inequalities for strong-mixing arrays (Doukhan, 1994, covariance inequality for mixing variables),

and the canonical space–time coupling to prove, uniformly over truth-rank fold fits,

$$\begin{aligned}
\sqrt{N} \max_i \|e'_i(\hat{U}_{m,-j} O_{U,m,j} - U_{0,m})\|_2 &= O_{p,0}(\delta_{\max,NT}), \\
\sqrt{T} \max_t \|e'_t(\hat{V}_{m,-j} O_{V,m,j} - V_{0,m})\|_2 &= O_{p,0}(\delta_{\max,NT}), \\
(NT)^{-1/2} \|O'_{U,m,j} \hat{\Sigma}_{m,-j} O_{V,m,j} - \Sigma_{0,m}\|_{\text{op}} &= O_{p,0}(\delta_{\max,NT}), \\
\max_m \|\hat{\Gamma}_{-j}^{(m),0} - \Gamma_0^{(m)}\|_{\max} + \|\hat{H}_{-j}^0 - H_0\|_{\max} &= O_{p,0}(\delta_{\max,NT}).
\end{aligned} \tag{6.6}$$

Thus the rowwise rate is a theorem, not an additional target assumption.

6.3 Step 3: Tangent spaces and debiasing weights

After rank selection, compute the truncated SVD of each preliminary component at the selected rank. These singular spaces define the estimated set of admissible local perturbations,

$$\hat{\mathcal{T}}_{-j} = \hat{\mathcal{T}}_{\Gamma,1,-j} \times \cdots \times \hat{\mathcal{T}}_{\Gamma,M,-j} \times \hat{\mathcal{T}}_{H,-j},$$

the sample counterpart of $\mathcal{T}_0$ from [Appendix A.3.5](#). Throughout, $\mathcal{P}_{\hat{\mathcal{T}}_{-j}}$ denotes the Hilbert-space orthogonal projection onto this estimated tangent space. For target ν , the feasible local information map on the buffered training set and the feasible debiasing weights are

$$\hat{\mathcal{G}}_{\nu,-j} = \alpha_j \mathcal{P}_{\hat{\mathcal{T}}_{-j}} \mathcal{A}^* \Pi_{-j}^{\text{buf}} \mathcal{A} \mathcal{P}_{\hat{\mathcal{T}}_{-j}}, \quad \hat{\mathbf{q}}_{\nu,-j} = \hat{\mathcal{G}}_{\nu,-j}^+ \mathcal{P}_{\hat{\mathcal{T}}_{-j}} \mathbf{D}_{\nu}, \tag{6.7}$$

the exact analogue of solving the normal equations $(X'X)\mathbf{q} = (\text{target direction})$ in a linear regression, restricted to the estimated local perturbation space. Here $\hat{\mathcal{G}}_{\nu,-j}^+$ denotes a numerically truncated inverse used only as a computational safeguard: under the empirical restricted-eigenvalue bound of [Assumption 5](#) the buffered training information map is nonsingular with probability tending to one, the truncation is inactive, and $\hat{\mathcal{G}}_{\nu,-j}^+$ equals the genuine inverse, so the exact feasible debiasing-weight identity holds with no range projection and no induced bias. The observation-space weight is

$$\hat{\Psi}_{\nu,-j} = \mathcal{A} \hat{\mathbf{q}}_{\nu,-j}.$$

Invertibility of $\hat{\mathcal{G}}_{\nu,-j}$ on the identified subspace follows from the buffered restricted-eigenvalue condition together with stability of the estimated perturbation spaces: under the scattered folds the true buffered information map is uniformly positive definite on $\mathcal{T}_0$, and $\|\mathcal{P}_{\hat{\mathcal{T}}_{-j}} - \mathcal{P}_{\mathcal{T}_0}\|_{\text{op}} = o_{p,0}(1)$ implies that the estimated map inherits the same lower eigenvalue for all large T, N . The held-out fold is used only in the final residual correction; the debiasing direction itself is computed from the forward-excluded complement. The implemented direction can still depend on other observations from the held-out time period through the estimated geometry, so it is not declared exactly predictable. [Appendix A.4](#) compares it with a leave-one-slice, forward-coupled direction and then with the genuinely predictable population-information weight. This is the dynamic analogue of

cross-fitted debiased machine learning and is the step that removes both future descendant leakage and hidden within-period independence requirements.

6.4 Step 4: The cross-fitted one-step estimator

Let

$$\widehat{\mathbf{R}}_{-j}^0 = \mathbf{Y} - \mathcal{A}(\widehat{\boldsymbol{\Theta}}_{-j}^0(\widehat{\mathbf{r}})), \quad \widehat{\phi}_{\nu,-j}^0 = \left\langle \mathbf{D}_{\nu}, \widehat{\boldsymbol{\Theta}}_{-j}^0(\widehat{\mathbf{r}}) \right\rangle_{\mathbb{H}},$$

the out-of-fold residual panel and the plug-in value of the target. Recall $p_{j,NT} = |\mathcal{I}_j|/(TN)$, so $\sum_{j=1}^{J_{NT}} p_{j,NT} = 1$ and, by fold balance, $\max_{1 \leq j \leq J_{NT}} |J_{NT} p_{j,NT} - 1| = o_{p,0}(1)$. The cross-fitted one-step estimator adds to each plug-in a weighted average of its own held-out residuals,

$$\check{\phi}_{\nu} = \sum_{j=1}^{J_{NT}} p_{j,NT} \left[\widehat{\phi}_{\nu,-j}^0 + p_{j,NT}^{-1} \left\langle \Pi_j \widehat{\boldsymbol{\Psi}}_{\nu,-j}, \Pi_j \widehat{\mathbf{R}}_{-j}^0 \right\rangle \right]. \quad (6.8)$$

The weights $p_{j,NT}$ average the fold plug-ins ($\sum_j p_{j,NT} = 1$), while the factor $p_{j,NT}^{-1}$ restores the full-sample scale of the held-out inner product, which uses only the $p_{j,NT}$ fraction of cells in fold j . With growing and possibly unequal folds this replaces the fixed $1/J$ and J of equal-fold cross-fitting. For a dynamic group mean target,

$$\check{\theta}_{G,t,\ell}^A = \sum_{j=1}^{J_{NT}} p_{j,NT} \left[\boldsymbol{\pi}'_G \widehat{\mathbf{A}}_{-j}^{(\ell),0} \mathbf{e}_t + p_{j,NT}^{-1} \left\langle \Pi_j \widehat{\boldsymbol{\Psi}}_{G,t,A,\ell,-j}, \Pi_j \widehat{\mathbf{R}}_{-j}^0 \right\rangle \right].$$

For a static group mean target,

$$\check{\theta}_{G,t,k}^B = \sum_{j=1}^{J_{NT}} p_{j,NT} \left[\boldsymbol{\pi}'_G \widehat{\mathbf{B}}_{-j}^{(k),0} \mathbf{e}_t + p_{j,NT}^{-1} \left\langle \Pi_j \widehat{\boldsymbol{\Psi}}_{G,t,B,k,-j}, \Pi_j \widehat{\mathbf{R}}_{-j}^0 \right\rangle \right].$$

6.5 Step 5: Standard errors and sensitivity intervals

Define cellwise cross-fitted weights and residuals by

$$\widehat{\boldsymbol{\Psi}}_{\nu,it}^{\text{cf}} = [\widehat{\boldsymbol{\Psi}}_{\nu,-j(i,t)}]_{it}, \quad \widehat{u}_{it}^{\text{cf}} = [\widehat{\mathbf{R}}_{-j(i,t)}^0]_{it}.$$

The maintained weak-dependence framework uses the within-period spatial-kernel quadratic form

$$\widehat{s}_{\nu,\text{sp}}^2 = \sum_{t=1}^T \sum_{i,j=1}^N K\left(\frac{d_N(i,j)}{h_N}\right) \widehat{\boldsymbol{\Psi}}_{\nu,it}^{\text{cf}} \widehat{\boldsymbol{\Psi}}_{\nu,jt}^{\text{cf}} \widehat{u}_{it}^{\text{cf}} \widehat{u}_{jt}^{\text{cf}}. \quad (6.9)$$

Here K is bounded, symmetric, compactly supported, and continuous at zero, and h_N is selected subject to the conditions stated immediately before Theorem 5. Conditional cross-sectional independence is the nested special case in which the off-diagonal covariances vanish; then (6.9) simplifies to

the diagonal form:

$$\hat{s}_{\nu,\text{diag}}^2 = \sum_{i,t} (\hat{\Psi}_{\nu,it}^{\text{cf}})^2 (\hat{u}_{it}^{\text{cf}})^2. \quad (6.10)$$

When a credible metric and an admissible bandwidth are available, the spatial-kernel form is the theorem-aligned dependence-robust studentizer. When no credible cross-sectional metric is available but units can be partitioned into known clusters $\mathcal{C}_{g,N}$, Appendix A.2.4 gives an exact known-cluster alternative. The estimator is unchanged, the space buffer is replaced by a cluster buffer, and the studentizer becomes the cluster-score quadratic form

$$\hat{s}_{\nu,\text{cl}}^2 = \sum_{t=1}^T \sum_{g=1}^{G_N} \left(\sum_{i \in \mathcal{C}_{g,N}} \hat{\Psi}_{\nu,it}^{\text{cf}} \hat{u}_{it}^{\text{cf}} \right)^2.$$

In empirical tables that also report by-period clustered intervals, those intervals are sensitivity diagnostics only; they are not the theorem's spatial-kernel object and are not guaranteed conservative without additional saturated-bandwidth replacement conditions.

7 Asymptotic Theory

This section states the main inferential results in the order in which the estimator is justified. The framework remains the maintained framework of Section 4: weak exogeneity rather than sequential exogeneity, finite $(8 + \eta)$ moments, weak contemporaneous dependence in the main model, space-time-buffered cross-fitting, a full-covariance influence representation, and dependence-robust studentization. The sequence below makes the proof chain explicit. First, an infeasible benchmark with the true local geometry has a predictable full-covariance influence representation. Second, the rank selector and the rank-constrained first stage recover the local low-rank geometry. Third, feasible cross-fitted Riesz weights are first-order equivalent to their benchmark counterparts. Fourth, the feasible one-step estimator inherits the benchmark central limit theorem. Fifth, the spatial-kernel studentizer consistently estimates the full contemporaneous variance. The maintained statements use spatial mixing. If no meaningful cross-sectional distance is available, Appendix A.2.4 states an exact known-cluster alternative under which the same five results hold after replacing the spatial buffer and spatial-kernel studentizer by their cluster analogues. The proofs of the five main theorems in this section are collected in Appendix A.1; the comments following each theorem explain the meaning and econometric role of the result.

Recall the benchmark objects defined before Assumption 6: $\bar{\mathcal{G}}_0$, $\bar{\mathbf{q}}_\nu$, $\tilde{\Psi}_\nu$, $\xi_{\nu,t}$, $V_{\nu,NT}$, and $s_{\nu,NT}^2$. The cellwise benchmark weight depends on the current observation only through $(x_{it}, y_{i,t-1}, \dots, y_{i,t-P})$ and is therefore $\mathcal{P}_{it}$ -measurable. Weak exogeneity is used coordinate by coordinate:

$$\mathbb{E}_0[\xi_{\nu,t} \mid \mathcal{F}_{t-1}] = \sum_i \mathbb{E}_0 \left[\tilde{\Psi}_{\nu,it} \mathbb{E}_0(u_{it} \mid x_{it}, \mathcal{F}_{t-1}) \mid \mathcal{F}_{t-1} \right] = 0.$$

Thus no vector restriction of the form $\mathbb{E}_0[\mathbf{u}_t \mid \mathbf{X}_t, \mathcal{F}_{t-1}] = 0$ is imposed. The variance scale is the

full predictable quadratic-variation scale

$$s_{\nu,NT}^2 = \mathbb{E}_0 \sum_{t=1}^T \mathbb{E}_0 \left[\left(\sum_{i=1}^N \tilde{\Psi}_{\nu,it} u_{it} \right)^2 \middle| \mathcal{F}_{t-1} \right],$$

or, equivalently,

$$s_{\nu,NT}^2 = \mathbb{E}_0 \sum_{t=1}^T \sum_{i,j=1}^N \mathbb{E}_0 \left[\tilde{\Psi}_{\nu,it} \tilde{\Psi}_{\nu,jt} u_{it} u_{jt} \middle| \mathcal{F}_{t-1} \right].$$

This expression keeps the current-regressor-measurable weights inside the conditional expectation and therefore does not treat $\tilde{\Psi}_{\nu,it}$ as $\mathcal{F}_{t-1}$ -measurable. Thus contemporaneous cross-sectional covariance is part of the main limit theory, not an add-on. Assumption 6 supplies target relevance, full-covariance noncancellation, predictable-variance stabilization, and influence dispersion.

All information-transport operators in the next display are normalized as averages over the NT cell design, and all operator norms are taken on the Hilbert-unit tangent sphere $\{\Delta \in \mathcal{T}_0 : \|\Delta\|_{\mathbb{H}} = 1\}$. With fixed ranks, the effective dimension of this sphere is $O(N + T)$, not $O(NT)$. The factor J_{NT} enters only because a validation fold contains a fraction $p_{j,NT} \asymp J_{NT}^{-1}$ of the cells and the fold operator is rescaled by $p_{j,NT}^{-1}$. Thus the validation concentration term is

$$\sqrt{\frac{J_{NT}(N + T) \log(NT J_{NT})}{NT}},$$

with the accompanying Bernstein linear term. The buffer contributes B_{NT}/J_{NT} because, for any fixed training cell, the probability of being deleted by the validation cells in one fold is bounded by the size of its complete space-time buffer neighborhood divided by the number of folds; the overlap-dependence fluctuation is controlled separately through D_{NT} . This normalization is the reason fold transport depends on the low-rank complexity $N + T$ and the deleted share B_{NT}/J_{NT} , rather than on a full NT -dimensional parameter count.

Define the derived fold-transport rate

$$\begin{aligned} \varepsilon_{G,NT} := & b_{NT}^2 \left[\sqrt{\frac{J_{NT}(N + T) \log(NT J_{NT})}{NT}} + \frac{J_{NT}(N + T) \log(NT J_{NT})}{NT} \right] \\ & + b_{NT}^2 \left[\sqrt{\frac{D_{NT}(N + T) \log(NT J_{NT})}{NT}} + \frac{D_{NT}(N + T) \log(NT J_{NT})}{NT} \right] \\ & + \frac{B_{NT}}{J_{NT}} + \rho_*^{qT} + \Omega_x(r_N) + \bar{\alpha}(r_N). \end{aligned}$$

and the finite-moment feasible-remainder envelope

$$r_{\nu,NT} := b_{NT} \delta_{\max,NT} + \varepsilon_{G,NT} + b_{NT}^2 \zeta_{NT} + \rho_*^{qT} \sqrt{NT} + \Omega_x(r_N) \sqrt{NT} + \bar{\alpha}(r_N) \sqrt{NT} \longrightarrow 0. \quad (7.1)$$

Appendices A.3, A.3.1, and A.4.2 derive the rowwise recovery, fold transport, Lyapunov control,

conditional variance regularity, and no-dominant-neighborhood leverage conditions for the standard linear target class: entries, group means, full means, contrasts, and cumulative persistence.

Theorem 1 (Infeasible-benchmark influence representation). *Let ν be a regular scalar target in the sense of Assumption 6, and define the benchmark linear term*

$$\tilde{L}_\nu = \sum_{i=1}^N \sum_{t=1}^T \tilde{\Psi}_{\nu,it} u_{it}.$$

Under Assumptions 1–6,

$$\frac{\tilde{L}_\nu}{s_{\nu,NT}} \xrightarrow{d} N(0, 1).$$

For every fixed finite vector of regular targets, the corresponding vector of benchmark linear terms converges, after deterministic covariance normalization, to a mean-zero Gaussian vector with the full contemporaneous covariance matrix.

Comment. The theorem isolates the infeasible object: population Riesz weights and the true tangent space are known, so no first-stage or numerical optimization error is present. The result is the martingale central limit theorem for the target-specific weighted idiosyncratic error array, with the full contemporaneous covariance retained.

Theorem 2 (Rank and first-stage verification). *Under Assumptions 1–5, the cross-fitting design of Section 6.1, and the rate regime (5.2), uniformly over folds, and, when a finite numerical iterate is used, additionally under the implementation requirement (6.5),*

$$\mathbb{P}_0(\hat{\mathbf{r}} = \mathbf{r}_0) \rightarrow 1, \quad \frac{1}{NT} \|\mathcal{A}(\hat{\Theta}_{-j}^0 - \Theta_0)\|_F^2 = O_{p,0}(\zeta_{NT}),$$

$$\frac{1}{NT} \|\hat{\Theta}_{-j}^0 - \Theta_0\|_{\mathbb{H}}^2 = O_{p,0}(\zeta_{NT}), \quad \|\hat{\mathcal{P}}_{\mathcal{T},-j} - \mathcal{P}_{\mathcal{T}_0}\|_{\text{op}} = O_{p,0}(\delta_{NT}),$$

and the aligned row, column, and entrywise errors are $O_{p,0}(\delta_{\max,NT})$. The statement concerns the ideal measurable global minimizer in (6.2); a computed nonconvex iterate requires a separately verified global objective gap.

Comment. The theorem is the bridge between the ideal low-rank model and the debiasing procedure: it states that the selected rank, fitted coefficient matrices, and estimated tangent spaces are accurate enough for inference. Its computational qualification is deliberately separate. Multistart agreement, rank stability, and first-order residual checks are reassuring numerical diagnostics, but they do not certify global optimality or the objective-gap condition.

Theorem 3 (Feasible Riesz-weight replacement). *Let ν be a regular target. Under Assumptions 1–6, the rate regime (5.2), and the feasible-remainder condition (7.1), the cross-fitted feasible Riesz*

weights are first-order equivalent to the benchmark weights:

$$\left\{ \sum_j \|\Pi_j(\widehat{\Psi}_{\nu,-j} - \Psi_{\nu,j}^{\text{or}})\|_F^2 \right\}^{1/2} = o_{p,0}(\|\Psi_{\nu}^{\text{or}}\|_F), \quad (7.2)$$

and the residual first-stage bias left after the one-step correction is negligible at the target scale:

$$\sum_j p_{j,NT} \left| \widehat{\mathbf{r}}_{\nu,-j}(\widehat{\Theta}_{-j}^0 - \Theta_0) \right| = o_{p,0}(s_{\nu,NT}). \quad (7.3)$$

Consequently, replacing the infeasible benchmark weights by the feasible space-time-buffered weights changes the leading score by $o_{p,0}(s_{\nu,NT})$.

Comment. The theorem is the formal leakage-removal statement. The feasible fold-specific Riesz weights can be transported to the benchmark weights because the validation cells are separated from their temporal descendants and nearby contemporaneous neighbors, and because the low-rank tangent directions are normalized in the global Hilbert metric.

Theorem 4 (Feasible central limit theorem). *Let ν be a regular target in the sense of Assumption 6. Under Assumptions 1–6, the rate regime (5.2), the remainder condition (7.1), and, when a finite numerical iterate is used, the implementation requirement (6.5),*

$$\check{\phi}_{\nu} - \phi_{\nu}(\Theta_0) = \widetilde{L}_{\nu} + o_{p,0}(s_{\nu,NT}), \quad \frac{\check{\phi}_{\nu} - \phi_{\nu}(\Theta_0)}{s_{\nu,NT}} \xrightarrow{d} N(0, 1).$$

For every fixed finite target vector, deterministic covariance normalization yields the corresponding mean-zero Gaussian limit with the full contemporaneous covariance matrix.

Comment. The theorem is the main feasible inference result. It combines first-stage localization, feasible Riesz-weight replacement, and the benchmark martingale CLT so that the actual one-step estimator has the same first-order distribution as the infeasible weighted score.

Studentization rate requirement. The kernel K is bounded, symmetric, compactly supported, continuous at zero, and satisfies $K(0) = 1$. The bandwidth $h_N \rightarrow \infty$ is selected so that

$$v_N(h_N)r_{\nu,NT} \rightarrow 0, \quad v_N(h_N)^2 b_{NT}^4 a_{NT} \ell_{NT}^2 \rightarrow 0, \quad \sum_{r>h_N} (1+r)^{d_s-1} \alpha(r)^{\chi_{\eta}} \rightarrow 0. \quad (7.4)$$

Appendix A.4.1 verifies that (7.4) yields ratio-consistent spatial-kernel studentization.

Theorem 5 (Dependence-robust studentization). *Under the assumptions of Theorem 4 and the studentization rate requirement (7.4),*

$$\widehat{s}_{\nu,\text{sp}}^2 / s_{\nu,NT}^2 \xrightarrow{p} 1, \quad \frac{\check{\phi}_{\nu} - \phi_{\nu}(\Theta_0)}{\widehat{s}_{\nu,\text{sp}}} \xrightarrow{d} N(0, 1).$$

The same result holds jointly with the full estimated contemporaneous covariance matrix. Cumulative persistence is covered whenever it is included as a linear combination of lag-coefficient targets satisfying Assumption 6.

Comment. The theorem identifies the dependence-robust standard error covered by the paper: a spatial-kernel covariance estimator under a specified metric and admissible bandwidth. The diagonal sandwich is the conditional-independence special case; by-period clustered intervals reported in the applications are sensitivity diagnostics, not consequences of this theorem.

What is theorem-backed. The standard error in Theorem 5 is the spatial-kernel standard error with a specified metric and bandwidth satisfying (7.4). The diagonal cell-heteroskedastic sandwich is the conditional-cross-sectional-independence special case. Appendix A.2.4 also gives a known-cluster route: with arbitrary within-cluster dependence and conditional independence across clusters after conditioning on period- t aggregate/common information, the same feasible CLT is studentized by the cluster-score variance estimator. Under Alternative Assumption 1', Assumption 2 read with the enlarged pre-idiosyncratic-innovation field, Alternative Assumption 3', and Assumptions 4–6, the five main theorems continue to hold with the cluster-buffer and cluster-studentization substitutions. A by-period clustered standard error corresponds to the saturated endpoint that gives every unit in a period positive covariance weight. It is therefore reported below only as a by-period clustered sensitivity interval unless additional high-level conditions verify the same replacement rate at that saturated bandwidth; it is not theorem-backed in the present framework, not guaranteed conservative, and not a co-equal consequence of Theorem 5.

Theorem 5 applies to entries, group means, full means, between-group contrasts, and fixed finite vectors of such standard linear targets once the corresponding target directions satisfy Assumption 6. Cumulative persistence is included because it is a linear combination of lag-coefficient targets. The formal leverage calculations, variance formulas, and proofs are in Appendix A.4.2. In particular, contrasts are studentized with their own covariance term rather than by comparing two marginal intervals. Nonlinear fixed-coefficient impulse responses and long-run multipliers are deliberately not main targets of the paper. They can be studied as smooth transforms only under additional stability, differentiability, and studentization conditions that are not imposed in the maintained theory.

8 Monte Carlo Experiments

This section evaluates the finite-sample performance of the proposed estimator. We consider the following data-generating process (DGP) with one lagged outcome and a scalar covariate:

$$y_{it} = a_{it} y_{i,t-1} + \beta_{it} x_{it} + c_{\xi}(h_{it} + u_{it}), \quad i = 1, \dots, N; t = -49, -48, \dots, 0, 1, \dots, T, \quad (8.1)$$

with $y_{i,-50} = 0$. We discard the initial 50 burn-in periods and retain $t = 1, \dots, T$ for the estimation sample.

We generate the raw autoregressive coefficients as $a_{it}^{raw} = \lambda_{a,i} f_{a,t}$ and the slope coefficients as $\beta_{it} = \lambda_{b,i} f_{b,t}$. For $m \in \{a, b\}$, the latent factors follow

$$\begin{aligned} f_{m,t} &= \mu_{f,m} + \kappa_{f,m} g_{m,t}, \quad t = -49, \dots, 0, 1, \dots, T, \\ g_{m,t} &= \rho_g g_{m,t-1} + (1 - \rho_g^2)^{1/2} v_{m,t}, \quad v_{m,t} \stackrel{iid}{\sim} N(0, 1), \quad g_{m,-50} = 0, \end{aligned}$$

with $\rho_g = 0.5$. For the autoregressive coefficients, we set $\mu_{f,a} = 0.5$, $\kappa_{f,a} = 0.1$, and generate the factor loadings from $\lambda_{a,i} \stackrel{iid}{\sim} N(1, 0.1^2)$. To ensure dynamic stability, we scale a_{it}^{raw} by $a_{it} = c_a a_{it}^{raw}$, where $c_a = \min \left\{ 1, \frac{0.85}{\max_{i,t} |a_{it}^{raw}|} \right\}$. For the slope coefficients, we set $\mu_{f,b} = 0.6$, $\kappa_{f,b} = 0.2$, and generate the factor loadings from $\lambda_{b,i} \stackrel{iid}{\sim} N(1, 0.4^2)$.

The interactive fixed effect is generated as $h_{it} = c_h \lambda_{h,i} g_{h,t}$, where $\lambda_{h,i} \stackrel{iid}{\sim} N(0, 1)$, and

$$g_{h,t} = \rho_g g_{h,t-1} + (1 - \rho_g^2)^{1/2} v_{h,t}, \quad v_{h,t} \stackrel{iid}{\sim} N(0, 1), \quad g_{h,-50} = 0.$$

The initial 50 observations for all the factor processes are discarded. Let $\xi_{it} = h_{it} + u_{it}$ denote the composite error. The scalar c_h is introduced to control the share of the interactive fixed effect in the variance of the composite error and will be specified below.

By construction, the three unknown coefficient and interactive effect matrices $\mathbf{A}_0 = (a_{it})$, $\mathbf{B}_0 = (\beta_{it})$, and $\mathbf{H}_0 = (h_{it})$ are exact rank-one matrices, so the true rank vector is $(r_A, r_B, r_H) = (1, 1, 1)$.

We consider three designs that cover different error and covariate processes:

DGP 1: Independent heteroskedastic errors. $u_{it} = \sigma_i \varepsilon_{it}$, $\varepsilon_{it} \stackrel{iid}{\sim} N(0, 1)$, and $\sigma_i^2 \stackrel{iid}{\sim} U(0.5, 1.5)$.

DGP 2: Spatially dependent heteroskedastic errors. The second design introduces weak cross-sectional dependence. Units are arranged on a one-dimensional lattice, and the distance between units i and j is $d_N(i, j) = |i - j|$. Draw $\varepsilon_{it} \stackrel{iid}{\sim} N(0, 1)$, and $\sigma_i^2 \stackrel{iid}{\sim} U(0.5, 1.5)$. Let $\boldsymbol{\varepsilon}_t = (\varepsilon_{1t}, \dots, \varepsilon_{Nt})'$ and $\mathbf{D}_\sigma = \text{diag}(\sigma_1, \dots, \sigma_N)$. The idiosyncratic error vector is generated as $\mathbf{u}_t = \mathbf{D}_\sigma \mathbf{R}_N^{1/2} \boldsymbol{\varepsilon}_t$, where $\mathbf{R}_N = (R_{ij})_{i,j \leq N}$ is the spatial correlation matrix with the (i, j) th element given by $R_{ij} = \rho_s^{|i-j|}$, and the spatial dependence parameter is set to $\rho_s = 0.5$. Then, conditional on the unit-specific variances, we have $\text{Cov}(u_{it}, u_{jt} \mid \sigma_1, \dots, \sigma_N) = \sigma_i \sigma_j \rho_s^{|i-j|}$. Therefore, $\text{Var}(u_{it} \mid \sigma_1, \dots, \sigma_N) = \sigma_i^2$, while the cross-sectional covariance decays geometrically with the distance $d_N(i, j)$.

For DGP 1 and DGP 2, we generate x_{it} as follows:

$$x_{it} = \rho_x x_{i,t-1} + \delta_x \lambda_{x,i} f_{x,t} + (1 - \rho_x^2)^{1/2} e_{it}, \quad t = -49, \dots, 0, 1, \dots, T, \quad (8.2)$$

with $\rho_x = 0.5$, $\lambda_{x,i} \stackrel{iid}{\sim} N(0, 1)$, $e_{it} \stackrel{iid}{\sim} N(0, \sigma_{e,i}^2)$, $\sigma_{e,i}^2 \stackrel{iid}{\sim} U(0.5, 1.5)$, and $x_{i,-50} = 0$ for all i . The parameter δ_x controls the strength of the common component in x_{it} , and we set $\delta_x = 0.5$. The

factor $f_{x,t}$ is generated as

$$f_{x,t} = \rho_{fx} f_{x,t-1} + (1 - \rho_{fx}^2)^{1/2} v_{x,t}, \quad v_{x,t} \stackrel{iid}{\sim} N(0, 1), \quad f_{x,-50} = 0,$$

with $\rho_{fx} = 0.5$.

DGP 3: Predetermined covariates. The third design keeps the error process from DGP 2 but allows x_{it} to depend on lagged idiosyncratic shocks. Specifically,

$$x_{it} = \rho_x x_{i,t-1} + \delta_x \lambda_{x,i} f_{x,t} + \eta_x u_{i,t-1} + (1 - \rho_x^2)^{1/2} e_{it}, \quad t = -49, \dots, 0, 1, \dots, T, \quad (8.3)$$

where $\eta_x = 0.3$. The other components are generated in the same way as those in (8.2).

In all experiments, we calibrate the parameter c_h so that the population share of the interactive effect in the variance of the composite disturbance $\xi_{it} = h_{it} + u_{it}$ equals $\pi_h = 0.3$. In all DGPs, $E(\sigma_i^2) = 1$, and the unconditional variance of u_{it} is one. Moreover, h_{it} and u_{it} are independent and mean zero. Therefore,

$$\frac{\text{Var}(h_{it})}{\text{Var}(h_{it} + u_{it})} = \frac{c_h^2}{c_h^2 + 1}.$$

Setting this share equal to $\pi_h = 0.3$ gives $c_h = \left(\frac{\pi_h}{1-\pi_h}\right)^{1/2} \approx 0.655$. Finally, we control the overall signal-to-noise ratio through the scalar c_ξ so that the pooled R-squared (PR^2) equals a target value, where

$$PR^2 = 1 - \frac{c_\xi^2 \sum_{i,t} (\xi_{it} - \bar{\xi})^2}{\sum_{i,t} [y_{it}(c_\xi) - \bar{y}(c_\xi)]^2}, \quad (8.4)$$

where $\bar{\xi} = (NT)^{-1} \sum_{i,t} \xi_{it}$ and $\bar{y} = (NT)^{-1} \sum_{i,t} y_{it}$. For each DGP and panel size, before running the Monte Carlo replications, we use a separate deterministic calibration sample to choose the scalar c_ξ . Specifically, c_ξ is chosen so that the average value of PR^2 over the calibration draws equals 0.5. This calibration sample is not part of the reported Monte Carlo replications. After c_ξ is fixed for that DGP–panel size cell, the random coefficient matrices, covariates, and shocks are redrawn with replication-specific seeds in each Monte Carlo replication. Consequently, the realized PR^2 varies across replications around the target value.

For DGP 1, inference uses the cross-sectionally independent heteroskedasticity-robust variance estimator given by (6.10). For DGP 2 and DGP 3, inference uses the spatial-kernel covariance estimator in (6.9) based on the lattice distance $d_N(i, j) = |i - j|$. We use the Bartlett kernel, and the baseline bandwidth is $h_N = \lfloor N^{1/3} \rfloor$, where $\lfloor \cdot \rfloor$ denotes the greatest integer less than or equal to its argument.

We consider combinations of sample sizes, $N, T = 50, 100, 200, 400$. The number of replications for all experiments is 1,000.

We evaluate rank-selection accuracy by recording the fraction of Monte Carlo replications in which the cross-fitted rank criterion selects the true rank vector $(1, 1, 1)$. Table 5 reports the rank recovery probabilities.

Reported Monte Carlo quantities. The fixed-rank performance tables use the true rank vector $(1, 1, 1)$. For each DGP, panel size, and displayed full-mean target, the manuscript-facing tables report the true value, bias, RMSE, mean estimated standard error, empirical size, and empirical coverage. Size is the empirical rejection probability of the nominal two-sided 5% test, and coverage is the empirical coverage of the nominal 95% confidence interval. Positive entries are reported without leading plus signs.

The manuscript-facing performance targets are full means of the autoregressive coefficient a_{it} and the slope coefficient β_{it} . The backend diagnostic CSV retains mean estimates, entry targets, group means, and between-group contrasts. We also report coefficient-generation diagnostics: the mean, standard deviation, minimum, and maximum of a_{it} , the mean and standard deviation of β_{it} , the realized PR^2 , and the calibrated value of c_ξ .

Finite fold floors. The asymptotic theory uses a slowly growing number of scattered folds $J = J_{NT}$. The simulation implementation uses the finite-sample floor rule (5.3), with $J_{\min} = 10$ as the baseline. We also report realized fold counts and retained training shares. These calculations are interpreted as finite-sample implementations of a growing- J_{NT} rule, not as evidence for a separate fixed- J theorem.

Simulation tables. The manuscript-facing Monte Carlo tables follow Cynthia’s requested reporting format. Table 2 reports fixed-rank inference performance using separate DGP tables. Within each DGP table, Panel A reports the full mean of the autoregressive coefficient and Panel B reports the full mean of the slope coefficient. Because the current DGPs do not impose economically meaningful unit groups, and because the coefficient matrices are redrawn in every replication, the manuscript-facing performance tables report only full-mean targets; individual entries, artificial group means, and group contrasts are kept only in the diagnostic CSV output. Table 5 reports rank-selection accuracy. Table 6 reports coefficient and signal summaries. Table 7 reports fold counts and retained training shares. Table 8 reports optimization diagnostics. The final numerical entries will be updated from the finalized simulation-output tables as the remaining `grid.v2` and `rank` production runs finish.

Table 2: Monte Carlo fixed-rank inference performance: DGP 1.

N	T	true	bias	RMSE	mean s.e.	size	coverage
<i>Panel A. Autoregressive coefficient full mean</i>							
100	100	0.448	0.002	0.098	0.102	0.049	0.951
<i>Panel B. Slope coefficient full mean</i>							
100	100	0.598	-0.011	0.160	0.165	0.041	0.959

Notes: Panel A reports the full mean of the autoregressive coefficient a_{it} , and Panel B reports the full mean of the slope coefficient β_{it} . The full mean target averages over all units at the midpoint time $t^* = \lfloor T/2 \rfloor + 1$. Coefficient matrices and shocks are redrawn independently across replications. Bias and RMSE are computed from replication-level errors $\hat{\theta}^{(r)} - \theta_0^{(r)}$. For DGP 1, inference uses the diagonal heteroskedasticity-robust standard error. Size is the empirical rejection probability of the nominal two-sided 5% test. Coverage is empirical coverage of the nominal 95% confidence interval. The number of replications is 1000.

Table 3: Monte Carlo fixed-rank inference performance: DGP 2.

N	T	true	bias	RMSE	mean s.e.	size	coverage
<i>Panel A. Autoregressive coefficient full mean</i>							
100	100	0.448	-0.001	0.116	0.108	0.065	0.935
200	200	0.429	0.001	0.072	0.069	0.068	0.932
<i>Panel B. Slope coefficient full mean</i>							
100	100	0.598	-0.011	0.158	0.163	0.051	0.949
200	200	0.602	-0.004	0.095	0.097	0.043	0.957

Notes: Panel A reports the full mean of the autoregressive coefficient a_{it} , and Panel B reports the full mean of the slope coefficient β_{it} . The full mean target averages over all units at the midpoint time $t^* = \lfloor T/2 \rfloor + 1$. Coefficient matrices and shocks are redrawn independently across replications. Bias and RMSE are computed from replication-level errors $\hat{\theta}^{(r)} - \theta_0^{(r)}$. For DGPs 2–3, inference uses the Bartlett spatial-kernel standard error with lattice distance $d_N(i, j) = |i - j|$ and bandwidth $h_N = \lfloor N^{1/3} \rfloor$. Size is the empirical rejection probability of the nominal two-sided 5% test. Coverage is empirical coverage of the nominal 95% confidence interval. The number of replications is 1000.

Table 4: Monte Carlo fixed-rank inference performance: DGP 3.

N	T	true	bias	RMSE	mean s.e.	size	coverage
<i>Panel A. Autoregressive coefficient full mean</i>							
50	50	0.466	0.067	4.523	0.431	0.068	0.932
100	100	0.448	-0.001	0.125	0.114	0.079	0.921
200	200	0.429	0.001	0.077	0.073	0.060	0.940
400	400	0.411	0.000	0.053	0.050	0.068	0.932
<i>Panel B. Slope coefficient full mean</i>							
50	50	0.592	-0.176	1.496	0.603	0.069	0.931
100	100	0.598	-0.011	0.197	0.191	0.060	0.940
200	200	0.602	-0.005	0.114	0.113	0.046	0.954
400	400	0.594	0.002	0.071	0.069	0.059	0.941

Notes: Panel A reports the full mean of the autoregressive coefficient a_{it} , and Panel B reports the full mean of the slope coefficient β_{it} . The full mean target averages over all units at the midpoint time $t^* = \lfloor T/2 \rfloor + 1$. Coefficient matrices and shocks are redrawn independently across replications. Bias and RMSE are computed from replication-level errors $\hat{\theta}^{(r)} - \theta_0^{(r)}$. For DGPs 2–3, inference uses the Bartlett spatial-kernel standard error with lattice distance $d_N(i, j) = |i - j|$ and bandwidth $h_N = \lfloor N^{1/3} \rfloor$. Size is the empirical rejection probability of the nominal two-sided 5% test. Coverage is empirical coverage of the nominal 95% confidence interval. The number of replications is 1000.

Table 5: Monte Carlo rank-selection accuracy.

DGP	T	N	$J_{\min}$	c_{κ}	retained non-val.	$P(\text{correct})$	$P(\text{under})$	$P(\text{over})$	modal rank	reps
No rank-selection production runs are currently available.										

Notes: The table is populated only by production rank-selection runs. Fixed-rank performance runs are omitted, so the placeholder row remains until `rank_*.jsonl` production files are completed.

Table 6: Monte Carlo coefficient and signal summaries.

DGP	N	T	mean a_{it}	sd a_{it}	min a_{it}	max a_{it}	max $ a_{it} $	mean β_{it}	sd β_{it}	min β_{it}	max β_{it}	mean PR^2	sd PR^2	c_{ξ}	reps
DGP 1	100	100	—	—	—	—	—	—	—	—	—	—	—	1.479	1000
DGP 2	100	100	—	—	—	—	—	—	—	—	—	—	—	1.480	1000
DGP 2	200	200	—	—	—	—	—	—	—	—	—	—	—	1.397	1000
DGP 3	50	50	0.466	0.101	0.033	0.850	0.850	0.598	0.313	-0.738	3.085	0.505	0.050	1.827	1000
DGP 3	100	100	0.449	0.099	0.036	0.850	0.850	0.600	0.320	-0.806	3.302	0.505	0.039	1.706	1000
DGP 3	200	200	0.431	0.096	0.034	0.850	0.850	0.599	0.320	-0.886	3.464	0.500	0.032	1.619	1000
DGP 3	400	400	—	—	—	—	—	—	—	—	—	—	—	1.497	1000

Notes: Coefficient and signal summaries are computed over Monte Carlo replications when the JSONL records contain DGP diagnostics. Older completed runs that predate these per-replication diagnostics display “—” for unavailable quantities.

Table 7: Fold counts and retained training shares for completed production cells.

DGP	N	T	q_T	r_N	h_N	$J_{\min}$	Realized J	Retained non-val.	Retained total
DGP 1	100	100	2	0	4	10	10	0.813	0.732
DGP 2	100	100	2	0	4	10	10	0.813	0.732
DGP 2	200	200	2	0	5	10	10	0.811	0.730
DGP 3	50	50	2	0	3	10	10	0.815	0.734
DGP 3	100	100	2	0	4	10	10	0.813	0.732
DGP 3	200	200	2	0	5	10	10	0.811	0.730
DGP 3	400	400	2	0	7	10	10	0.811	0.730

Notes. This table reports cross-fitting quantities only. The spatial-kernel bandwidth h_N does not affect the retained training shares. For DGPs 2–3, inference uses Bartlett spatial-kernel standard errors with $h_N = \lfloor N^{1/3} \rfloor$. The quantity q_T is the time buffer used in cross-fitting, and r_N is the unit or spatial buffer. The quantity J is the realized number of folds generated by the finite-fold rule unless manually overridden. “Retained non-validation” is the average share of non-validation cells remaining after buffer deletion. “Retained total” is the average number of retained training cells divided by the total number of cells, NT . The retained shares are averaged over the realized scattered-fold assignments and production replications. Consequently, small differences across runs or master seeds are expected before rounding.

Table 8: Monte Carlo optimization diagnostics.

DGP	N	T	mean sweeps	sweep-cap hit rate	monotone rate	final decrease $< 10^{-6}$	restart improvement $> 10^{-6}$	mean total sec.	mean first-stage sec.	mean Riesz sec.
DGP 1	100	100	—	—	—	—	—	—	—	—
DGP 2	100	100	24.936	—	1.000	—	0.454	—	—	—
DGP 2	200	200	49.862	0.995	1.000	1.000	0.922	—	—	—
DGP 3	50	50	21.743	0.296	1.000	0.990	0.234	7.708	3.260	4.430
DGP 3	100	100	33.494	0.593	1.000	1.000	0.502	20.534	10.449	10.046
DGP 3	200	200	49.840	0.994	1.000	1.000	0.918	61.105	31.379	29.627
DGP 3	400	400	49.382	0.981	1.000	1.000	0.998	—	—	—

Notes: Optimization diagnostics are numerical implementation diagnostics for the nonconvex first-stage problem. A sweep-cap hit is treated as a resource-use diagnostic rather than an automatic estimator failure, especially when the final relative objective decrease is small.

9 Empirical Application

We apply the estimator to two monthly U.S. metropolitan panels. The first uses metropolitan unemployment rates to estimate idiosyncratic persistence after removing common-cycle movements through the interactive block. The second uses metropolitan house-price tier series to estimate short-run house-price momentum and the top-minus-bottom price-tier contrast. The applications examine the common targets of interest: full means, group means, between-group contrasts, and cumulative persistence. They also illustrate how predetermined controls x_{it} can be included. In the unemployment panel, the control specification adds lagged metropolitan population growth and real-GDP growth. In the housing panel, the control specification adds lagged building-permit growth, population growth, and real-GDP growth. Detailed data construction, control definitions, rank diagnostics, maps, and robustness tables are in Appendix B. For standard errors, the diagonal cell-heteroskedastic sandwich is the theorem-backed conditional-independence benchmark. The theorem-aligned dependence-robust interval is the spatial-kernel interval when a credible metric and an admissible bandwidth are specified. When a meaningful cross-sectional distance is unavailable but a known cluster partition is credible, the cluster-score standard error in Appendix A.2.4 can be computed from the same cross-fitted scores and reported alongside the spatial-kernel calculation. The period-cluster interval reported below is only a by-period clustered sensitivity diagnostic; it is not theorem-backed in the present framework and is not guaranteed conservative.

Regional unemployment. The unemployment application uses monthly, not-seasonally-adjusted metropolitan unemployment rates, deseasonalized metro by metro. The headline sample covers 315 metros over 240 months, 2005–2024, the window for which the metropolitan controls are complete. The rank criterion selects $(r_1, r_2, r_H) = (1, 0, 1)$, so the idiosyncratic dynamics are a heterogeneous AR(1) after the common cycle is absorbed. High- and low-unemployment metros are determined by the median of each metro’s sample-average unemployment rate; this is a time-invariant pre-analysis split and does not use the estimated autoregressive coefficients.

Table 9: Idiosyncratic persistence in metro unemployment.

Target	Estimate	s.e.	period-cluster sens.	s.e.	95% CI	95% period-cluster sens. CI
Lag-1 persistence, global mean	+0.909	0.033		0.063	[+0.845, +0.972]	[+0.784, +1.033]
Lag-1 persistence, high-unemployment metros	+0.867	0.053		0.103	[+0.764, +0.970]	[+0.665, +1.069]
Lag-1 persistence, low-unemployment metros	+0.915	0.029		0.057	[+0.857, +0.972]	[+0.803, +1.026]
Lag-1 persistence, high–low contrast	−0.012	0.010		0.013	[−0.031, +0.007]	[−0.038, +0.014]

Notes: Headline unemployment specification: $N = 315$, $T = 240$, 2005–2024, ranks $(1, 0, 1)$. “s.e.” is the diagonal cell-heteroskedastic sandwich standard error (the heteroskedasticity-robust estimator that treats cells as conditionally independent). “period-cluster” is a by-period clustered sensitivity standard error, reported as a diagnostic; it is not the theorem’s spatial-kernel object and is not guaranteed conservative unless additional saturated-bandwidth replacement conditions are verified. The high/low split is the median split of sample-average unemployment rates. The separately debiased high-minus-low contrast has its own Riesz representer and is reported as the preferred contrast target.

House-price momentum. The housing application stacks the top and bottom house-price tiers for each metro as separate panel units. The full sample has 610 metro-tier units over 315 months.

The outcome is monthly log price growth, and the cross-fitted criterion selects $(r_1, r_2, r_H) = (1, 1, 1)$. The estimated dynamics exhibit strong one-month momentum and partial reversal in the second month. Although the first lag exceeds one, the sample-average AR(2) summary has companion radius below one. This is a stability statement about the reported average dynamic summary, not a certificate of the pointwise uniform AR(2) bound in Assumption 1. The top-minus-bottom contrast is small relative to the common momentum pattern. It is only suggestive under the by-period clustered sensitivity diagnostic, but it is bounded away from zero in the theorem-aligned geographic spatial-kernel calculation reported in Table 11.

Table 10: House-price dynamics in the metro house-price panel.

Target	Estimate	s.e.	period-cluster sens. s.e.	95% CI	95% period-cluster sens. CI
Lag-1 coefficient, all segments	+1.250	0.004	0.006	[+1.243, +1.257]	[+1.239, +1.261]
Lag-2 coefficient, all segments	-0.419	0.004	0.006	[-0.427, -0.412]	[-0.431, -0.407]
Lag-1 coefficient, top price tier	+1.259	0.005	0.010	[+1.250, +1.268]	[+1.240, +1.278]
Lag-1 coefficient, bottom price tier	+1.241	0.004	0.006	[+1.233, +1.249]	[+1.229, +1.254]
Lag-1 coefficient, top-bottom contrast	+0.018	0.005	0.012	[+0.008, +0.027]	[-0.005, +0.040]
Cumulative persistence ($a_1 + a_2$)	+0.831	0.002	0.002	[+0.827, +0.834]	[+0.827, +0.834]

Notes: Headline housing specification: $N = 610$ metro-tier units, $T = 315$, ranks $(1, 1, 1)$. “s.e.” is the diagonal cell-heteroskedastic sandwich standard error. “period-cluster” is a by-period clustered sensitivity standard error reported as a diagnostic, not the theorem-backed spatial-kernel object and not guaranteed conservative without additional saturated-bandwidth replacement conditions. Cumulative persistence is the linear target $a_1 + a_2$. Nonlinear long-run multipliers and full response paths are treated as descriptive quantities in Appendix B.

Theorem-aligned geographic spatial-kernel inference. Table 11 reports the dependence-robust spatial-kernel standard error of Theorem 5 using a credible geographic metric: great-circle distances between metro centroids and Bartlett bandwidths of 300 and 600 km, following Conley (1999). The geographic spatial-kernel standard errors are close to the diagonal standard errors, which suggests that residual spatial dependence is limited at the reported bandwidths. The by-period clustered interval remains a wider sensitivity diagnostic; it is not theorem-backed in this framework and is not generally guaranteed conservative. For unemployment, all 315 headline metros match a centroid, so the geographic interval is the operative theorem-aligned dependence-robust interval. For housing, the calculation is reported on the coordinate-matched subsample; it is therefore a theorem-aligned robustness check for spatial dependence, not the exact standard error for the full $N = 610$ headline estimate.

Samples, seasonality, and estimation settings. Both panels are monthly. The raw housing series runs from January 2000 through May 2026. With two lags, the usable post-lag estimation sample has $T = 315$ monthly observations. The headline unemployment panel runs from January 2005 to December 2024 ($T = 240$ months, the window over which the metropolitan controls are complete; the full unemployment sample extends to May 2026). Both samples therefore span the COVID-19 period, whose imprint on the dynamics is visible in the time path of the period-by-period average coefficients (Figure 4): house-price momentum compresses sharply at the onset of the 2020 pandemic, while the unemployment coefficient-path diagnostic has a few excursions above one, including a

Table 11: Geographic spatial-kernel standard errors for both applications.

Target	Estimate	diagonal	period-cluster	geo. (300 km)	geo. (600 km)
<i>Unemployment (spec B, N = 315 metros)</i>					
Lag-1 mean	+0.909	0.033	0.063	0.033	0.036
Lag-1, high-unemployment	+0.867	0.053	0.103	0.053	0.059
Lag-1, low-unemployment	+0.915	0.029	0.057	0.030	0.032
Lag-1 high–low contrast	−0.012	0.010	0.013	0.010	0.011
<i>Housing (spec A, N = 390 coord.-matched metro-tier units)</i>					
Lag-1 mean	+1.264	0.005	0.007	0.006	0.006
Lag-2 mean	−0.413	0.005	0.008	0.006	0.006
Lag-1, top tier	+1.272	0.006	0.012	0.007	0.007
Lag-1, bottom tier	+1.256	0.005	0.007	0.006	0.006
Lag-1 top–bottom contrast	+0.016	0.006	0.012	0.005	0.006

Notes: Standard errors for the headline targets or, for housing, for the coordinate-matched subsample described below. “diagonal” is the cell-heteroskedastic sandwich. “period-cluster” is the by-period clustered sensitivity standard error; it is not theorem-backed in the present framework and is not generally guaranteed conservative. “geo.” is the geographic spatial-kernel standard error with Bartlett weights $W_{ij} = \max(0, 1 - d_{ij}/b)$ over great-circle distance d_{ij} between CBSA centroids and bandwidth $b \in \{300, 600\}$ km. All 315 unemployment metros, and 390 of the 610 housing metro-tier units, match a Census CBSA centroid. The housing rows are therefore a coordinate-matched subsample, so their point estimates differ slightly from the full-sample headline. For housing, the table is a theorem-aligned robustness check for spatial dependence, not the exact standard error for the full $N = 610$ headline estimate.

large March–April 2020 break and smaller excursions in other high-volatility periods. This path is a cross-sectional average of estimated coefficients around a near-integrated idiosyncratic persistence, not a pointwise estimate of the true stability envelope. Assumption 1 constrains the true time-varying companion products; the reported empirical targets are sample and group-average linear summaries. The pre-pandemic sensitivity check in Appendix B is therefore used as a stability stress test for the target estimates, not as proof that every fitted cell coefficient satisfies the maintained pointwise bound. In that check, excluding the COVID-era months raises the unemployment persistence estimate to a near-integrated 0.988 (s.e. 0.008), so spec D should be read as near-integrated rather than as strong evidence of a strict subunit AR(1) margin. For housing, the corresponding pre-COVID check leaves cumulative persistence essentially unchanged (0.832 versus 0.831), indicating that the headline house-price momentum estimates are not driven by the 2020 episode. The unemployment rate is *not* seasonally adjusted and is deseasonalized metro-by-metro by subtracting each area’s month-of-year means; the house-value index is entered as monthly log price growth. The applications use the same finite-fold-floor rule as the simulations, with the realized J , the forward exclusion window q , and the retained training shares reported in Appendix B. These diagnostics are used to document that the buffer does not remove an excessive fraction of the candidate training cells. Full data construction, seasonal treatment, and diagnostics are in Appendix B.

The robustness exercises in Appendix B show that the main empirical conclusions are not driven by the omission of observed local controls. For unemployment, adding lagged population growth and real-GDP growth changes the lag-1 mean from 0.909 to 0.937, and both control coefficients are small and statistically insignificant. For house prices, adding lagged permit growth, population growth, and real-GDP growth moves cumulative persistence from 0.831 to 0.801, with all three

control coefficients statistically indistinguishable from zero. In both panels, the low-rank interactive block absorbs the dominant common movements, while the observed predetermined controls provide a transparent check that the dynamic coefficients are not simply proxying for local fundamentals.

10 Conclusion

This paper develops inference for dynamic heterogeneous-coefficient panels in which autoregressive coefficient matrices, slope coefficient matrices, and interactive-effect matrices are all low rank. The central device is to keep the original dynamic panel equation as an operator inverse problem rather than reducing it to an additive score matrix. This preserves strong dependence in observed regressors and lagged outcomes inside the design operator and leaves the leading stochastic term as a weighted array of idiosyncratic errors, handled by a hybrid temporal-martingale and within-period spatial-mixing limit theory. The main methodological contribution is a cross-fitted debiasing correction with a space–time exclusion buffer. The fold-specific debiasing directions solve the target-specific local least-squares normal equations on buffered training sets and are then evaluated on held-out folds; the feasible directions use local perturbation spaces estimated from those same training sets. Under scattered cell-level cross-fitting the forward exclusion is a bias–variance trade-off: a moderate window removes the temporal descendants and nearby contemporaneous cells through which u_{it} would otherwise enter its own estimated weight, while an over-long window starves the first stage of training information. The effect is most visible for the pooled cross-sectional means, where own-error leakage is coherent across units and does not average out; the Monte Carlo evidence in [Section 8](#) exhibits the resulting interior optimum at the data-driven window, confirms nominal coverage of the feasible intervals, and shows that the pooled cross-sectional means under-cover without the window; two applications—idiosyncratic persistence in a regional unemployment panel and price-tier momentum in a house-price panel ([Section 9](#))—illustrate the entry, mean, cumulative persistence, and contrast targets. The main asymptotic theory separates the argument into an infeasible-benchmark influence representation, rank and first-stage verification, feasible Riesz-weight replacement, feasible CLT, and dependence-robust studentization. Under Assumptions 1–6, the main-text rate requirements, and the sufficient conditions stated in the appendix, the appendix establishes the rowwise, information-transport, decoupling, and nuisance-product bounds needed for asymptotic normality of entries, group and full means, cumulative persistence, and contrasts that satisfy the target condition. The rank, fold, and exclusion-window choices are made operational by the data-driven recipe of [Section 5.2](#). The cross-sectional-mean theory of [Appendix A.4.2](#) is likely to be the part most used in practice: [Proposition 10](#) shows that group and full means are estimated at a faster rate than entries whenever the group aligns with the loading space, and [Proposition 10](#) gives the correct contrast variance, including the cross term that overlapping group intervals would miss. The key limitation is also clear. The least-squares operator procedure is valid for predetermined lagged outcomes under the weak-exogeneity moment restriction. If idiosyncratic errors are serially correlated in a way that violates that restriction, the lagged outcome becomes endogenous and a

HAC variance correction does not repair the bias: the problem is in the first-stage moment condition, not in the standard error. Such cases require a different instrumented or GMM first-stage criterion rather than only a different standard error or a different debiasing weight. The present paper therefore keeps the formal theory focused on the least-squares weak-exogeneity framework, exact fixed low rank, and the linear regular targets covered by Assumption 6.

References

- Abbe, E., Fan, J., Wang, K., and Zhong, Y. (2020). Entrywise eigenvector analysis of random matrices with low expected rank. *Annals of Statistics*, 48(3), 1452–1474. doi: [10.1214/19-AOS1854](https://doi.org/10.1214/19-AOS1854).
- Agarwal, A., Dahleh, M., Shah, D., and Shen, D. (2023). Causal matrix completion. *Proceedings of the 36th Conference on Learning Theory*, PMLR 195, 3821–3826.
- Anderson, T. W., and Hsiao, C. (1981). Estimation of dynamic models with error components. *Journal of the American Statistical Association*, 76(375), 598–606. doi: [10.1080/01621459.1981.10477691](https://doi.org/10.1080/01621459.1981.10477691).
- Anderson, T. W., and Hsiao, C. (1982). Formulation and estimation of dynamic models using panel data. *Journal of Econometrics*, 18(1), 47–82. doi: [10.1016/0304-4076\(82\)90095-1](https://doi.org/10.1016/0304-4076(82)90095-1).
- Arellano, M., and Bond, S. (1991). Some tests of specification for panel data: Monte Carlo evidence and an application to employment equations. *Review of Economic Studies*, 58(2), 277–297. doi: [10.2307/2297968](https://doi.org/10.2307/2297968).
- Athey, S., Bayati, M., Doudchenko, N., Imbens, G. W., and Khosravi, K. (2021). Matrix completion methods for causal panel data models. *Journal of the American Statistical Association*, 116(536), 1716–1730. doi: [10.1080/01621459.2021.1891924](https://doi.org/10.1080/01621459.2021.1891924).
- Bai, J. (2003). Inferential theory for factor models of large dimensions. *Econometrica*, 71(1), 135–171. doi: [10.1111/1468-0262.00392](https://doi.org/10.1111/1468-0262.00392).
- Bai, J. (2009). Panel data models with interactive fixed effects. *Econometrica*, 77(4), 1229–1279. doi: [10.3982/ECTA6135](https://doi.org/10.3982/ECTA6135).
- Bai, J., and Ng, S. (2002). Determining the number of factors in approximate factor models. *Econometrica*, 70(1), 191–221. doi: [10.1111/1468-0262.00273](https://doi.org/10.1111/1468-0262.00273).
- Bai, J., and Ng, S. (2021). Matrix completion, counterfactuals, and factor analysis of missing data. *Journal of the American Statistical Association*, 116(536), 1746–1763. doi: [10.1080/01621459.2021.1967163](https://doi.org/10.1080/01621459.2021.1967163).
- Belloni, A., Chernozhukov, V., and Hansen, C. (2014). Inference on treatment effects after selection among high-dimensional controls. *Review of Economic Studies*, 81(2), 608–650. doi: [10.1093/restud/rdt044](https://doi.org/10.1093/restud/rdt044).
- Bickel, P. J., Klaassen, C. A. J., Ritov, Y., and Wellner, J. A. (1993). *Efficient and Adaptive Estimation for Semiparametric Models*. Johns Hopkins University Press, Baltimore.
- Bonhomme, S., and Manresa, E. (2015). Grouped patterns of heterogeneity in panel data. *Econometrica*, 83(3), 1147–1184. doi: [10.3982/ECTA11319](https://doi.org/10.3982/ECTA11319).

- Cameron, A. C., and Miller, D. L. (2015). A practitioner’s guide to cluster-robust inference. *Journal of Human Resources*, 50(2), 317–372. doi: [10.3368/jhr.50.2.317](https://doi.org/10.3368/jhr.50.2.317).
- Candès, E. J., and Recht, B. (2009). Exact matrix completion via convex optimization. *Foundations of Computational Mathematics*, 9(6), 717–772. doi: [10.1007/s10208-009-9045-5](https://doi.org/10.1007/s10208-009-9045-5).
- Chen, Y., Cheng, C., and Fan, J. (2021). Asymmetry helps: Eigenvalue and eigenvector analyses of asymmetrically perturbed low-rank matrices. *Annals of Statistics*, 49(1), 435–458. doi: [10.1214/20-AOS1963](https://doi.org/10.1214/20-AOS1963).
- Chen, Y., Chi, Y., Fan, J., Ma, C., and Yan, Y. (2020). Noisy matrix completion: Understanding statistical guarantees for convex relaxation via nonconvex optimization. *SIAM Journal on Optimization*, 30(4), 3098–3121. doi: [10.1137/19M1290000](https://doi.org/10.1137/19M1290000).
- Bolthausen, E. (1982). On the central limit theorem for stationary mixing random fields. *Annals of Probability*, 10(4), 1047–1050.
- Chen, Y., Fan, J., Ma, C., and Yan, Y. (2019). Inference and uncertainty quantification for noisy matrix completion. *Proceedings of the National Academy of Sciences*, 116(46), 22931–22937. doi: [10.1073/pnas.1910053116](https://doi.org/10.1073/pnas.1910053116).
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1), C1–C68. doi: [10.1111/ectj.12097](https://doi.org/10.1111/ectj.12097).
- Chernozhukov, V., Hansen, C., Liao, Y., and Zhu, Y. (2023). Inference for low-rank models. *Annals of Statistics*, 51(3), 1309–1330. doi: [10.1214/23-AOS2293](https://doi.org/10.1214/23-AOS2293).
- Choi, J., Kwon, H., and Liao, Y. (2024). Inference for low-rank completion without sample splitting with application to treatment effect estimation. *Journal of Econometrics*, 240(1), Article 105682. doi: [10.1016/j.jeconom.2024.105682](https://doi.org/10.1016/j.jeconom.2024.105682).
- Choi, J., Kwon, H., and Liao, Y. (2025). Inference for low-rank models without estimating the rank. *Journal of the American Statistical Association* (forthcoming). doi: [10.1080/01621459.2025.2538272](https://doi.org/10.1080/01621459.2025.2538272).
- Conley, T. G. (1999). GMM estimation with cross sectional dependence. *Journal of Econometrics*, 92(1), 1–45. doi: [10.1016/S0304-4076\(98\)00084-0](https://doi.org/10.1016/S0304-4076(98)00084-0).
- Davis, C., and Kahan, W. M. (1970). The rotation of eigenvectors by a perturbation. III. *SIAM Journal on Numerical Analysis*, 7(1), 1–46. doi: [10.1137/0707001](https://doi.org/10.1137/0707001).
- Wedin, P.-Å. (1972). Perturbation bounds in connection with singular value decomposition. *BIT Numerical Mathematics*, 12(1), 99–111. doi: [10.1007/BF01932678](https://doi.org/10.1007/BF01932678).

- Yu, Y., Wang, T., and Samworth, R. J. (2015). A useful variant of the Davis–Kahan theorem for statisticians. *Biometrika*, 102(2), 315–323. doi: [10.1093/biomet/asv008](https://doi.org/10.1093/biomet/asv008).
- Doukhan, P. (1994). *Mixing: Properties and Examples*. Lecture Notes in Statistics 85, Springer, New York. doi: [10.1007/978-1-4612-2642-0](https://doi.org/10.1007/978-1-4612-2642-0).
- Hall, P., and Heyde, C. C. (1980). *Martingale Limit Theory and Its Application*. Academic Press, New York.
- Hsiao, C., Pesaran, M. H., and Tahmiscioglu, A. K. (1999). Bayes estimation of short-run coefficients in dynamic panel data models. In C. Hsiao, L.-F. Lee, K. Lahiri, and M. H. Pesaran (eds.), *Analysis of Panels and Limited Dependent Variable Models*, 268–296. Cambridge University Press, Cambridge.
- Jenish, N., and Prucha, I. R. (2009). Central limit theorems and uniform laws of large numbers for arrays of random fields. *Journal of Econometrics*, 150(1), 86–98. doi: [10.1016/j.jeconom.2008.12.008](https://doi.org/10.1016/j.jeconom.2008.12.008).
- Koltchinskii, V., Lounici, K., and Tsybakov, A. B. (2011). Nuclear-norm penalization and optimal rates for noisy low-rank matrix completion. *Annals of Statistics*, 39(5), 2302–2329. doi: [10.1214/11-AOS894](https://doi.org/10.1214/11-AOS894).
- Liang, K.-Y., and Zeger, S. L. (1986). Longitudinal data analysis using generalized linear models. *Biometrika*, 73(1), 13–22. doi: [10.1093/biomet/73.1.13](https://doi.org/10.1093/biomet/73.1.13).
- Moon, H. R., and Weidner, M. (2015). Linear regression for panel with unknown number of factors as interactive fixed effects. *Econometrica*, 83(4), 1543–1579. doi: [10.3982/ECTA9382](https://doi.org/10.3982/ECTA9382).
- Moon, H. R., and Weidner, M. (2017). Dynamic linear panel regression models with interactive fixed effects. *Econometric Theory*, 33(1), 158–195. doi: [10.1017/S0266466615000328](https://doi.org/10.1017/S0266466615000328).
- Newey, W. K. (1990). Semiparametric efficiency bounds. *Journal of Applied Econometrics*, 5(2), 99–135. doi: [10.1002/jae.3950050202](https://doi.org/10.1002/jae.3950050202).
- Newey, W. K. (1994). The asymptotic variance of semiparametric estimators. *Econometrica*, 62(6), 1349–1382. doi: [10.2307/2951752](https://doi.org/10.2307/2951752).
- Pesaran, M. H. (2006). Estimation and inference in large heterogeneous panels with a multifactor error structure. *Econometrica*, 74(4), 967–1012. doi: [10.1111/j.1468-0262.2006.00692.x](https://doi.org/10.1111/j.1468-0262.2006.00692.x).
- Pesaran, M. H., and Smith, R. (1995). Estimating long-run relationships from dynamic heterogeneous panels. *Journal of Econometrics*, 68(1), 79–113. doi: [10.1016/0304-4076\(94\)01644-F](https://doi.org/10.1016/0304-4076(94)01644-F).
- Recht, B., Fazel, M., and Parrilo, P. A. (2010). Guaranteed minimum-rank solutions of linear matrix equations via nuclear norm minimization. *SIAM Review*, 52(3), 471–501. doi: [10.1137/070697835](https://doi.org/10.1137/070697835).
- Rudelson, M., and Vershynin, R. (2013). Hanson–Wright inequality and finite-moment concentration. *Electronic Communications in Probability*, 18, 1–9. doi: [10.1214/ECP.v18-2865](https://doi.org/10.1214/ECP.v18-2865).

- Semenova, V., Goldman, M., Chernozhukov, V., and Taddy, M. (2018). Estimation and inference on heterogeneous treatment effects in high-dimensional dynamic panels under weak dependence. *arXiv preprint arXiv:1712.09988*. <https://arxiv.org/abs/1712.09988>.
- Semenova, V., and Chernozhukov, V. (2021). Debiased machine learning of conditional average treatment effects and other causal functions. *The Econometrics Journal*, 24(2), 264–289. doi: [10.1093/ectj/utaa027](https://doi.org/10.1093/ectj/utaa027).
- Statulevičius, V., and Jakimavičius, D. (1988). Estimates of semiinvariants and centered moments of stochastic processes with mixing, I. *Lithuanian Mathematical Journal*, 28(3), 226–238. doi: [10.1007/BF00969984](https://doi.org/10.1007/BF00969984).
- Swamy, P. A. V. B. (1970). Efficient inference in a random coefficient regression model. *Econometrica*, 38(2), 311–323. doi: [10.2307/1913012](https://doi.org/10.2307/1913012).
- Talagrand, M. (2014). *Upper and Lower Bounds for Stochastic Processes: Modern Methods and Classical Problems*. Springer, Heidelberg. doi: [10.1007/978-3-642-54075-2](https://doi.org/10.1007/978-3-642-54075-2).
- Tropp, J. A. (2011). Freedman’s inequality for matrix martingales. *Electronic Communications in Probability*, 16, 262–270. doi: [10.1214/ECP.v16-1624](https://doi.org/10.1214/ECP.v16-1624).
- Tropp, J. A. (2015). An introduction to matrix concentration inequalities. *Foundations and Trends in Machine Learning*, 8(1–2), 1–230. doi: [10.1561/22000000048](https://doi.org/10.1561/22000000048).
- Uschmajew, A. (2015). A new convergence proof for the higher-order power method and generalizations. *Pacific Journal of Optimization*, 11(2), 309–321.
- van der Vaart, A. W. (1998). *Asymptotic Statistics*. Cambridge University Press, Cambridge. doi: [10.1017/CBO9780511802256](https://doi.org/10.1017/CBO9780511802256).

A Mathematical Appendix

This appendix contains the mathematical verification for the main text. It is organized in four layers. Appendix A.1 gives only the proofs of the five main theorems. Appendix A.2 records assumption interpretation, the known-cluster alternative to spatial mixing, feasible rate regimes, and notation. Appendix A.3 collects the paper-specific auxiliary results for rank selection, first-stage recovery, fold transport, dynamic leakage, and feasible Riesz-weight replacement. Appendix A.4 collects the CLT, studentization, and target-verification arguments. External probability, perturbation, and spatial-covariance tools are cited at the proof steps where they enter, so the appendix focuses on the paper-specific arguments. Appendix B contains the simulation and empirical material.

A.1 Proofs of the Five Main Theorems

This subsection contains only the proofs of the five main theorems stated in Section 7. Auxiliary lemmas and propositions used in these proofs are stated and proved in the later appendix subsections, where they are grouped by purpose.

Proof of Theorem 1. Let

$$\xi_{\nu,t} = \sum_{i=1}^N \tilde{\Psi}_{\nu,it} u_{it}, \quad \tilde{L}_{\nu} = \sum_{t=1}^T \xi_{\nu,t}.$$

By Lemma 3, the benchmark weights are predictable and the resulting weighted score satisfies the martingale CLT conditions. In particular, $\tilde{\Psi}_{\nu,it}$ is measurable with respect to the current-cell predictable sigma field generated by $(x_{it}, y_{i,t-1}, \dots, y_{i,t-P}, \mathcal{F}_{t-1})$. Hence the coordinatewise predetermined-regressor weak exogeneity condition in Assumption 2 gives

$$\mathbb{E}_0(\xi_{\nu,t} \mid \mathcal{F}_{t-1}) = \sum_{i=1}^N \mathbb{E}_0[\tilde{\Psi}_{\nu,it} \mathbb{E}_0(u_{it} \mid x_{it}, \mathcal{F}_{t-1}) \mid \mathcal{F}_{t-1}] = 0.$$

Thus $\{\xi_{\nu,t}, \mathcal{F}_t\}_{t \leq T}$ is a martingale-difference array. Its predictable quadratic variation is

$$V_{\nu,NT} = \sum_{t=1}^T \mathbb{E}_0(\xi_{\nu,t}^2 \mid \mathcal{F}_{t-1}) = \sum_{t=1}^T \sum_{i,j=1}^N \mathbb{E}_0[\tilde{\Psi}_{\nu,it} \tilde{\Psi}_{\nu,jt} u_{it} u_{jt} \mid \mathcal{F}_{t-1}],$$

and $s_{\nu,NT}^2 = \mathbb{E}_0 V_{\nu,NT}$. The same lemma verifies, under Assumption 6, the variance stabilization $V_{\nu,NT}/s_{\nu,NT}^2 \rightarrow_p 1$ and the conditional Lindeberg–Lyapunov bound

$$s_{\nu,NT}^{-(2+\delta)} \sum_{t=1}^T \mathbb{E}_0(|\xi_{\nu,t}|^{2+\delta} \mid \mathcal{F}_{t-1}) \rightarrow_p 0$$

for some $\delta > 0$ allowed by the finite $(8+\eta)$ -moment assumption. The martingale central limit theorem for triangular martingale arrays (Hall and Heyde, 1980, martingale central limit theorem under conditional variance and Lindeberg conditions) then yields $\tilde{L}_{\nu}/s_{\nu,NT} \xrightarrow{d} N(0, 1)$. For a fixed vector of

targets, apply the same argument to any fixed linear combination of the corresponding $\xi_{\nu,t}$'s and use the Cramér–Wold device. The covariance appearing in that argument is the full contemporaneous covariance because the display for $V_{\nu,NT}$ retains all pairs (i, j) within each period. $\square$

Proof of Theorem 2. Let $\widehat{\mathbf{r}}$ be the cross-fitted rank selector and let $\widehat{\boldsymbol{\Theta}}_{-j}^0$ denote the ideal measurable minimizer on the purged training set for fold j . Proposition 5 gives

$$\mathbb{P}_0(\widehat{\mathbf{r}} = \mathbf{r}_0) \rightarrow 1.$$

On the event $\{\widehat{\mathbf{r}} = \mathbf{r}_0\}$, Proposition 5 proves, uniformly over folds,

$$\frac{1}{NT} \|\mathcal{A}(\widehat{\boldsymbol{\Theta}}_{-j}^0 - \boldsymbol{\Theta}_0)\|_F^2 = O_{p,0}(\zeta_{NT}), \quad \frac{1}{NT} \|\widehat{\boldsymbol{\Theta}}_{-j}^0 - \boldsymbol{\Theta}_0\|_{\mathbb{H}}^2 = O_{p,0}(\zeta_{NT}),$$

$$\|\widehat{\mathcal{P}}_{\mathcal{T},-j} - \mathcal{P}_{\mathcal{T}_0}\|_{\text{op}} = O_{p,0}(\delta_{NT}), \quad \max\{\text{aligned row, column, and entrywise errors}\} = O_{p,0}(\delta_{\max,NT}).$$

These are exactly the displayed rates in the theorem. If the estimator is a finite ALS or Riemannian iterate rather than the ideal minimizer, let $\widehat{\boldsymbol{\Theta}}_{-j}^{\text{alg}}$ denote the reported iterate and write

$$Q_{-j}(\boldsymbol{\Theta}) := \frac{1}{2} \left\| \Pi_{-j}^{\text{buf}} \{\mathbf{Y} - \mathcal{A}(\boldsymbol{\Theta})\} \right\|_F^2$$

for the fold objective in (6.2). The implementation requirement (6.5) gives, uniformly in j ,

$$Q_{-j}(\widehat{\boldsymbol{\Theta}}_{-j}^{\text{alg}}) \leq Q_{-j}(\widehat{\boldsymbol{\Theta}}_{-j}^0) + o_{p,0}(NT\zeta_{NT}),$$

where $\widehat{\boldsymbol{\Theta}}_{-j}^0$ is the ideal fold minimizer. Since $Q_{-j}(\widehat{\boldsymbol{\Theta}}_{-j}^0) \leq Q_{-j}(\boldsymbol{\Theta}_0)$ on the true rank class, the reported iterate also satisfies the explicit basic inequality

$$Q_{-j}(\widehat{\boldsymbol{\Theta}}_{-j}^{\text{alg}}) - Q_{-j}(\boldsymbol{\Theta}_0) \leq o_{p,0}(NT\zeta_{NT}).$$

Combining this near-argmin inequality with the same quadratic expansion used in Proposition 5 adds only an $o_{p,0}(NT\zeta_{NT})$ term to the stochastic expansion of the localized objective. The restricted-curvature lower bound in Assumption 5 then forces $\mathcal{A}(\widehat{\boldsymbol{\Theta}}_{-j}^{\text{alg}} - \boldsymbol{\Theta}_0)$ and $\widehat{\boldsymbol{\Theta}}_{-j}^{\text{alg}} - \boldsymbol{\Theta}_0$ into the same $O_{p,0}(\zeta_{NT}^{1/2})$ neighborhood as the ideal minimizer, so the perturbation-space and aligned row/column bounds transfer unchanged. Without that objective-gap condition, multistart agreement, rank stability, and first-order residual checks remain numerical diagnostics only and do not certify the theorem's global-minimizer requirement. $\square$

Proof of Theorem 3. For each fold j , let $\widehat{G}_j$ be the feasible empirical information operator on the estimated tangent space and let G_j^{or} be the corresponding benchmark fold operator. The feasible and benchmark Riesz representers solve

$$\widehat{G}_j \widehat{q}_{\nu,-j} = \widehat{W}_{\nu,j}, \quad G_j^{\text{or}} q_{\nu,j}^{\text{or}} = W_{\nu,j}^{\text{or}}.$$

Subtracting the two normal equations gives

$$\widehat{q}_{\nu,-j} - q_{\nu,j}^{\text{or}} = \widehat{G}_j^{-1}(\widehat{W}_{\nu,j} - W_{\nu,j}^{\text{or}}) - \widehat{G}_j^{-1}(\widehat{G}_j - G_j^{\text{or}})q_{\nu,j}^{\text{or}}.$$

The restricted-eigenvalue part of Assumption 5 and the first-stage localization in Theorem 2 keep $\widehat{G}_j^{-1}$ uniformly bounded on the relevant tangent space. Proposition 7 combines this resolvent identity with the fold-transport bounds

$$\|\widehat{G}_j - G_j^{\text{or}}\|_{\text{op}} = O_{p,0}(\varepsilon_{G,NT} + b_{NT}\delta_{\max,NT} + b_{NT}^2\zeta_{NT})$$

and the target-direction perturbation bound to obtain

$$\left\{ \sum_j \|\Pi_j(\widehat{\Psi}_{\nu,-j} - \Psi_{\nu,j}^{\text{or}})\|_F^2 \right\}^{1/2} = o_{p,0}(\|\Psi_{\nu}^{\text{or}}\|_F),$$

which is (7.2). The same proposition applies the population normal equation for the Riesz representer to the first-stage error and uses the envelope (7.1) to show

$$\sum_j p_{j,NT} \left| \widehat{\mathbf{r}}_{\nu,-j}(\widehat{\Theta}_{-j}^0 - \Theta_0) \right| = o_{p,0}(s_{\nu,NT}),$$

which is (7.3). Finally, by Cauchy–Schwarz and the finite-moment score bounds from the Davydov-type covariance inequalities for strong-mixing arrays in Doukhan (1994, Davydov-type covariance inequality for strong-mixing variables), the difference between the feasible and benchmark weighted scores is bounded by the product of the left side of (7.2) and a stochastic score norm of order $O_{p,0}(s_{\nu,NT}/\|\Psi_{\nu}^{\text{or}}\|_F)$. Hence the score replacement error is $o_{p,0}(s_{\nu,NT})$. $\square$

Proof of Theorem 4. Proposition 8 gives the exact decomposition

$$\check{\phi}_{\nu} - \phi_{\nu}(\Theta_0) = \widetilde{L}_{\nu} + R_{1,\nu,NT} + R_{2,\nu,NT} + R_{3,\nu,NT},$$

where $R_{1,\nu,NT}$ is the first-stage localization/Taylor remainder, $R_{2,\nu,NT}$ is the feasible Riesz-weight replacement error, and $R_{3,\nu,NT}$ is the residual dynamic-leakage and numerical-transport error. Theorem 2 controls $R_{1,\nu,NT}$, Theorem 3 controls $R_{2,\nu,NT}$, and the rate envelope (7.1), together with Proposition 8, controls $R_{3,\nu,NT}$. Therefore

$$R_{1,\nu,NT} + R_{2,\nu,NT} + R_{3,\nu,NT} = o_{p,0}(s_{\nu,NT})$$

and

$$\check{\phi}_{\nu} - \phi_{\nu}(\Theta_0) = \widetilde{L}_{\nu} + o_{p,0}(s_{\nu,NT}).$$

The scalar central limit theorem follows by dividing by $s_{\nu,NT}$ and using Theorem 1 plus Slutsky's theorem. The finite-vector statement follows by applying the same expansion to any fixed linear

combination of the target vector and then invoking the Cramér–Wold device. $\square$

Proof of Theorem 5. Let $s_{\nu,\text{sp},0}^2$ be the infeasible spatial-kernel variance computed with benchmark weights and true errors. Decompose

$$\left| \frac{\hat{s}_{\nu,\text{sp}}^2}{s_{\nu,NT}^2} - 1 \right| \leq \left| \frac{\hat{s}_{\nu,\text{sp}}^2 - s_{\nu,\text{sp},0}^2}{s_{\nu,NT}^2} \right| + \left| \frac{s_{\nu,\text{sp},0}^2 - s_{\nu,NT}^2}{s_{\nu,NT}^2} \right|.$$

Proposition 9 proves that the second term is $o_{p,0}(1)$ under the metric, bandwidth, local fourth-moment, and covariance-tail conditions in (7.4). Proposition 9 shows that replacing benchmark score vectors by feasible cross-fitted score vectors changes each local covariance sum by an amount controlled by $v_N(h_N)r_{\nu,NT}$. Proposition 9 combines this replacement rate with (7.4) and yields the first term as $o_{p,0}(1)$. Hence

$$\hat{s}_{\nu,\text{sp}}^2 / s_{\nu,NT}^2 \rightarrow_p 1.$$

Combining this ratio consistency with the feasible CLT in Theorem 4 and applying Slutsky’s theorem gives the studentized normal limit. The covariance-matrix statement follows by repeating the same argument for fixed linear combinations of target scores; the limiting covariance is the full contemporaneous covariance retained in the benchmark variance. $\square$

A.2 Assumptions, rate regimes, and notation

This section explains the role and restrictiveness of the main maintained assumptions. It also separates primitive DGP restrictions from primitive stochastic, identification, procedural, target-specific, and functional restrictions. The cross-fitting growth regime (5.2), the studentization rates (7.4), the numerical objective gap (6.5), and all derived stochastic envelopes are not additional maintained assumptions.

A.2.1 Identification check

The following sufficient condition is used in the main text to make Assumption 5 econometrically checkable. It is an appendix result, not one of the five main theorems.

Proposition 1 (Canonical sufficient check for Assumption 5). *Let $\mathcal{F}_H$ denote the row-column spaces spanned by the interactive component and by the common components of the regressors, and write $\mathbf{Z}^{(m)} = \mathbf{Z}_F^{(m)} + \mathbf{E}_Z^{(m)}$ with $\mathbf{Z}_F^{(m)} \in \mathcal{F}_H$ and $\mathbf{E}_Z^{(m)} \perp \mathcal{F}_H$. Suppose that, uniformly over the fixed candidate rank box, the residual regressor components have nondegenerate row and column energy on admissible coefficient perturbations,*

$$\mathbb{E}_0 \left\| \sum_m \mathbf{E}_Z^{(m)} \odot \boldsymbol{\Delta}^{(m)} \right\|_F^2 \geq c \sum_m \|\boldsymbol{\Delta}^{(m)}\|_{\mathbb{H}}^2,$$

up to the fixed finite-dimensional rotations used to normalize the low-rank factors. Suppose also that the common-factor fitted-value component cannot cancel this residual component through an

admissible interactive-effect perturbation:

$$\mathbb{E}_0 \left\| \sum_m \mathbf{Z}^{(m)} \odot \Delta^{(m)} + \Delta^H \right\|_F^2 \geq c' \left\{ \mathbb{E}_0 \left\| \sum_m \mathbf{E}_Z^{(m)} \odot \Delta^{(m)} \right\|_F^2 + \|\Delta^H\|_{\mathbb{H}}^2 \right\}.$$

If, in addition, the corresponding normalized row and column chart Hessians have eigenvalues bounded away from zero and infinity and satisfy the block-dominance bound in Assumption 5, then Assumption 5 holds.

Proof of Proposition 1. Let $\Delta = (\{\Delta^{(m)}\}_m, \Delta^H)$ be an admissible secant in the fixed rank box. The fitted-value perturbation can be decomposed as

$$\mathcal{A}(\Delta) = \sum_m \mathbf{Z}^{(m)} \odot \Delta^{(m)} + \Delta^H = \sum_m \mathbf{E}_Z^{(m)} \odot \Delta^{(m)} + \left\{ \sum_m \mathbf{Z}_F^{(m)} \odot \Delta^{(m)} + \Delta^H \right\}.$$

The no-cancellation inequality assumed in the proposition gives

$$\mathbb{E}_0 \|\mathcal{A}(\Delta)\|_F^2 \geq c' \left\{ \mathbb{E}_0 \left\| \sum_m \mathbf{E}_Z^{(m)} \odot \Delta^{(m)} \right\|_F^2 + \|\Delta^H\|_{\mathbb{H}}^2 \right\}.$$

Using the residual-energy lower bound in the first term, and using the product Hilbert norm $\|\Delta\|_{\mathbb{H}}^2 \asymp \sum_m \|\Delta^{(m)}\|_{\mathbb{H}}^2 + \|\Delta^H\|_{\mathbb{H}}^2$ under the fixed normalization of the factor spaces, there is a constant $c_S > 0$ such that

$$\mathbb{E}_0 \|\mathcal{A}(\Delta)\|_F^2 \geq c_S \|\Delta\|_{\mathbb{H}}^2.$$

The corresponding upper bound follows from the fixed rank box, bounded entries, uniform finite moments of the regressors, and equivalence of the chart and Hilbert norms on the finite-dimensional normalized factor charts: for some $C_S < \infty$,

$$\mathbb{E}_0 \|\mathcal{A}(\Delta)\|_F^2 \leq C_S \|\Delta\|_{\mathbb{H}}^2.$$

These two displays are exactly the population secant inequality (4.4). The remaining conditions stated in the proposition are the row-Hessian, column-Hessian, block-dominance, and local Lipschitz clauses appearing in Assumption 5. Therefore all parts of Assumption 5 hold. $\square$

A.2.2 Assumption roles

No.	Type	Restrictiveness	Econometric role and status
1	Primitive dynamic DGP	Moderate	Controls propagation through lagged outcomes and stochastic predetermined covariates. It permits heterogeneous, time-varying coefficient paths but excludes local-to-unity and long-memory feedback at the maintained normalization.
2	Primitive exogeneity and time structure	Mixed	Preserves unit-level weak exogeneity and allows future covariates to respond to current idiosyncratic errors. Independence of primitive slices across time periods is the stronger temporal part; no vector conditional-mean restriction given the full current covariate panel is imposed.
3	Primitive moments and contemporaneous dependence	Moderate to strong	Finite $8 + \eta$ moments are weaker than exponential tails. Conditional spatial mixing allows unrestricted local contemporaneous dependence but excludes sufficiently persistent long-range dependence and requires a meaningful unit metric.
4	Low-rank signal strength	Substantive	Fixed strong rank and incoherence support singular-space and rowwise recovery. It excludes weak factors, highly localized singular vectors, and growing-rank asymptotics.
5	Population identification and local curvature	Substantive but verifiable	The secant inequality is the low-rank analogue of full column rank: coefficient-matrix and interactive-effect perturbations must not cancel in fitted-value space. Local row and column information support stable factor inversion. Appendix A.3.1 gives sufficient verification routes based on fitted-value no-cancellation and residual identifying variation.
6	Procedure definition	Analyst controlled	Defines the auxiliary scattered folds and the single canonical space–time buffer. Balance, retention, decoupling, information transport, and all rate envelopes are proved rather than assumed.

No.	Type	Restrictiveness	Econometric role and status
7	Regular targets	Target-specific and partly irreducible	Relevance, variance noncancellation, and influence dispersion cannot hold for arbitrary user-chosen functionals. The assumption defines the class of targets for which the one-step inference is valid. Appendices A.4 and A.4.2 give sufficient verification conditions for entries, group means, full means, contrasts, and cumulative persistence.

A.2.3 Assumption details

Assumption 1: stable dynamics and stochastic predetermined covariates. The companion-product bound controls the descendants of a held-out idiosyncratic error through lagged outcomes. The time-slice physical-dependence coupling for x_{it} handles temporal memory in the covariate process, and the unit-specific space-time coupling handles the second leakage channel, in which the same idiosyncratic error enters future predetermined regressors at the same or nearby units. Geometric decay is stronger than polynomial decay, but it delivers a transparent logarithmic forward-exclusion radius and allows the covariate path to remain stochastic rather than conditioned away. The assumption does not require constant coefficients or stationarity of the full panel. The innovation slice ε_t^x is vector-valued and can include unit-specific, aggregate, and latent common-factor innovations affecting covariates, provided the induced process continues to satisfy weak exogeneity and conditional spatial mixing.

Assumption 2: weak exogeneity and temporal error structure. The central restriction is $\mathbb{E}_0(u_{it} \mid x_{it}, \mathcal{F}_{t-1}) = 0$. It neither conditions on future covariates nor imposes $\mathbb{E}_0(\mathbf{u}_t \mid \mathbf{X}_t, \mathcal{F}_{t-1}) = 0$. The proof therefore constructs cell-specific predictable approximations to the Riesz weights. Conditional independence of primitive slices across time periods supplies the temporal error structure; current cross-sectional dependence is left to Assumption 3.

Assumption 3: finite moments and weak contemporaneous dependence. The $8 + \eta$ moment order is used for truncation, row and column maxima, fourth-order spatial quadratic forms, and residual replacement. No sub-Gaussian, sub-exponential, or sub-Weibull condition is part of the baseline theory. The weighted mixing series is calibrated to the ordered quadruples that arise in the spatial-kernel variance proof. Conditional independence is a nested special case, while dense long-range dependence is excluded.

Assumption 4: pervasive, well-spread low-rank signal. The $\sqrt{NT}$ singular-value scale is the strong-factor normalization. Incoherence prevents a latent direction from being supported on a vanishing number of time periods or units. Together these restrictions convert global coefficient-

matrix error into stable singular-space and rowwise recovery. The fixed candidate caps are procedural bounds that contain the fixed truth; the population underfitting gap is derived from Assumptions 4 and 5 rather than maintained separately.

Assumption 5: population identification and local curvature. The global secant inequality rules out observational cancellation among the coefficient-matrix perturbations and the interactive-effect component. It is the low-rank counterpart of a full-rank regressor condition. The local Hessian bounds and block-dominance restriction make row and column factor updates locally invertible and stable. This is the principal high-level identification assumption, but it is not left uninterpreted: Appendix A.3.1 verifies it through two sufficient routes. The first route is fitted-value no-cancellation. The second route uses residual identifying variation after common factor components are removed.

Scattered space–time-buffered cross-fitting. The auxiliary randomization and exact training-set construction are part of the estimator. The researcher chooses (J_{NT}, q_T, r_N) subject to the main-text rate regime (5.2). Lemma 1 proves row and column coverage and retained training share; Proposition 4 proves information and quadratic-variation transport; Lemma 3 proves dynamic and spatial decoupling. These are results, not maintained assumptions.

Regular targets. Assumption 6 is target-specific. It should not be read as a primitive restriction on the entire data-generating process. No data-generating process can make every possible functional regular: some functionals may have too little tangent-space projection, some may have asymptotically negligible variance, and some may be too localized for the finite-moment central limit theorem. The assumption therefore defines the class of targets for which this paper reports inference. Appendix A.4.2 gives explicit leverage formulas for entries, group means, full means, contrasts, and cumulative persistence, while Appendix A.4 gives sufficient covariance and influence-dispersion conditions.

A.2.4 Known-cluster alternative

The main theory uses spatial mixing because it allows weak dependence that decays with a specified cross-sectional distance. In applications where no credible distance is available but units can be assigned to known clusters, the same estimator and proof strategy can be used under exact clustered dependence. This alternative follows the standard cluster-robust logic of allowing arbitrary within-cluster dependence while treating clusters as independent after conditioning on aggregate/common information (Liang and Zeger, 1986; Cameron and Miller, 2015). It is stated as an alternative to the spatial parts of Assumptions 1 and 3. Assumption 2 remains the same predetermined-regressor weak-exogeneity restriction, but under the clustered alternative it is imposed with respect to the enlarged period- t pre-idiosyncratic-innovation information $\mathcal{H}_t$ defined below. Assumptions 4–6 are otherwise unchanged, with Assumption 6(iv) read using cluster blocks as described below.

Alternative Assumption 1' (aggregate-conditioned predetermined structure). Let $\mathcal{A}_t$ denote the time- t aggregate/common information, including aggregate covariate shocks and primitive common-factor innovations that may create cross-cluster dependence in the observed regressors. Define the enlarged conditioning field

$$\mathcal{H}_t = \mathcal{F}_{t-1} \vee \mathcal{A}_t.$$

The field $\mathcal{H}_t$ is interpreted as period- t pre-idiosyncratic-innovation information: aggregate/common shocks are conditioned on before the cluster-specific idiosyncratic innovations u_t are realized. The stability, temporal forgetting, and initial-moment clauses of Assumption 1 continue to hold. The spatial-propagation coupling in Assumption 1(ii) is replaced by cluster-local propagation: after conditioning on the aggregate/common blocks, replacing the unit-specific primitive block ζ_{js} can affect future predetermined covariates only inside the known cluster containing j (defined in Assumption 3' below), and the effect decays geometrically in the forward horizon,

$$\sup_{s,j,\ell,k,h \geq 0} \left\| x_{\ell,s+h,k} - x_{\ell,s+h,k}^{(s,j)} \right\|_{q,0} \leq C_* \rho_*^h \mathbf{1}\{\ell \in \mathcal{C}_N(j)\}.$$

Consequently, Assumption 2 is read under this alternative as the same coordinatewise predetermined-regressor weak-exogeneity condition, but with the enlarged pre-idiosyncratic-innovation information:

$$\mathbb{E}_0[u_{it} \mid x_{it}, \mathcal{H}_t] = 0. \quad (\text{A.1})$$

Thus the regressors may share aggregate or common shocks across clusters. The restriction is only that, after conditioning on the current regressor and those common components, the remaining idiosyncratic innovation has zero conditional mean.

Alternative Assumption 3' (exact known-cluster dependence). For each N , the units are partitioned into known clusters

$$\mathcal{C}_{1,N}, \dots, \mathcal{C}_{G_N,N}, \quad \{1, \dots, N\} = \bigcup_{g=1}^{G_N} \mathcal{C}_{g,N},$$

with $G_N \rightarrow \infty$ and maximum cluster size $M_N = \max_{g \leq G_N} |\mathcal{C}_{g,N}|$ allowed to grow slowly. Conditional on $\mathcal{H}_t$, arbitrary dependence is allowed within each cluster, and the cluster-specific residual sigma-fields generated by $\{(u_{it}, x_{it}) : i \in \mathcal{C}_{g,N}\}$ are independent across g . This conditional statement permits the observed regressors to have aggregate or common components shared across clusters through $\mathcal{A}_t$; only the remaining cluster-specific components are independent across clusters after conditioning on $\mathcal{H}_t$. The conditional moment bounds in Assumption 3 hold with $\mathcal{F}_{t-1}$ replaced by $\mathcal{H}_t$. Under this alternative the contemporaneous dependence block used in Assumption 6(iv) is

$$\mathcal{N}_{NT}(k) = \mathcal{C}_N(k),$$

where $\mathcal{C}_N(k)$ is the cluster containing unit k .

Cluster target dispersion. For a regular target, define the benchmark cluster score

$$S_{\nu,tg} = \sum_{i \in \mathcal{C}_{g,N}} \tilde{\Psi}_{\nu,it} u_{it}.$$

Under exact conditional cluster independence, the benchmark variance can be written as

$$s_{\nu,NT}^2 = \mathbb{E}_0 \sum_{t=1}^T \sum_{g=1}^{G_N} \mathbb{E}_0(S_{\nu,tg}^2 \mid \mathcal{H}_t).$$

The cluster analogue of Assumption 6(iv) requires no dominant cluster and a conditional cluster-array Lyapunov condition. For some $\delta_{\text{cl}} > 0$,

$$\max_{t,g} \frac{\mathbb{E}_0(S_{\nu,tg}^2)}{s_{\nu,NT}^2} \longrightarrow 0, \quad \frac{1}{s_{\nu,NT}^{2+\delta_{\text{cl}}}} \sum_{t=1}^T \sum_{g=1}^{G_N} \mathbb{E}_0 \left[|S_{\nu,tg}|^{2+\delta_{\text{cl}}} \mid \mathcal{H}_t \right] \xrightarrow{p} 0. \quad (\text{A.2})$$

The first condition is the no-dominant-cluster analogue of the local-block dispersion condition in Assumption 6(iv). The second condition is the conditional Lindeberg/Lyapunov condition used by the cluster-array CLT. When cluster sizes are uniformly bounded, it is implied by the finite-moment conditions and the no-dominant-cluster bound. Assumption 6 remains a target-regularity condition, not a second dependence assumption.

Cluster buffer and cluster studentizer. For a held-out cell (i, t) , replace the spatial ball by the whole cluster $\mathcal{C}_N(i)$. The cluster-buffered training set for fold j is

$$\mathcal{I}_{-j}^{\text{cl}} = \left\{ (\ell, s) \notin \mathcal{I}_j : \nexists (i, t) \in \mathcal{I}_j \text{ with } 0 \leq s - t \leq q_T, \ell \in \mathcal{C}_N(i) \right\}.$$

The buffer-volume terms are replaced by

$$B_{NT}^{\text{cl}} = (q_T + 1)M_N, \quad D_{NT}^{\text{cl}} = (2q_T + 1)M_N,$$

and the spatial tail terms $\Omega_x(r_N)$ and $\bar{\alpha}(r_N)$ are zero because different clusters are conditionally independent after conditioning on $\mathcal{H}_t$. The corresponding cluster-score variance estimator is

$$\hat{s}_{\nu,\text{cl}}^2 = \sum_{t=1}^T \sum_{g=1}^{G_N} \left(\sum_{i \in \mathcal{C}_{g,N}} \hat{\Psi}_{\nu,it}^{\text{cf}} \hat{u}_{it}^{\text{cf}} \right)^2. \quad (\text{A.3})$$

The cluster analogue of the fold-transport rate is obtained from $\varepsilon_{G,NT}$ by replacing B_{NT}, D_{NT} by $B_{NT}^{\text{cl}}, D_{NT}^{\text{cl}}$ and setting $\Omega_x(r_N) = \bar{\alpha}(r_N) = 0$. The rate regime therefore requires

$$\frac{B_{NT}^{\text{cl}}}{J_{NT}} \rightarrow 0, \quad b_{NT}^4 \frac{(J_{NT} + D_{NT}^{\text{cl}})(N + T) \log(NTJ_{NT})}{NT} \rightarrow 0, \quad \sqrt{NT} \rho_*^{q_T} \rightarrow 0,$$

with the remaining first-stage and Riesz-replacement rates unchanged.

Proposition 2 (Known-cluster alternative to spatial mixing). *Under Alternative Assumption 1', Assumption 2 read with the enlarged pre-idiosyncratic-innovation field $\mathcal{H}_t$ as in (A.1), Alternative Assumption 3', and Assumptions 4–6, with Assumption 6(iv) read through the cluster conditions in (A.2), the five main theorems continue to hold after the cluster replacements above. The estimator, target definition, rank selector, tangent-space construction, and Riesz equations are unchanged; $\widehat{s}_{\nu, \text{sp}}$ in Theorem 5 is replaced by $\widehat{s}_{\nu, \text{cl}}$.*

Proof. Only the decoupling and studentization arguments change. In the benchmark CLT, the within-period score is grouped as

$$S_{\nu, tg} = \sum_{i \in \mathcal{C}_{g, N}} \tilde{\Psi}_{\nu, it} u_{it}, \quad \xi_{\nu, t} = \sum_{g=1}^{G_N} S_{\nu, tg}.$$

Conditional independence across clusters given $\mathcal{H}_t$ gives conditional variance additivity across g . The no-dominant-cluster part of (A.2) prevents any single cluster from carrying a nonvanishing share of the benchmark variance, and the Lyapunov part of (A.2) gives the cluster-array Lindeberg condition. Equation (A.1) supplies the same martingale difference property used in the benchmark proof. In fold transport and leakage control, the spatial-neighborhood counts are replaced by M_N , producing B_{NT}^{cl} and D_{NT}^{cl} , and the spatial-mixing tails disappear because the full held-out cluster is removed. The rank-selection, first-stage, tangent-space, Riesz-replacement, and one-step expansion arguments depend on the dependence structure only through these transport and leakage rates, so they are unchanged after the stated substitutions. Finally, the spatial-kernel covariance proof is replaced by consistency of the cluster-score quadratic form $\widehat{s}_{\nu, \text{cl}}^2$; generated-weight replacement is the same because the feasible weights and residuals are still cross-fitted and predictable for held-out cluster scores. This proves the five main theorems under the known-cluster alternative. $\square$

A.2.5 Derived requirements

The common cross-fitting rates in (5.2) restrict analyst-chosen tuning sequences. The kernel and bandwidth requirements in (7.4) belong specifically to the variance estimator. The global objective-gap requirement (6.5) belongs only to a finite nonconvex iterate and is vacuous for the ideal minimizer. Finally, λ_{NT} , ζ_{NT} , δ_{NT} , $\delta_{\max, NT}$, $\varepsilon_{G, NT}$, and $r_{\nu, NT}$ are conclusions of the finite-moment localization, rowwise recovery, fold-transport, and one-step-expansion arguments below.

A.2.6 Feasible rates

Let $q = 8 + \eta$, suppose $N \asymp T = m$, and write $n = NT \asymp m^2$. Assume geometric temporal decay, geometric spatial propagation, and geometric spatial mixing decay. Choose

$$q_T = C_q \log n, \quad r_N = C_r \log n, \quad J_{NT} = \lceil m^\gamma \rceil, \quad 0 < \gamma < 1 - \frac{8}{q},$$

which is feasible for every $q > 8$, and let $L_{NT} = \log \log(n + e^e)$. Since the finite-moment localization envelope is $b_{NT} = n^{1/q} L_{NT}$,

$$b_{NT} = O(m^{2/q} L_{NT}), \quad \zeta_{NT} = O\{m^{4/q-1} L_{NT}^2 \log n\},$$

$$\delta_{\max, NT} = O\{m^{2/q-1/2} L_{NT} \sqrt{\log n} + m^{4/q-1} L_{NT}^2 \log n + m^{4/q-1+o(1)}\} = m^{2/q-1/2+o(1)}.$$

The validation-fold and overlap terms in the information-transport rate obey

$$\begin{aligned} \varepsilon_{G, NT} = O\big(& m^{4/q+(\gamma-1)/2} L_{NT}^2 \sqrt{\log n} + m^{4/q+\gamma-1} L_{NT}^2 \log n \\ & + m^{4/q-1/2} L_{NT}^2 (\log n)^{(d_s+2)/2} + m^{-\gamma} (\log n)^{d_s+1} \big) = o(1). \end{aligned}$$

The polynomial parts of the inference envelope are therefore

$$b_{NT} \delta_{\max, NT} = m^{4/q-1/2+o(1)}, \quad b_{NT}^2 \zeta_{NT} = m^{8/q-1+o(1)},$$

plus $\varepsilon_{G, NT} = o(1)$. All polynomial exponents are negative under the displayed interval and $q > 8$. The slice-deletion term is $b_{NT}^2 \{q_T/T + v_N(r_N)/N\} = m^{4/q-1+o(1)}$, and is already included in $\delta_{\max, NT}$; geometric decay of ρ_*^q , $\omega_x(r)$, and $\alpha(r)^{\chi_\eta}$ makes the temporal, spatial-propagation, and spatial-mixing tails negligible. Thus the finite-moment fold, rank-selection, first-stage, and inference restrictions are jointly nonempty for every fixed $\eta > 0$.

Rate-regime feasibility check. Under the balanced geometric-decay regime above, the cross-fitting design in [Section 6.1](#), the rate regime [\(5.2\)](#), the explicit penalty [\(5.7\)](#), and the remainder requirement [\(7.1\)](#) hold simultaneously. If the target satisfies [Assumption 6](#), a bandwidth satisfying the studentization rate requirement [\(7.4\)](#) also exists. The verification is direct substitution: the buffer volume is polylogarithmic, whereas $J_{NT} = m^\gamma$, so the retained share tends to one and every fold contains a diverging number of cells. The penalty separation follows from the orders in [Lemma 2](#); the inference remainder contains only negative powers of m . Finally, choose $h_N \rightarrow \infty$ sufficiently slowly that the mixing tail, local fourth-moment, and feasible-replacement requirements in [\(7.4\)](#) vanish.

A.2.7 Implementation notes

Remark 2 (Finite fold floors and growing J_{NT}). The asymptotic results of [Sections 7–A.4.2](#) require $(q_T + 1)v_N(r_N)/J_{NT} \rightarrow 0$, $n_{NT}/J_{NT} \rightarrow \infty$, and a retained training share bounded away from zero.

These restrictions rule out treating a fixed fold count as a separate general asymptotic theorem when the space–time buffer volume diverges.

The implementation rule (5.3) reconciles the asymptotic requirement with finite-sample practice. The term $\lceil c_J B_{NT} L_{NT}^J \rceil$ grows slowly and dominates the buffer volume, while $J_{\min}$ is a conventional finite-sample floor. Common cross-fitting floors such as $J_{\min} = 5$ or $J_{\min} = 10$ are therefore implementation choices, not additional asymptotic assumptions. The empirical and simulation sections should report the realized J , the retained training share, and robustness to nearby floors such as $J_{\min} \in \{5, 6, 8, 10, 12\}$.

If the retained training share is low, the fold-Gram matrices are unstable, or the target estimates are sensitive to nearby $J_{\min}$ values, the finite-sample fold design should be treated as unreliable for that target. The remedy is to increase J , reconsider the buffer, or report the calculation as a sensitivity analysis rather than as theorem-backed inference.

The criterion (5.8) uses the fixed finite candidate box (4.3) and explicit penalty (5.7). In practice one must choose the number of folds J , the forward exclusion window q_T , and a residual-scale multiplier for the penalty. This subsection gives an explicit, data-guided recipe. Its theorem-covered version uses a one-sided persistence bound, lets J_{NT} grow so that the fold conditions hold, searches the complete fixed box, and uses the penalty order in (5.7). The commonly used fixed- J version is a finite-sample approximation governed by Remark 2; it is not asserted to inherit Proposition 5 without the stated diagnostics.

Step 0: preliminary stable fit and a one-sided persistence safeguard. Compute a full-sample regularized fit at a deliberately generous working rank vector $\mathbf{r}^{\text{wk}}$, form the fitted companion matrices $\hat{\mathbf{C}}_{it}$, and use finite-horizon companion products to obtain the point diagnostic

$$\hat{\rho}_{\text{pt}} = \max_{1 \leq h \leq H_{NT}} \max_{i,t} \left\| \hat{\mathbf{C}}_{it} \hat{\mathbf{C}}_{t-1,i} \cdots \hat{\mathbf{C}}_{t-h+1,i} \right\|_{\text{op}}^{1/h},$$

where the inner maximum uses only products observed within the sample. A point diagnostic is not enough for the purge theorem: finite-horizon approximation, estimation error, and a near-unit-root boundary can all make it undershoot the true decay modulus. Inflate it by a one-sided uncertainty buffer (from analytic perturbation bounds or a stability-respecting resampling calibration) to obtain $\bar{\rho}_{NT}$. The theorem-covered roadmap requires

$$\mathbb{P}_0(\rho_* \leq \bar{\rho}_{NT} < 1) \rightarrow 1.$$

If no upper bound strictly below one is credible, the stable-dynamics assumption is not empirically certified and the paper’s dynamic interval should not be presented as theorem-backed. This step also returns the residual scale $\hat{\sigma}^2$ used in the rank penalty.

Step 1: length of the space–time exclusion buffer. For a constant $c_q > 1/2$, set

$$q_T = \left\lceil \frac{c_q \log(TN)}{|\log \bar{\rho}_{NT}|} \right\rceil.$$

On the one-sided-bound event $\rho_* \leq \bar{\rho}_{NT} < 1$, this gives $\rho_*^{q_T} \sqrt{TN} \leq (TN)^{1/2-c_q} \rightarrow 0$. The spatial radius must be selected, or verified on a sensitivity grid, so that $\{\Omega_x(r_N) + \bar{\alpha}(r_N)\} \sqrt{NT} \rightarrow 0$. The default $c_q = 1$ leaves a temporal margin. The selected window must also be compatible with training retention; Step 2 increases J when needed. A hard finite-sample cap on q may be reported as a sensitivity analysis, but unless the capped sequence still diverges fast enough to satisfy the temporal-tail restriction in (5.2), the resulting interval is not covered by the dynamic leakage theorem.

Step 2: number of scattered folds. Choose J through the finite-sample floor rule (5.3). The realized retained shares are

$$\tau_{\text{nv}} = \frac{1}{J} \sum_{j=1}^J \frac{|\mathcal{I}_{-j}^{\text{buf}}|}{NT - |\mathcal{I}_j|}, \quad \tau_{\text{tot}} = \frac{1}{J} \sum_{j=1}^J \frac{|\mathcal{I}_{-j}^{\text{buf}}|}{NT},$$

where τ_{nv} is the fraction of candidate non-validation cells retained after buffering and τ_{tot} is the average fraction of all cells used in each fold-specific training problem. A useful implementation keeps τ_{nv} bounded away from zero and reports it in the simulation and empirical diagnostics. We also report robustness to nearby finite-sample floors $J_{\min}$, because $J_{\min}$ is an implementation choice while the growing term in (5.3) is what connects the procedure to the asymptotic theorem. Scattered cell-level folds generated by the auxiliary stratified randomization, *not* contiguous time blocks, are used: the forward exclusion removes only future space–time descendants, and scattering the held-out cells keeps every time period and every unit represented in training, so the local information map stays well conditioned on the buffered training set (Assumption 5). A contiguous time-block hold-out would leave the held-out block’s time-specific heterogeneity unidentified from training and make the training-set information map singular in time-localized directions; the scattered construction avoids this. Remark 2 explains the fixed-versus-growing J trade-off in full.

Step 3: fixed candidate box with spectral ordering. The theorem uses the deterministic box $\mathcal{R}$ in (4.3). A preliminary fit may order candidates by the blockwise singular-value profiles $\hat{\sigma}_{m,1} \geq \hat{\sigma}_{m,2} \geq \dots$, but it does not delete any rank from the theorem-covered box. If screening is used to prune the box in computation, the resulting search is treated as a finite-sample approximation unless the retained set is independently shown to contain $\mathbf{r}_0$. Fixed caps keep $|\mathcal{R}|$ small, so the full search is usually feasible.

Step 4: penalty constant. Use the residual-scaled operational version of (5.7),

$$\hat{\kappa}_{NT} = c_\kappa \hat{\sigma}^2 b_{NT}^2 \ell_{NT}^2 \zeta_{NT}^{-1/2},$$

with default $c_\kappa = 1$. If $\hat{\sigma}^2$ is consistent and bounded away from zero, it does not change the two separation orders in Lemma 2. Report the selected ranks over $c_\kappa \in \{0.5, 1, 2\}$ as a sensitivity check.

Step 5: solve and diagnose. Evaluate (5.8) over $\mathcal{R}$ using the scattered folds and exclusion window from Steps 1–2. Report the selected $\hat{\mathbf{r}}$, the criterion value in a neighborhood of $\hat{\mathbf{r}}$, and the stability diagnostic below.

Algorithm 1 (Data-driven rank, fold, and exclusion-window selection). **Input:** panel $\mathbf{Y}$, regressors $\{\mathbf{Z}^{(m)}\}$, fixed rank caps $\bar{\mathbf{r}}$, and grid $\{0.5, 1, 2\}$ for c_κ .

1. Fit the full-sample regularized estimator at the generous working rank; obtain $\hat{\rho}_{\text{pt}}$, a one-sided bound $\bar{\rho}_{NT} < 1$, and $\hat{\sigma}^2$ (Step 0).
2. Set $q_T = \lceil c_q \log(TN) / |\log \bar{\rho}_{NT}| \rceil$ and record whether the persistence safeguard is credible (Step 1).
3. Choose J to meet the retention and validation-size diagnostics; label the output theorem-covered only when the growing-fold conditions are also met (Step 2).
4. Retain the fixed product box $\mathcal{R}$ in (4.3); use the preliminary singular values only to order the candidate fits (Step 3).
5. For each $\mathbf{r} \in \mathcal{R}$ and fold j , compute the regularized fit on the buffered training set and the out-of-fold loss (5.5).
6. Set $\hat{\kappa}_{NT} = c_\kappa \hat{\sigma}^2 b_{NT}^2 \ell_{NT}^2 \zeta_{NT}^{-1/2}$ and return $\hat{\mathbf{r}}$ from (5.8) with $\hat{\kappa}_{NT}$ substituted for κ_{NT} (Steps 4–5).
7. Report $\hat{\mathbf{r}}$ for each c_κ in the grid and the stability diagnostic of Definition 1.

Output: $\hat{\mathbf{r}}$, q_T , J , and a stability report.

Definition 1 (Rank stability diagnostic). Re-run Steps 2–6 of Algorithm 1 for each J' in the feasible set and each $c_\kappa \in \{0.5, 1, 2\}$. Report the fraction of these configurations that return the same $\hat{\mathbf{r}}$. A selection that is stable across folds and penalty constants is numerically reassuring; instability flags either a weak rank gap (small $\hat{\sigma}_{m, r_{0,m}}$, violating signal strength) or a too-persistent panel ($\bar{\rho}_{NT}$ near one, so the certified q_T consumes too much of the sample).

Computational cost. Each candidate-fold pair requires one alternating-ridge fit, which by [Proposition 6](#) converges in $O(\log(TN))$ sweeps, each sweep costing $O(TN\bar{r}^2)$ for the block updates [\(6.3\)–\(6.4\)](#). The total cost is $O(|\mathcal{R}|JTN\bar{r}^2\log(TN))$. Because the candidate box has only a handful of values per coordinate and J is moderate in implementation, this is linear in the panel size up to logarithmic factors and is dominated by the cost of the final inference step.

Overselection robustness. By [Proposition 5](#) the selector is consistent, so the event of a tie vanishes asymptotically; the question is only which error is safer in finite samples when the stability diagnostic reports a near-tie between $\mathbf{r}_0$ and an overfitted neighbor. Underselection is fatal—a missed rank direction is omitted-variable bias that no debiasing step can repair—whereas mild overselection leaves the prediction risk at the frontier and inflates only the effective dimension $d(\mathbf{r})$. We therefore recommend taking the larger of two near-tied rank vectors when the diagnostic is ambiguous. We are explicit about what is and is not proved: the limit theory of [Sections 4–7](#) is stated at the true rank vector, and on the vanishing overselection event it applies via rank consistency; a formal treatment of inference *conditional* on a fixed overfitted rank would require extending the tangent-space analysis of [Propositions 5 and 7](#) to the enlarged (rank-inflated) tangent spaces. Because this requires a separate rank-inflated inference theory, we do not state it as part of the present theorem.

A.2.8 Notation

All notation is introduced in the text where it is first needed; this table collects it in one place for reference. Throughout, $X_{NT} = O_{p,0}(a_{NT})$ means X_{NT}/a_{NT} is bounded in $\mathbb{P}_0$ -probability and $X_{NT} = o_{p,0}(a_{NT})$ means $X_{NT}/a_{NT} \rightarrow 0$ in $\mathbb{P}_0$ -probability, both under the conditional law $\mathbb{P}_0$ defined in [Section 4](#); $\mathbf{1}\{A\}$ is the indicator of A .

Table 13: Notation glossary.

Symbol	Meaning
<i>Sample, rates, and probability</i>	
$\mathcal{I}_{NT}, T$	Usable post-lag sample $\{1, \dots, N\} \times \{1, \dots, T\}$ and its time length (Section 3.1).
$n_{NT}, a_{NT}, b_{NT}, \zeta_{NT}, \delta_{NT}$	Cell count and basic rates: $n_{NT} = NT$, $a_{NT} = T^{-1} + N^{-1}$, b_{NT} is the finite-moment localization sequence, $\zeta_{NT} = b_{NT}^2 \ell_{NT}^2 a_{NT}$, and $\delta_{NT} = b_{NT} \ell_{NT} \sqrt{a_{NT}}$ (Section 5).
$\tau_{U,NT}, \tau_{V,NT}$	LOO deletion rates: $\tau_{U,NT} = b_{NT}^2 \{N^{-1} \vee (TN)^{-1/2}\}$ controls deletion effects for unit-side singular vectors, and $\tau_{V,NT} = b_{NT}^2 \{T^{-1} \vee (TN)^{-1/2}\}$ controls deletion effects for time-side singular vectors.

Symbol	Meaning
$\delta_{\max, NT}$	Uniform row-/column-wise singular-vector, singular-value, and entrywise first-stage envelope in (6.6); it dominates the global subspace rate δ_{NT} .
$\mathbb{P}_0, \mathbb{E}_0, \text{Var}_0$	Probability, expectation, and variance conditional on the primitive frame $\mathcal{G}_0$, which does not condition away the stochastic covariate path (Section 3.4).
$j(a)$	Unique validation-fold index containing cell a .
<i>Model objects</i>	
$\mathbf{\Gamma}_0^{(m)}$	m th coefficient matrix: autoregressive coefficient for $m \leq P$, slope for $m > P$ (Section 3.1).
$\mathbf{H}_0$	Interactive-effect (nuisance) low-rank matrix.
$\mathbf{C}_{it}$	Heterogeneous companion matrix built from the autoregressive coefficients, (3.2).
$\mathbb{H}_{NT}$	Product parameter space containing all coefficient blocks and the nuisance block.
$\mathcal{A}$	Design operator, $\mathcal{A}(\boldsymbol{\Theta}) = \sum_m \mathbf{Z}^{(m)} \odot \mathbf{\Gamma}^{(m)} + \mathbf{H}$, (A.27).
ρ_*	Geometric decay (stability) rate of the heterogeneous dynamics, Assumption 1.
<i>Local geometry and identification</i>	
$\mathcal{T}_0$	Set of admissible local perturbations of the low-rank matrices (product tangent space), Appendix A.3.5.
$\mathcal{G}_0$	Full-sample local information map, $\mathcal{G}_0 = \mathcal{P}_{\mathcal{T}_0} \mathcal{A}^* \mathcal{A} \mathcal{P}_{\mathcal{T}_0}$ (analogue of $X'X$), (A.31).
$\mathcal{L}_{\Gamma, m}, \mathcal{L}_H$	Fitted-value changes generated by local perturbations: $\mathbf{Z}^{(m)} \odot \mathcal{T}_{\Gamma, m}$ and $\mathcal{T}_H$, (3.6).
$\mathcal{S}_{\mathcal{R}}, \mathfrak{S}_{\mathcal{R}}$	Between-candidate (rank-class) differences and their normalized fitted contrasts, Assumption 5.
<i>Targets and debiasing weights</i>	
$\mathbf{D}_\nu, \mathbf{W}_\nu$	Target direction in the product space and its local projection $\mathbf{W}_\nu = \mathcal{P}_{\mathcal{T}_0} \mathbf{D}_\nu$, (A.30).
$\mathbf{q}_{\nu, j}^{\text{or}}, \boldsymbol{\Psi}_{\nu, j}^{\text{or}}$	Benchmark (infeasible) fold-specific debiasing direction and observation-space weight for validation fold j , (A.34).
$\hat{\mathbf{q}}_{\nu, -j}, \hat{\boldsymbol{\Psi}}_{\nu, -j}$	Feasible debiasing direction and weight estimated from the buffered complement of fold j , (6.7).
$L_\nu^{\text{or}}, s_{\nu, NT}^2$	Benchmark leading term and its variance (Lemma 3).
$\check{\phi}_\nu, \hat{s}_{\nu, \text{sp}}, \hat{s}_{\nu, \text{cl}}$	One-step estimator (6.8), spatial-kernel standard error (6.9), and known-cluster standard error (A.3).
<i>Cross-fitting and the exclusion window</i>	
$J_{NT}, p_{j, NT}$	Number of scattered folds and realized fold share $ \mathcal{I}_j /(TN)$, with $p_{j, NT} \asymp J_{NT}^{-1}$ uniformly by fold balance, Section 6.1.
q_T, r_N	Temporal and spatial radii of the space–time exclusion buffer, Section 6.1.

Symbol	Meaning
$\mathcal{I}_{-j}^{\text{buf}}$	Buffered training set removing each held-out cell's temporal descendants and contemporaneous spatial neighborhood, (6.1).
$\varepsilon_{G,NT}$	Common envelope for training-validation information transport, expected-Gram concentration, and leave-one-slice transport.
$R_{\nu,NT}^{\text{cf}}$	Target-specific full-covariance variation is handled in Appendix A.4. Derived target-local remainder in (A.37), including rowwise, fold-transport, temporal-tail, and spatial-mixing terms.
<i>Rank selection</i>	
$\mathcal{R}, \mathcal{M}(\mathbf{r}), d(\mathbf{r})$	Candidate set, rank class, and effective dimension (5.6).
$L_0(\mathbf{r}), \hat{L}(\mathbf{r}), \kappa_{NT}$	Population risk (5.4), cross-fitted loss (5.5), and penalty level.

A.3 Rank selection, first-stage recovery, fold transport, and Riesz replacement

A.3.1 Operators and identification

Proposition 3 (Operator identities, identification, entropy, and local information). *For $\mathcal{A}$ in (A.27), the adjoint is (A.28). Hence, for every $\Delta \in \mathcal{T}_0$,*

$$\langle \Delta, \mathcal{G}_0 \Delta \rangle_{\mathbb{H}} = \|\mathcal{A}(\Delta)\|_F^2.$$

If the purged-training fold Gram operator in (A.34) is nonsingular, then (A.36) holds. If, in addition, the fold-transport bound of Proposition 4 holds, then, uniformly over $\Delta \in \mathcal{T}_0$,

$$\left| \langle \mathbf{D}_{\nu}, \Delta \rangle_{\mathbb{H}} - p_{j,NT}^{-1} \langle \Pi_j \Psi_{\nu,j}^{\text{or}}, \Pi_j \mathcal{A}(\Delta) \rangle \right| \leq \varepsilon_{G,NT} \|\mathbf{q}_{\nu,j}^{\text{or}}\|_{\mathbb{H}} \|\Delta\|_{\mathbb{H}}.$$

Proof. For any $\mathbf{R} \in \mathbb{R}^{N \times T}$,

$$\langle \mathcal{A}(\Theta), \mathbf{R} \rangle = \sum_{m=1}^M \left\langle \mathbf{Z}^{(m)} \odot \Gamma^{(m)}, \mathbf{R} \right\rangle + \langle \mathbf{H}, \mathbf{R} \rangle = \sum_{m=1}^M \left\langle \Gamma^{(m)}, \mathbf{Z}^{(m)} \odot \mathbf{R} \right\rangle + \langle \mathbf{H}, \mathbf{R} \rangle,$$

which gives the adjoint. If $\Delta \in \mathcal{T}_0$, then $\mathcal{P}_{\mathcal{T}_0} \Delta = \Delta$, so

$$\langle \Delta, \mathcal{G}_0 \Delta \rangle_{\mathbb{H}} = \langle \Delta, \mathcal{A}^* \mathcal{A} \Delta \rangle_{\mathbb{H}} = \|\mathcal{A}(\Delta)\|_F^2.$$

For the fold identity, let $\mathbf{W}_{\nu} = \mathcal{P}_{\mathcal{T}_0} \mathbf{D}_{\nu}$. For $\Delta \in \mathcal{T}_0$,

$$\begin{aligned} \alpha_j \left\langle \mathcal{A}_{-j}^{\text{buf,pred}} \mathbf{q}_{\nu,j}^{\text{or}}, \mathcal{A}_{-j}^{\text{buf,pred}} \Delta \right\rangle &= \left\langle \mathbf{q}_{\nu,j}^{\text{or}}, \alpha_j \mathcal{P}_{\mathcal{T}_0} (\mathcal{A}_{-j}^{\text{buf,pred}})^* \mathcal{A}_{-j}^{\text{buf,pred}} \mathcal{P}_{\mathcal{T}_0} \Delta \right\rangle_{\mathbb{H}} \\ &= \langle \mathbf{W}_{\nu}, \Delta \rangle_{\mathbb{H}} = \langle \mathbf{D}_{\nu}, \Delta \rangle_{\mathbb{H}}. \end{aligned}$$

The validation-fold display follows by subtracting the validation Gram from the predictable purged-training Gram and applying the validation component of (A.12) with $\Xi = \mathbf{q}_{\nu,j}^{\text{or}} / \|\mathbf{q}_{\nu,j}^{\text{or}}\|_{\mathbb{H}}$. $\square$

Fitted-value no-cancellation condition. There are constants $c_L, C_L > 0$ and $\rho_L < 1$ such that, on the full sample and on each rescaled fold, every coefficient fitted-value change satisfies

$$c_L \|\Delta^{(m)}\|_F^2 \leq \|\mathbf{Z}^{(m)} \odot \Delta^{(m)}\|_F^2 \leq C_L b_{NT}^2 \|\Delta^{(m)}\|_F^2,$$

and, for arbitrary images $\mathbf{L}_m = \mathbf{Z}^{(m)} \odot \Delta^{(m)}$ and $\mathbf{L}_H = \Delta^H$,

$$\sum_{a \neq b} |\langle \mathbf{L}_a, \mathbf{L}_b \rangle| \leq \rho_L \sum_a \|\mathbf{L}_a\|_F^2. \quad (\text{A.4})$$

The same inequalities may be imposed on the candidate low-rank difference spaces to verify the full Assumption 5 rather than only its tangent restriction.

Residual identifying-variation condition. For the decomposition (3.8), there are $c_E > 0$ and $\rho_E < 1$ such that, uniformly over relevant secants,

$$\left\| \sum_{m=1}^M \mathbf{E}_Z^{(m)} \odot \Delta^{(m)} \right\|_F^2 \geq c_E \sum_{m=1}^M \|\Delta^{(m)}\|_F^2, \quad (\text{A.5})$$

and, with $\mathbf{L}_E = \sum_m \mathbf{E}_Z^{(m)} \odot \Delta^{(m)}$ and $\mathbf{C}_F = \sum_m \mathbf{Z}_F^{(m)} \odot \Delta^{(m)} + \Delta^H$,

$$|\langle \mathbf{L}_E, \mathbf{C}_F \rangle| \leq \rho_E \{\|\mathbf{L}_E\|_F^2 + \|\mathbf{C}_F\|_F^2\} / 2. \quad (\text{A.6})$$

The common-factor image is also bounded above by $C b_{NT}^2 \|\Delta\|_{\mathbb{H}}^2$. Fold analogues are imposed after the normalizations in Assumption 5.

Auxiliary result (Two primitive identification checks). *The fitted-value no-cancellation condition above implies the local tangent restricted-eigenvalue bounds in Assumption 5; when imposed uniformly on $\mathcal{S}_{\mathcal{R}}$, it verifies the full assumption. The residual identifying-variation condition likewise implies the local tangent restricted-eigenvalue bounds; when imposed uniformly on the candidate low-rank difference classes, it verifies the global secant part of Assumption 5.*

Proof. For the no-cancellation check, take $\Delta \in \mathcal{T}_0$, write $\mathbf{L}_m = \mathbf{Z}^{(m)} \odot \Delta^{(m)}$ and $\mathbf{L}_H = \Delta^H$. Then $\mathcal{A}(\Delta) = \sum_a \mathbf{L}_a$. Expanding the squared norm and using (A.4),

$$\|\mathcal{A}(\Delta)\|_F^2 \geq (1 - \rho_L) \sum_a \|\mathbf{L}_a\|_F^2 \geq c \|\Delta\|_{\mathbb{H}}^2.$$

The upper bound follows from the within-image upper bounds and the triangle inequality. Fold versions are identical after replacing the full inner product with the rescaled fold inner product.

For the residual-variation check, write

$$\mathbf{L}_E = \sum_{m=1}^M \mathbf{E}_Z^{(m)} \odot \Delta^{(m)}, \quad \mathbf{C}_F = \sum_{m=1}^M \mathbf{Z}_F^{(m)} \odot \Delta^{(m)} + \Delta^H,$$

so that $\mathcal{A}(\Delta) = \mathbf{L}_E + \mathbf{C}_F$. The residual identifying-variation condition gives a lower bound for coefficient perturbations through residual variation, while the angle condition prevents this residual component from being canceled by the common-factor and nuisance components. In particular,

$$\|\mathcal{A}(\Delta)\|_F^2 = \|\mathbf{L}_E\|_F^2 + \|\mathbf{C}_F\|_F^2 + 2\langle \mathbf{L}_E, \mathbf{C}_F \rangle \geq (1 - \rho_E)\{\|\mathbf{L}_E\|_F^2 + \|\mathbf{C}_F\|_F^2\}.$$

By (A.5), $\sum_m \|\Delta^{(m)}\|_F^2 \leq c_E^{-1} \|\mathbf{L}_E\|_F^2$. Since $\Delta^H = \mathbf{C}_F - \sum_m \mathbf{Z}_F^{(m)} \odot \Delta^{(m)}$ and the common components are bounded on the localization event,

$$\|\Delta^H\|_F^2 \leq C \|\mathbf{C}_F\|_F^2 + C b_{NT}^2 \sum_m \|\Delta^{(m)}\|_F^2.$$

Combining these bounds gives the required population restricted-eigenvalue lower bound. The upper bound follows from bounded regressors, fixed number of coefficient blocks, and the triangle inequality. The same argument applies to candidate low-rank difference classes and to rescaled folds. $\square$

Lemma 1 (Fold balance, local dependence, and norm transport). *Under the cross-fitting design in Section 6.1 and the rate regime (5.2), let*

$$N_{jt} = \sum_{i=1}^N \mathbf{1}\{(i, t) \in \mathcal{I}_j\}, \quad T_{ji} = \sum_{t=1}^T \mathbf{1}\{(i, t) \in \mathcal{I}_j\}, \quad n_j = |\mathcal{I}_j|.$$

Then, uniformly over folds, time periods, and units,

$$\max_{j,t} \left| \frac{J_{NT} N_{jt}}{N} - 1 \right| + \max_{j,i} \left| \frac{J_{NT} T_{ji}}{T} - 1 \right| + \max_j \left| \frac{J_{NT} n_j}{NT} - 1 \right| = o_{p,0}(1). \quad (\text{A.7})$$

If

$$N_{js}^{\text{tr}} = |\{\ell : (\ell, s) \in \mathcal{I}_{-j}^{\text{buf}}\}|, \quad T_{j\ell}^{\text{tr}} = |\{s : (\ell, s) \in \mathcal{I}_{-j}^{\text{buf}}\}|,$$

then

$$\min_{j,s} \frac{N_{js}^{\text{tr}}}{N} \wedge \min_{j,\ell} \frac{T_{j\ell}^{\text{tr}}}{T} \wedge \min_j \frac{|\mathcal{I}_{-j}^{\text{buf}}|}{NT} \geq 1 - \{1 + o_{p,0}(1)\} \frac{B_{NT}}{J_{NT}} \rightarrow_p 1. \quad (\text{A.8})$$

Consequently every validation fold contains diverging numbers of observations from every time period and every unit, while every buffered training sample retains asymptotically all rows, columns, and cells.

Proof. Conditional on the panel, the auxiliary labels are independent. For fixed (j, t) , $N_{jt} \sim \text{Bin}(N, J_{NT}^{-1})$; for fixed (j, i) , $T_{ji} \sim \text{Bin}(T, J_{NT}^{-1})$; and $n_j \sim \text{Bin}(NT, J_{NT}^{-1})$. For any fixed $\epsilon \in (0, 1)$,

the multiplicative Chernoff bound and a union bound give

$$\Pr_0 \left(\max_{j,t} \left| \frac{J_{NT} N_{jt}}{N} - 1 \right| > \epsilon \right) \leq 2J_{NT}T \exp\{-c\epsilon^2 N/J_{NT}\},$$

with analogous bounds $2J_{NT}N \exp\{-c\epsilon^2 T/J_{NT}\}$ and $2J_{NT} \exp\{-c\epsilon^2 NT/J_{NT}\}$. The validation-size implication of (5.2) makes all three probabilities vanish, proving (A.7).

Fix a fold j and a time s . A unit ℓ can be removed from time s only by a validation cell $(i, t) \in \mathcal{I}_j$ with $s - q_T \leq t \leq s$ and $d_N(i, \ell) \leq r_N$. Hence, deterministically,

$$N - N_{js}^{\text{tr}} \leq v_N(r_N) \sum_{t=(s-q_T) \vee 1}^s N_{jt} \leq B_{NT} \max_t N_{jt}.$$

Likewise, for a fixed unit ℓ ,

$$T - T_{j\ell}^{\text{tr}} \leq (q_T + 1) \sum_{i: d_N(i, \ell) \leq r_N} T_{ji} \leq B_{NT} \max_i T_{ji}.$$

Finally each validation cell deletes at most B_{NT} cells, so

$$NT - |\mathcal{I}_{-j}^{\text{buf}}| \leq B_{NT} n_j.$$

Substituting (A.7) into these deterministic bounds and using $B_{NT}/J_{NT} \rightarrow 0$ proves (A.8) uniformly over folds. $\square$

Matrix concentration under local fold dependence. We use the graph-coloring reduction to independent color classes, combined with Tropp's matrix Freedman inequality (Tropp, 2011, Theorem 1.2) and matrix Bernstein inequality (Tropp, 2015, matrix Bernstein inequality for independent self-adjoint sums). Thus, if centered self-adjoint matrices Y_a have a dependency graph of maximum degree D , operator envelope R , conditional variance proxy σ^2 , and matrix dimension d , then

$$\left\| \sum_a Y_a \right\|_{\text{op}} = O_{p,0} \{ \sqrt{(D+1)\sigma^2 \log d} + (D+1)R \log d \}.$$

Proposition 4 (Fold-information transport from auxiliary randomization). *Suppose Assumptions 4 and 3, the cross-fitting design in Section 6.1, and the rate regime (5.2) hold, and let $\mathcal{G}^{\text{full}} = \mathcal{P}_{\mathcal{T}_0} \mathcal{A}^* \mathcal{A} \mathcal{P}_{\mathcal{T}_0}$. Then, uniformly in j ,*

$$\|\mathcal{G}_j^{\text{val}} - \mathcal{G}^{\text{full}}\|_{\text{op}(\mathcal{T}_0)} = O_{p,0} \left[b_{NT}^2 \left\{ \sqrt{\frac{J_{NT}(N+T) \log(NTJ_{NT})}{NT}} + \frac{J_{NT}(N+T) \log(NTJ_{NT})}{NT} \right\} \right],$$

$$\|\mathcal{G}_j^{\text{tr}} - \mathcal{G}^{\text{full}}\|_{\text{op}(\mathcal{T}_0)} = O_{p,0}(\varepsilon_{G,NT}).$$

The same order holds for bounded variance-weighted bilinear forms and for arguments in estimated

tangent spaces at operator distance $O_{p,0}(\delta_{NT})$. If $\mathcal{G}_j^{(-t)}$ deletes the complete space-time dependence neighborhood of time t , then

$$\max_{j,t} \|\mathcal{G}_j^{(-t)} - \mathcal{G}_j^{\text{tr}}\|_{\text{op}(\mathcal{T}_0)} = O_{p,0} \left[b_{NT}^2 \left\{ \frac{q_T + 1}{T} + \frac{v_N(r_N)}{N} \right\} + \rho_*^{q_T} + \Omega_x(r_N) + \bar{\alpha}(r_N) \right] = o_{p,0}(1). \quad (\text{A.9})$$

No separate population fold-homogeneity assumption is used.

Proof. Write $d_0 = \dim(\mathcal{T}_0) \leq C(N+T)$ and let $\varphi_a \in \mathcal{T}_0$ be the normalized tangent feature satisfying

$$\langle \varphi_a, \Delta \rangle_{\mathbb{H}} = \sqrt{NT} [\mathcal{A}\Delta]_a.$$

Thus a Hilbert-unit tangent direction $\|\Delta\|_{\mathbb{H}} = 1$ has prediction norm measured by the averaged quadratic form

$$(NT)^{-1} \sum_a \langle \varphi_a, \Delta \rangle_{\mathbb{H}}^2 = \|\mathcal{A}\Delta\|_F^2.$$

The random matrices below therefore live in the $d_0 = O(N+T)$ tangent dimension; the ambient cell count NT appears only as the number of averaged observations. On the finite-moment localization event, incoherence, fixed ranks, and the bounded coefficient envelope give

$$\max_a \|\varphi_a\|_{\mathbb{H}}^2 \leq C b_{NT}^2 (N+T), \quad \left\| \frac{1}{NT} \sum_a \varphi_a \otimes \varphi_a \right\|_{\text{op}} \leq C. \quad (\text{A.10})$$

The second inequality is the sample consequence of Assumption 5, proved first on a fixed net by Davydov-type covariance and blocking inequalities for strong-mixing arrays (Doukhan, 1994, covariance inequality for mixing variables) and then on the tangent sphere by Proposition 3.

Validation folds. Conditional on the panel, the indicators $I_{a,j} = \mathbf{1}\{\sigma_a = j\}$ are independent Bernoulli variables with mean J_{NT}^{-1} . The validation operator is

$$\mathcal{G}_j^{\text{val}} = \frac{1}{p_{j,NT} NT} \sum_a I_{a,j} \varphi_a \otimes \varphi_a.$$

Lemma 1 gives $p_{j,NT} = J_{NT}^{-1} \{1 + o_{p,0}(1)\}$, uniformly over folds, time periods, and units. Conditional matrix Bernstein, using (A.10), therefore gives

$$\max_j \left\| \mathcal{G}_j^{\text{val}} - \mathcal{G}^{\text{full}} \right\|_{\text{op}} \leq C b_{NT}^2 \left\{ \sqrt{\frac{J_{NT} d_0 \log(NT J_{NT})}{NT}} + \frac{J_{NT} d_0 \log(NT J_{NT})}{NT} \right\}.$$

After a union bound over j . This calculation keeps $p_{j,NT}^{-1} \asymp J_{NT}$ explicitly: it converts the raw held-out sum into an average over NT/J_{NT} cells.

Unbuffered training complement. The normalized unbuffered complement is an affine combination of the full and validation operators. Its discrepancy is therefore of the validation order.

Overlapping buffer deletion. Let $R_{a,j}$ indicate that cell a survives the buffer for fold j . It is

a function only of labels in the backward space–time neighborhood of a , which contains at most $B_{NT} + 1$ cells. Two survival indicators are independent whenever their label neighborhoods are disjoint; their dependency graph has degree at most CD_{NT} . Apply Lemma 1 to

$$\frac{1}{NT} \sum_a \{R_{a,j} - E(R_{a,j})\} \varphi_a \otimes \varphi_a.$$

Using (A.10) gives the two terms involving D_{NT} in $\varepsilon_{G,NT}$. Moreover, $1 - E(R_{a,j}) \leq CB_{NT}/J_{NT}$ uniformly, including boundary cells: a fixed cell is deleted only if one of the at most B_{NT} cells in its complete space–time buffer neighborhood falls in validation fold j , and each such label has probability J_{NT}^{-1} . Hence the deterministic deletion bias is at most CB_{NT}/J_{NT} in operator norm. Renormalization by the retained share changes these bounds only by $1 + o_{p,0}(1)$.

Recursive and spatial comparison. Replace the surviving generated regressors by their fold-coupled versions. Assumption 1 and the resolvent/telescoping expansion of the dynamic recursion give an operator perturbation $O_{p,0}\{\rho_*^{q_T} + \Omega_x(r_N)\}$. Conditional Davydov inequality and the spatial buffer give $O_{p,0}\{\bar{\alpha}(r_N)\}$. Combining these bounds with the preceding three steps gives the training result.

Slice deletion and extensions. A row-local tangent perturbation has energy $O(T^{-1})$ per time period, and a column-local perturbation has energy $O(N^{-1})$ per unit, by incoherence and Assumption 4. Deleting $q_T + 1$ time periods and at most $v_N(r_N)$ neighboring units therefore gives (A.9); the coupling tails are the same as above. Bounded variance weights enter only as scalar multipliers. Polarization gives bilinear forms, and the projector identity $P\mathcal{G}P - \tilde{P}\mathcal{G}\tilde{P} = (P - \tilde{P})\mathcal{G}P + \tilde{P}\mathcal{G}(P - \tilde{P})$ handles estimated tangent spaces. This completes the proof. $\square$

Auxiliary result (Population projection inequality). *Fix a candidate rank $\mathbf{r}$, and let $\Theta_{\mathbf{r}}^*$ minimize $L_0(\mathbf{r})$ over the compact class $\mathcal{M}_B(\mathbf{r})$. Then, for every $\Theta \in \mathcal{M}_B(\mathbf{r})$,*

$$2\mathbb{E}_0 \langle \mathcal{A}(\Theta_0 - \Theta_{\mathbf{r}}^*), \mathcal{A}(\Theta - \Theta_{\mathbf{r}}^*) \rangle \leq \mathbb{E}_0 \|\mathcal{A}(\Theta - \Theta_{\mathbf{r}}^*)\|_F^2. \quad (\text{A.11})$$

No strict projection-angle margin is claimed. The rank-selection proof uses uniform risk consistency for underfitted candidates and the fast basic inequality only for candidates that contain the truth.

Proof. Let $R_0(\Theta) = (NT)^{-1}E_0\|\mathcal{A}(\Theta - \Theta_0)\|_F^2$. Global minimality gives $R_0(\Theta) - R_0(\Theta_{\mathbf{r}}^*) \geq 0$. Expand the squared loss and rearrange. This yields (A.11) exactly. A coefficient strictly smaller than one on the right would require additional curvature of the nonconvex rank projection and is neither assumed nor used. $\square$

Auxiliary result (Fixed-rank entropy of the normalized operator secant class). *Under bounded dynamic-design localization and the rank-secant lower eigenvalue in Assumption 5,*

$$\log N\{\epsilon, \mathfrak{S}_{\mathcal{R}}, \|\cdot\|_F\} \leq C(N + T) \log(Cb_{NT}/\epsilon), \quad 0 < \epsilon < 1.$$

Proof. The candidate set is finite and all ranks are fixed. For a single $N \times T$ rank- r matrix, the factorization $\mathbf{M} = \mathbf{A}\mathbf{F}'$ with a rotation-removing normalization gives a covering number bounded by $\exp\{Cr(N+T)\log(CR/\epsilon)\}$ in Frobenius norm on the ball of radius R . Multiplying the component nets over the $M+1$ fixed-rank blocks gives $\log N\{\epsilon, \mathcal{S}_{\mathcal{R}} \cap \{\|\mathbf{\Delta}\|_{\mathbb{H}} \leq R\}, \|\cdot\|_{\mathbb{H}}\} \leq C(N+T)\log(CR/\epsilon)$. On the localization event $\mathcal{A}$ is Cb_{NT} -Lipschitz from product Frobenius norm to outcome Frobenius norm. The secant lower eigenvalue implies any representative with $\|\mathcal{A}(\mathbf{\Delta})\|_F = 1$ has $\|\mathbf{\Delta}\|_{\mathbb{H}} \leq C$. Thus an $\epsilon/(Cb_{NT})$ -net of the bounded product secant slice maps to an ϵ -net of the normalized operator images, giving the stated bound. $\square$

A.3.2 Folds and local information

The following formal statements are placed in the appendix so that the main paper can present the proof chain in five readable blocks: infeasible-benchmark influence representation, rank and first-stage verification, feasible Riesz replacement, feasible CLT, and dependence-robust studentization.

Derived fold operators. On $\mathcal{T}_0$, define

$$\mathcal{G}_j^{\text{tr}} = \alpha_j \mathcal{P}_{\mathcal{T}_0} (\mathcal{A}_{-j}^{\text{buf,pred}})^* \mathcal{A}_{-j}^{\text{buf,pred}} \mathcal{P}_{\mathcal{T}_0}, \quad \mathcal{G}_j^{\text{val}} = p_{j,NT}^{-1} \mathcal{P}_{\mathcal{T}_0} \mathcal{A}^* \Pi_j \mathcal{A} \mathcal{P}_{\mathcal{T}_0}, \quad \bar{\mathcal{G}}_j = \mathbb{E}_0[\mathcal{G}_j^{\text{tr}}]. \quad (\text{A.12})$$

Quadratic variation is target-specific because it depends on the full within-period covariance operator and the target-specific Riesz direction. It is therefore treated directly in Appendix A.4, rather than folded into a uniform information-transport envelope. In particular, no diagonal variance weight or unnormalized direct-sum fluctuation is maintained here.

Let $\mathcal{G}_j^{(-t)}$ be the buffered training information operator after deleting the dependence neighborhood of time t and replacing surviving descendants by the recursive comparison process. Define

$$\varepsilon_{G,NT}^{\text{slice}} = \max_{j,t} \|\mathcal{G}_j^{(-t)} - \mathcal{G}_j^{\text{tr}}\|_{\text{op}(\mathcal{T}_0)}. \quad (\text{A.13})$$

The common deterministic envelope of the training, validation, and deletion discrepancies is $\varepsilon_{G,NT}$.

Local collinearity diagnostic. If the target direction is locally collinear with the nuisance image, the population information map for the Riesz equation is singular and the target is not regular. In that case there exists a nonzero local perturbation whose coefficient-target contrast is nonzero but whose fitted-value image is zero. This is the standard rank/identification failure behind Assumption 5: the Riesz normal equation has no bounded solution and the benchmark variance is not bounded away from zero. The statement follows directly from the adjoint identity in Proposition 3 and the Fredholm alternative for the finite-dimensional tangent-space normal equation.

Lemma 2 (Cross-fitted risk expansion, rank selection, and fast rates). *Under Assumptions 1–5,*

the cross-fitting design in [Section 6.1](#), and the rate regime [\(5.2\)](#),

$$\sup_{\mathbf{r} \in \mathcal{R}} \left| \widehat{L}(\mathbf{r}) - \widehat{\sigma}_{u,NT}^2 - L_0(\mathbf{r}) \right| = o_{p,0}(1), \quad \widehat{\sigma}_{u,NT}^2 = (NT)^{-1} \sum_{i,t} u_{it}^2. \quad (\text{A.14})$$

For $\mathcal{R}_{\geq 0} = \{\mathbf{r} \in \mathcal{R} : \mathbf{r} \geq \mathbf{r}_0\}$,

$$\sup_{\mathbf{r} \in \mathcal{R}_{\geq 0}} \left| \widehat{L}(\mathbf{r}) - \widehat{\sigma}_{u,NT}^2 - L_0(\mathbf{r}) \right| = O_{p,0}(\zeta_{NT}).$$

The two-rate formulation avoids a nonconvex projection-angle margin for underfitted pseudo-true parameters: a uniform law of large numbers suffices for those candidates because their population risk gap is fixed.

Proposition 5 (Rank consistency and first-stage rates). *Under Assumptions [4](#) and [5](#), the cross-fitting design in [Section 6.1](#), and the rate regime [\(5.2\)](#), and Lemma [2](#), $\mathbb{P}_0(\widehat{\mathbf{r}} = \mathbf{r}_0) \rightarrow 1$.*

Auxiliary result (First-stage and rowwise rates). *Under Assumptions [1–5](#), the cross-fitting design in [Section 6.1](#), and the rate regime [\(5.2\)](#), on the truth-rank event and uniformly over folds,*

$$(NT)^{-1} \|\mathcal{A}(\widehat{\Theta}_{-j}^0 - \Theta_0)\|_F^2 = O_{p,0}(\zeta_{NT}), \quad (\text{A.15})$$

$$(NT)^{-1} \|\widehat{\Theta}_{-j}^0 - \Theta_0\|_{\mathbb{H}}^2 = O_{p,0}(\zeta_{NT}), \quad (\text{A.16})$$

$$\Delta_{\mathcal{T},-j} = O_{p,0}(\delta_{NT}), \quad (\text{A.17})$$

and

$$\|\widehat{\Theta}_{-j}^0 - \Theta_0\|_{\max} = O_{p,0}(\delta_{\max,NT}).$$

A.3.3 Rank selection

Auxiliary result (Automatic penalty separation). *Under the rate regime [\(5.2\)](#), $\zeta_{NT} \rightarrow 0$, and the explicit penalty [\(5.7\)](#) satisfies*

$$\frac{\kappa_{NT}}{b_{NT}^2 \ell_{NT}^2} = c_{\kappa} \zeta_{NT}^{-1/2} \rightarrow \infty, \quad a_{NT} \kappa_{NT} = c_{\kappa} \zeta_{NT}^{1/2} \rightarrow 0.$$

Thus the penalty dominates the $O_{p,0}(\zeta_{NT})$ difference between correctly specified and overfitted validation risks after multiplication by the effective-dimension increment, while its total contribution is negligible relative to the constant underfitting gap.

Proof. The last restriction in [\(5.2\)](#) implies

$$b_{NT}^4 \frac{(N+T) \log(NT J_{NT})}{NT} \rightarrow 0,$$

because $J_{NT} + D_{NT} \geq 1$. Since $b_{NT} \geq 1$ eventually,

$$\zeta_{NT} = b_{NT}^2 \ell_{NT}^2 \frac{N+T}{NT} \leq b_{NT}^4 \frac{(N+T) \log(NT J_{NT})}{NT} \rightarrow 0.$$

The two displayed separation relations follow by direct substitution into (5.7). Because fixed rank caps imply $d(\mathbf{r}) - d(\mathbf{r}_0) \asymp N + T$ for every strict overfit and $d(\mathbf{r}) = O(N + T)$ uniformly, the corresponding criterion increments have the claimed orders. $\square$

Auxiliary result (Buffered and validation norm transport). *Under Assumptions 3, 4, 5, and the cross-fitting design in Section 6.1, and the rate regime (5.2), uniformly over folds,*

$$\sup_{\substack{\Delta \in \mathcal{S}_{\mathcal{R}} \\ \|\Delta\|_{\mathbb{H}}=1}} \left| \alpha_j \|\Pi_{-j}^{\text{buf}} \mathcal{A}(\Delta)\|_F^2 - \mathbb{E}_0 \|\mathcal{A}(\Delta)\|_F^2 \right| = O_{p,0}(\varepsilon_{G,NT}), \quad (\text{A.18})$$

$$\sup_{\substack{\Delta \in \mathcal{S}_{\mathcal{R}} \\ \|\Delta\|_{\mathbb{H}}=1}} \left| p_{j,NT}^{-1} \|\Pi_j \mathcal{A}(\Delta)\|_F^2 - \mathbb{E}_0 \|\mathcal{A}(\Delta)\|_F^2 \right| = O_{p,0}(\varepsilon_{G,NT}).$$

Consequently, with probability tending to one, every buffered-training and validation form has lower and upper restricted eigenvalues bounded by fixed positive constants on the differences between candidate low-rank matrices.

Proof. Proposition 4 was stated for the tangent space because that is the geometry used by the Riesz map. Its proof depends only on fixed rank, the feature bound (A.10), and the entropy of the normalized image class. Apply the same argument to the finite union of between-differences between candidate low-rank matrices; Proposition 3 supplies the required entropy bound. The population centers in both displays are identical because the fold labels are auxiliary and independent of the panel. Assumption 5, Weyl's inequality, and $\varepsilon_{G,NT} = o(1)$ then give the sample restricted eigenvalues. No foldwise restricted eigenvalue is inserted into Assumption 5. $\square$

Auxiliary result (Localized finite-moment noise process over fixed-rank secants). *Under Assumptions 1 to 5, the cross-fitting design in Section 6.1, and the rate regime (5.2), uniformly over folds,*

$$\max_{j \leq J_{NT}} \sup_{\substack{\Delta \in \mathcal{S}_{\mathcal{R}} \\ \mathcal{A}(\Delta) \neq 0}} \frac{|\langle \Pi_{-j}^{\text{buf}} U, \Pi_{-j}^{\text{buf}} \mathcal{A}(\Delta) \rangle|}{\|\Pi_{-j}^{\text{buf}} \mathcal{A}(\Delta)\|_F} = O_{p,0}\{b_{NT} \ell_{NT} (\sqrt{T} + \sqrt{N})\}. \quad (\text{A.19})$$

The validation-fold analogue, aggregated with the validation shares, has the same normalized order.

Proof. For coefficient block m , write the masked score matrix as $\Pi_{-j}^{\text{buf}}(Z^{(m)} \odot U)$. Its period-level increments are matrix martingale differences after conditioning on the current own predetermined regressors; their predictable quadratic variation retains the complete within-period covariance matrix. Tropp's rectangular matrix Freedman inequality (Tropp, 2011, Corollary 1.3), applied after

self-adjoint dilation, yield

$$\max_{j \leq J_{NT}} \left\{ \max_{m \leq M} \|\Pi_{-j}^{\text{buf}}(Z^{(m)} \odot U)\|_{\text{op}} + \|\Pi_{-j}^{\text{buf}} U\|_{\text{op}} \right\} = O_{p,0}\{b_{NT}\ell_{NT}(\sqrt{T} + \sqrt{N})\}.$$

The truncation tail is controlled by the conditional q th moments, and the within-period covariance proxy by the Davydov-type covariance inequalities for strong-mixing variables in [Doukhan \(1994, Davydov-type covariance inequality for strong-mixing variables\)](#). Every block of a candidate low-rank difference has fixed rank, so operator–nuclear duality bounds the score by the displayed matrix norm times $\|\Delta\|_{\mathbb{H}}$. Buffered norm transport converts this to the denominator in [\(A.19\)](#). A net of the normalized image class and [Proposition 3](#) give uniformity. For validation folds, work with the average over NT/J_{NT} cells rather than treating $p_{j,NT}^{-1}$ as bounded; the same calculation gives the stated aggregate order. $\square$

Auxiliary result (Candidate-risk consistency and fast rates on classes containing the truth). *Under [Assumptions 1 to 5](#), the cross-fitting design in [Section 6.1](#), and the rate regime [\(5.2\)](#), let $\mathcal{R}_{\geq 0} = \{\mathbf{r} \in \mathcal{R} : \mathbf{r} \geq \mathbf{r}_0\}$. Then*

$$\max_{\mathbf{r} \in \mathcal{R}_{\geq 0}} \max_{j \leq J_{NT}} \frac{1}{NT} \|\mathcal{A}(\hat{\Theta}_{-j}^0(\mathbf{r}) - \Theta_0)\|_F^2 = O_{p,0}(\zeta_{NT}), \quad (\text{A.20})$$

$$\max_{\mathbf{r} \in \mathcal{R}} \max_{j \leq J_{NT}} \{L_0(\hat{\Theta}_{-j}^0(\mathbf{r})) - L_0(\mathbf{r})\} = o_{p,0}(1). \quad (\text{A.21})$$

Moreover,

$$\sup_{\mathbf{r} \in \mathcal{R}} \left| \frac{1}{NT} \sum_j \|\Pi_j \mathcal{A}(\hat{\Theta}_{-j}^0(\mathbf{r}) - \Theta_0)\|_F^2 - L_0(\mathbf{r}) \right| = o_{p,0}(1), \quad (\text{A.22})$$

with $O_{p,0}(\zeta_{NT})$ uniformly over $\mathcal{R}_{\geq 0}$.

Proof. Step 1: a uniform law for all candidate classes. For $\Theta \in \mathcal{M}_B(\mathbf{r})$, define the population criterion $R(\Theta) = (NT)^{-1} E_0 \|\mathcal{A}(\Theta - \Theta_0)\|_F^2$ and the normalized buffered criterion $R_j^{\text{buf}}(\Theta)$. The compact entry envelope and fixed ranks imply

$$\log N\{\epsilon, \cup_{\mathbf{r} \in \mathcal{R}} \mathcal{M}_B(\mathbf{r}), \|\cdot\|_{\mathbb{H}}\} \leq C(N+T) \log(C/\epsilon).$$

Truncate the squared fitted residuals at the finite-moment level justified by the truncation and uniform-integrability argument in [van der Vaart \(1998, Sec. 2.4, truncation and uniform-integrability argument\)](#). The bounded part is uniformly equicontinuous; fold transport controls the design quadratic form, and the score process in [Lemma 2](#) controls the linear term. The tail part is uniformly integrable because $q > 8$. Hence

$$\max_{j, \mathbf{r}} \sup_{\Theta \in \mathcal{M}_B(\mathbf{r})} |R_j^{\text{buf}}(\Theta) - R(\Theta)| = o_{p,0}(1).$$

The objective gap in the implementation requirement [\(6.5\)](#) is $o_{p,0}(NT\zeta_{NT}) = o_{p,0}(NT)$, so the

argmin inequality gives (A.21). For the ideal minimizer the gap is zero.

Step 2: the fast rate when the class contains the truth. If $\mathbf{r} \in \mathcal{R}_{\geq 0}$, Θ_0 is feasible. Put $\hat{\Delta}_{j,\mathbf{r}} = \hat{\Theta}_{-j}^0(\mathbf{r}) - \Theta_0$. The basic inequality has no approximation-residual term:

$$\frac{1}{2} \|\Pi_{-j}^{\text{buf}} \mathcal{A} \hat{\Delta}_{j,\mathbf{r}}\|_F^2 \leq \langle \Pi_{-j}^{\text{buf}} U, \Pi_{-j}^{\text{buf}} \mathcal{A} \hat{\Delta}_{j,\mathbf{r}} \rangle + o_{p,0}(NT\zeta_{NT}).$$

Lemma 2 and a quadratic inequality give a buffered prediction error $O_{p,0}\{b_{NT}^2 \ell_{NT}^2(N+T)\} = O_{p,0}(NT\zeta_{NT})$. Lemma 1 transports it to the population/full-sample norm, proving (A.20).

Step 3: validation risk. Apply the validation part of Lemma 1 to the random candidate chord after conditioning on its buffered training sigma-field and then use a finite net. Together with (A.21), this yields (A.22). On $\mathcal{R}_{\geq 0}$, the fast prediction bound from Step 2 and the localized score bound sharpen the remainder to $O_{p,0}(\zeta_{NT})$. This argument deliberately does not seek a fast rate around an underfitted nonconvex projection. $\square$

Auxiliary result (Aggregate validation-score bound). *Under the hypotheses of Lemma 2,*

$$\sup_{\mathbf{r} \in \mathcal{R}} \left| \frac{2}{NT} \sum_j \langle \Pi_j \{\mathcal{A}(\Theta_0) - \mathcal{A}(\hat{\Theta}_{-j}^0(\mathbf{r}))\}, \Pi_j U \rangle \right| = o_{p,0}(1), \quad (\text{A.23})$$

and the left side is $O_{p,0}(\zeta_{NT})$ uniformly over $\mathbf{r} \in \mathcal{R}_{\geq 0}$.

Proof. For a fixed candidate, add and subtract a population minimizer $\Theta_{\mathbf{r}}^*$. The approximation residual $\mathcal{A}(\Theta_0 - \Theta_{\mathbf{r}}^*)$ is cellwise measurable with respect to $\sigma(x_{it}, \mathcal{F}_{t-1})$, because $\Theta_{\mathbf{r}}^*$ is $\mathcal{G}_0$ -measurable and the design operator is cellwise. Weak exogeneity therefore centers its period-level score. Covariance summability in Assumption 3 and the finite second moment of the residual give

$$(NT)^{-1} \sum_j \langle \Pi_j \mathcal{A}(\Theta_0 - \Theta_{\mathbf{r}}^*), \Pi_j U \rangle = O_{p,0}\{(NT)^{-1/2}\}.$$

For the estimation chord, use the validation secant process and (A.21); its normalized contribution is $o_{p,0}(1)$. If $\mathbf{r} \in \mathcal{R}_{\geq 0}$, take $\Theta_{\mathbf{r}}^* = \Theta_0$ and combine Lemma 2 with (A.20); the product divided by NT is $O_{p,0}(\zeta_{NT})$. The calculation is performed for the aggregate validation score, so overlap among training complements creates no fictitious independence or extra J_{NT} factor. $\square$

Proof of Lemma 2. Expand the out-of-fold loss as

$$\begin{aligned} \hat{L}(\mathbf{r}) &= \frac{1}{NT} \sum_j \|\Pi_j \mathcal{A}(\Theta_0 - \hat{\Theta}_{-j}^0(\mathbf{r}))\|_F^2 + \hat{\sigma}_{u,NT}^2 \\ &\quad + \frac{2}{NT} \sum_j \langle \Pi_j \mathcal{A}(\Theta_0 - \hat{\Theta}_{-j}^0(\mathbf{r})), \Pi_j U \rangle. \end{aligned}$$

Lemma 2 gives the stated uniform approximation for both the quadratic prediction component and the cross term. In particular, the first term is $L_0(\mathbf{r}) + o_{p,0}(1)$ uniformly, and $L_0(\mathbf{r}) + O_{p,0}(\zeta_{NT})$

on $\mathcal{R}_{\geq 0}$; the cross term has the corresponding two rates. The error-square term is common to all candidates and is left realized; no unnecessary law of large numbers for it is used. This proves both assertions of Lemma 2. $\square$

Proof of Proposition 5. Let $\widehat{C}(\mathbf{r}) = \widehat{L}(\mathbf{r}) + \kappa_{NT}d(\mathbf{r})/(NT)$. If $\mathbf{r} \not\geq \mathbf{r}_0$, the Eckart–Young theorem and the strong singular-value condition imply that the candidate class omits at least one non-negligible true direction; the restricted-eigenvalue condition then gives a fixed population risk gap, $L_0(\mathbf{r}) - L_0(\mathbf{r}_0) \geq c_{\text{gap}} > 0$. Lemma 2 contributes $o_{p,0}(1)$, and the penalty difference is $O\{\kappa_{NT}(N+T)/(NT)\} = o(1)$. Thus every underfitted candidate has a larger criterion with probability tending to one.

If $\mathbf{r} \geq \mathbf{r}_0$ and $\mathbf{r} \neq \mathbf{r}_0$, both population risks are zero. The fixed rank box implies $d(\mathbf{r}) - d(\mathbf{r}_0) \geq c_d(N+T)$, while the risk fluctuation is $O_{p,0}(\zeta_{NT})$. The penalty difference is at least $c_d\kappa_{NT}a_{NT}$, and

$$\frac{\kappa_{NT}a_{NT}}{\zeta_{NT}} = c_\kappa\zeta_{NT}^{-1/2} \rightarrow \infty$$

by Lemma 2, so it dominates the $O_{p,0}(\zeta_{NT})$ risk fluctuation. Finiteness of $\mathcal{R}$ completes the proof. $\square$

A.3.4 First-stage recovery

The next proposition collects the paper-specific first-stage consequences used in the five theorem proofs. Its proof relies on Tropp’s self-adjoint and rectangular matrix Freedman inequalities (Tropp, 2011, Theorem 1.2 and Corollary 1.3) and on the Davis–Kahan and Wedin singular-space perturbation bounds (Davis and Kahan, 1970, sin–theta theorem), (Wedin, 1972, Wedin singular-vector perturbation theorem), and (Yu et al., 2015, Theorems 1, 2, and 4); the nonstandard part is the combination with the scattered buffered folds, dynamic generated regressors, and the objective-gap condition for the computed nonconvex iterate.

Proposition 6 (Local optimization, row/column concentration, and rowwise recovery). *Under Assumptions 1–6, the fold design in Section 6.1, and the rate regime (5.2), the following statements hold uniformly over folds on an event with probability tending to one.*

- (i) *The row and column empirical Gram matrices in the balanced low-rank chart satisfy the concentration bounds*

$$\left\| N^{-1} \sum_i \{ \mathbf{d}_{0,it} \mathbf{d}'_{0,it} - E_0(\mathbf{d}_{0,it} \mathbf{d}'_{0,it} \mid \mathcal{F}_{t-1}) \} \right\|_{\text{op}} = o_{p,0}(1),$$

$$\left\| T^{-1} \sum_t \{ \mathbf{c}_{0,it} \mathbf{c}'_{0,it} - E_0(\mathbf{c}_{0,it} \mathbf{c}'_{0,it}) \} \right\|_{\text{op}} = o_{p,0}(1),$$

and hence the localized row and column Hessians have eigenvalues bounded above and away from zero.

- (ii) A warm start inside the local basin and satisfying the block-dominance condition of Assumption 5 is contracted by the alternating-ridge map. After $L \asymp \kappa_{\text{cond}} \log(NT)$ iterations the objective gap is $o_{p,0}(NT\zeta_{NT})$. This local result is a computational sufficient condition only; the statistical theory is stated for the ideal global minimizer or for an iterate satisfying the explicit objective-gap condition (6.5).
- (iii) The truth-rank minimizer satisfies the global prediction and Hilbert-norm rates, and its aligned factors satisfy the row, column, entrywise, and target projection perturbation bounds used in Theorem 2. In particular,

$$(NT)^{-1} \|\mathcal{A}(\hat{\Theta}_{-j}^0 - \Theta_0)\|_F^2 + (NT)^{-1} \|\hat{\Theta}_{-j}^0 - \Theta_0\|_{\mathbb{H}}^2 = O_{p,0}(\zeta_{NT}),$$

$$\|\hat{\mathcal{P}}_{\mathcal{T},-j} - \mathcal{P}_{\mathcal{T}_0}\|_{\text{op}} = O_{p,0}(\delta_{NT}), \quad \text{aligned row/column/entrywise errors} = O_{p,0}(\delta_{\max,NT}).$$

Proof. Part (i) applies finite-moment truncation (van der Vaart, 1998, Sec. 2.4, truncation and uniform-integrability argument) and matrix martingale concentration (Tropp, 2011, Theorem 1.2) to the fixed-dimensional row and column Gram arrays. The fold-balance and retention bounds in Lemma 1 ensure that deleting the validation fold and its buffer changes the row and column averages by only the retained-sample rate $B_{NT}/J_{NT} + o_{p,0}(1)$. Assumption 5 supplies the population lower and upper information bounds; Weyl’s inequality transports them to the sample Hessians.

For (ii), the localized objective has Hessian blocks with spectra bounded above and away from zero by part (i). The block-dominance condition makes the linearization of the row–column Gauss–Seidel map a contraction. The usual Newton/alternating-minimization contraction argument gives geometric decay of parameter error in the basin; restricted smoothness then converts this into the stated objective-gap order. This argument establishes only a sufficient local computational certificate. It does not replace the global-minimizer or objective-gap requirement used by the statistical theorems.

For (iii), the first-stage basic inequality follows from the rank-constrained least-squares optimality condition, the restricted-curvature part of Assumption 5, and the cross-fitted risk expansion in Lemma 2. The strong singular-value gap and incoherence in Assumption 4 convert prediction error into coefficient-matrix error. Davis–Kahan and Wedin perturbation bounds (Davis and Kahan, 1970, sin–theta theorem), (Wedin, 1972, Wedin singular-vector perturbation theorem), and (Yu et al., 2015, Theorems 2 and 4) then give the tangent-space and singular-vector perturbation rates. The rowwise and entrywise bounds follow from the same localized Hessian inversion applied to one row or one column at a time, with the leave-one-out score terms controlled by Davydov-type covariance and blocking inequalities for strong-mixing arrays (Doukhan, 1994, covariance inequality for mixing variables). Finally, applying the projection perturbation bound to the finite-dimensional target directions gives $\|\hat{\mathcal{P}}_{\mathcal{T},-j} \mathbf{D}_\nu - \mathcal{P}_{\mathcal{T}_0} \mathbf{D}_\nu\|_{\mathbb{H}} = O_{p,0}(\delta_{\max,NT} \|\mathcal{P}_{\mathcal{T}_0} \mathbf{D}_\nu\|_{\mathbb{H}})$. $\square$

Underfitting gap and balanced incoherent designs. The rank selector uses two elementary consequences of Assumptions 4 and 5. First, if a candidate rank omits a true singular direction,

then the Eckart–Young theorem and the strong singular-value condition imply a coefficient distance of order NT ; the restricted-eigenvalue inequality turns this into a strictly positive fitted-risk gap. Overfitted classes contain the truth and therefore have zero population approximation risk. Second, in the balanced incoherent design class with bounded off-block interaction, the row and column Hessian bounds above and a Neumann-series argument verify the local block-dominance condition. These facts are routine linear-algebra consequences of the displayed assumptions and are used only as sufficient checks, not as additional maintained assumptions.

A.3.5 Geometry and Riesz weights

This appendix develops in full the operator notation, the target directions, the local geometry, and the fold-based debiasing-weight construction summarized in Section 3.4.

Matrix form: regressor matrices and the design map The general analysis is easiest in matrix form. Let $M = P + K$ count the coefficient matrices and order them lags first: for $m = 1, \dots, P$ the m th *regressor matrix* and coefficient matrix are

$$\mathbf{Z}^{(m)} = (y_{i,t-m})_{t \leq T, i \leq N}, \quad \mathbf{\Gamma}_0^{(m)} = \mathbf{A}_0^{(m)},$$

and for $m = P + k$ with $k \leq K$,

$$\mathbf{Z}^{(P+k)} = (x_{it,k})_{i \leq N, t \leq T}, \quad \mathbf{\Gamma}_0^{(P+k)} = \mathbf{B}_0^{(k)}.$$

All matrices are $N \times T$, with rows indexed by units $i = 1, \dots, N$ and columns indexed by periods $t = 1, \dots, T$. The full parameter is the list

$$\boldsymbol{\Theta} = (\mathbf{\Gamma}^{(1)}, \dots, \mathbf{\Gamma}^{(M)}, \mathbf{H}) \in \mathbb{H}_{NT} := (\mathbb{R}^{N \times T})^{M+1}, \quad (\text{A.24})$$

a point in the product space $\mathbb{H}_{NT}$ of $M + 1$ matrices, equipped with the natural inner product

$$\langle \boldsymbol{\Theta}_1, \boldsymbol{\Theta}_2 \rangle_{\mathbb{H}} = \sum_{m=1}^M \langle \mathbf{\Gamma}_1^{(m)}, \mathbf{\Gamma}_2^{(m)} \rangle + \langle \mathbf{H}_1, \mathbf{H}_2 \rangle, \quad (\text{A.25})$$

where $\langle \cdot, \cdot \rangle$ is the entrywise (trace) inner product of matrices. Writing $\odot$ for the entrywise (Hadamard) product, the model (3.1) over the effective sample is exactly

$$\mathbf{Y} = \mathcal{A}(\boldsymbol{\Theta}_0) + \mathbf{U}, \quad \mathcal{A}(\boldsymbol{\Theta}) := \sum_{m=1}^M \mathbf{Z}^{(m)} \odot \mathbf{\Gamma}^{(m)} + \mathbf{H}, \quad (\text{A.26})$$

with $\mathbf{Y} = (y_{it})$ and $\mathbf{U} = (u_{it})$. We call $\mathcal{A}$ the *design map*: it plays the role of the regressor matrix $\mathbf{X}$ in linear regression, sending a coefficient configuration to the implied systematic part of the

outcome,

$$\mathcal{A} : \mathbb{H}_{NT} \rightarrow \mathbb{R}^{N \times T}. \quad (\text{A.27})$$

Its adjoint—the analogue of $\mathbf{X}'$ —multiplies a residual matrix back into each coefficient block,

$$\mathcal{A}^*(\mathbf{R}) = (\mathbf{Z}^{(1)} \odot \mathbf{R}, \dots, \mathbf{Z}^{(M)} \odot \mathbf{R}, \mathbf{R}). \quad (\text{A.28})$$

Keeping the equation in the form (A.26) is a deliberate choice: the generated lagged outcomes sit inside the *design*, and the error remains the idiosyncratic error $\mathbf{U}$. The alternative—reducing the model to an auxiliary equation whose error is a product of regressors and idiosyncratic errors—destroys the martingale structure that our inference exploits.

Each block is low rank,

$$\text{rank}(\mathbf{\Gamma}_0^{(m)}) = r_{0,m}, \quad \text{rank}(\mathbf{H}_0) = r_{0,H},$$

with rank vector $\mathbf{r}_0 = (r_{0,1}, \dots, r_{0,M}, r_{0,H})$, and we write the compact singular-value decompositions $\mathbf{\Gamma}_0^{(m)} = \mathbf{U}_{0,m} \mathbf{\Sigma}_{0,m} \mathbf{V}_{0,m}'$ and $\mathbf{H}_0 = \mathbf{U}_{0,H} \mathbf{\Sigma}_{0,H} \mathbf{V}_{0,H}'$. The left singular vectors encode latent unit exposures, and the right singular vectors encode latent time paths of each coefficient matrix.

Targets as linear functionals Every target of Section 3.1 is a linear functional of $\mathbf{\Theta}_0$, and it is convenient to encode each one by a *direction* in the parameter space. For a matrix $\mathbf{M} \in \mathbb{R}^{N \times T}$, let $\mathcal{E}_m(\mathbf{M})$ denote the point of $\mathbb{H}_{NT}$ with $\mathbf{M}$ in the m th coefficient block and zeros elsewhere, and $\mathcal{E}_H(\mathbf{M})$ the point with $\mathbf{M}$ in the nuisance block. Writing $\mathbf{e}_t$ and $\mathbf{e}_i$ for coordinate vectors, the entry targets correspond to the directions

$$\mathbf{D}_{it,A,\ell} = \mathcal{E}_\ell(\mathbf{e}_i \mathbf{e}_t'), \quad \mathbf{D}_{it,B,k} = \mathcal{E}_{P+k}(\mathbf{e}_i \mathbf{e}_t').$$

For a group G_N , define the group weight vector

$$\boldsymbol{\pi}_G = g_N^{-1} (\mathbf{1}\{i \in G_N\})_{i=1}^N, \quad (\text{A.29})$$

so that group means have directions $\mathbf{D}_{G,t,A,\ell} = \mathcal{E}_\ell(\boldsymbol{\pi}_G \mathbf{e}_t')$ and $\mathbf{D}_{G,t,B,k} = \mathcal{E}_{P+k}(\boldsymbol{\pi}_G \mathbf{e}_t')$; full means use $\boldsymbol{\pi}_N = N^{-1} \mathbf{1}_N$ and contrasts use $\boldsymbol{\pi}_\Delta = \boldsymbol{\pi}_{G_1} - \boldsymbol{\pi}_{G_2}$. We index a generic target by $\nu \in \mathcal{V}_{NT}$ and write

$$\phi_\nu(\mathbf{\Theta}_0) = \langle \mathbf{D}_\nu, \mathbf{\Theta}_0 \rangle_{\mathbb{H}}. \quad (\text{A.30})$$

Cumulative persistence is the direction obtained by summing the lag-block directions over ℓ , so it is one more member of the same family.

Local geometry: admissible perturbations and the information map Because the parameter is constrained to low rank, the relevant notion of a “small change” in the parameter is a change that stays within the low-rank model to first order. For a rank- r matrix $\mathbf{M} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}'$, these

admissible local perturbations form the linear space

$$\mathcal{T}(\mathbf{M}) = \{\mathbf{U}\mathbf{A}' + \mathbf{C}\mathbf{V}' : \mathbf{A} \in \mathbb{R}^{T \times r}, \mathbf{C} \in \mathbb{R}^{N \times r}\},$$

the set of matrices that perturb the existing unit exposure space or the existing time-factor space; in differential-geometric language this is the tangent space of the rank- r set at $\mathbf{M}$, and we adopt that name. Its orthogonal projection has the closed form

$$\mathcal{P}_{\mathcal{T}(\mathbf{M})}(\mathbf{A}) = \mathbf{P}_U \mathbf{A} + \mathbf{A} \mathbf{P}_V - \mathbf{P}_U \mathbf{A} \mathbf{P}_V, \quad \mathbf{P}_U = \mathbf{U}\mathbf{U}', \quad \mathbf{P}_V = \mathbf{V}\mathbf{V}',$$

and the admissible perturbations of the full parameter form the product $\mathcal{T}_0 = \mathcal{T}_{\Gamma,1} \times \cdots \times \mathcal{T}_{\Gamma,M} \times \mathcal{T}_H$ of the blockwise tangent spaces at the truth.

The local least-squares geometry is then summarized by one object, the *local information map*

$$\mathcal{G}_0 = \mathcal{P}_{\mathcal{T}_0} \mathcal{A}^* \mathcal{A} \mathcal{P}_{\mathcal{T}_0}, \tag{A.31}$$

the exact analogue of the design Gram matrix $\mathbf{X}'\mathbf{X}$ in linear regression, restricted to admissible perturbations: for $\Delta \in \mathcal{T}_0$, $\langle \Delta, \mathcal{G}_0 \Delta \rangle_{\mathbb{H}} = \|\mathcal{A}(\Delta)\|_F^2$ is the sum of squared fitted-value changes induced by the perturbation. As in regression, nonsingularity of $\mathcal{G}_0$ on $\mathcal{T}_0$ is the local identification condition, and the inverse of $\mathcal{G}_0$ prices how much information the sample carries about each direction.

Debiasing weights, cross-fitting, and the space–time exclusion buffer For a target direction $\mathbf{D}_\nu$, define its tangent projection and the associated full-sample design quantities

$$\mathbf{W}_\nu = \mathcal{P}_{\mathcal{T}_0} \mathbf{D}_\nu, \quad \mathbf{q}_\nu = \mathcal{G}_0^{-1} \mathbf{W}_\nu, \quad \Psi_\nu = \mathcal{A} \mathbf{q}_\nu.$$

The matrix Ψ_ν collects the *debiasing weights* for target ν : the observation-space weights whose inner product with any admissible fitted-value change reproduces the induced change in the target, exactly as in display (3.4) of the walkthrough. (They are the Riesz representer of ϕ_ν in the local least-squares geometry; we use the descriptive name.) In a static model one could work with Ψ_ν directly after conditioning on the design. In the dynamic model the full-sample design contains *future* outcomes of every unit, so a weight built from it need not be unpredictable for the idiosyncratic error it multiplies. The construction must therefore be subsample-specific.

Scattered cross-fitting folds. Let $\sigma : \mathcal{I}_{NT} \rightarrow \{1, \dots, J_{NT}\}$ be generated by the auxiliary randomization $\mathcal{R}_{\text{fold}}$, independently of the panel. The theory uses the i.i.d. uniform labels in the cross-fitting design in Section 6.1; stratified random permutations may be used in computation to improve finite-sample balance, but they are not used in the asymptotic concentration proof. The held-out (validation) set is $\mathcal{I}_j = \sigma^{-1}(j)$, and $|\sigma^{-1}(j)|/(NT) \asymp J_{NT}^{-1}$ uniformly with probability tending to one. The theory takes $J_{NT} \rightarrow \infty$ under the rate regime (5.2); a fixed moderate J is only a finite-sample approximation (Remark 2). Write $p_{j,NT} := |\mathcal{I}_j|/(NT)$ for fold j 's realized cell share.

The space–time exclusion buffer. For fold j , use exactly the training set in the cross-fitting design in [Section 6.1](#):

$$\mathcal{I}_{-j}^{\text{buf}} = \left\{ (\ell, s) \in \mathcal{I}_{NT} : (\ell, s) \notin \mathcal{I}_j, \nexists (i, t) \in \mathcal{I}_j \text{ with } 0 \leq s - t \leq q_T \text{ and } d_N(\ell, i) \leq r_N \right\}. \quad (\text{A.32})$$

Thus one definition is used by the first stage, rank selector, information operators, Riesz weights, and variance estimator. Let $\gamma_{j,NT} = |\mathcal{I}_{-j}^{\text{buf}}|/(NT)$, $n_{-j}^{\text{buf}} = |\mathcal{I}_{-j}^{\text{buf}}|$, and $\alpha_j = NT/n_{-j}^{\text{buf}}$.

Gram normalization convention. Every Gram in the paper is read on a single *per-cell-average* scale. For unit- $\mathbb{H}$ -norm $\Delta, \Xi \in \mathcal{T}_0$, write $X_a(\Delta) := \sqrt{TN}[\mathcal{A}\Delta]_a$ for the normalized design product, which is of order one per cell. The full-sample average is exactly the raw Frobenius inner product, $(TN)^{-1} \sum_{a \in \mathcal{I}_{NT}} X_a(\Delta) X_a(\Xi) = \langle \mathcal{A}\Delta, \mathcal{A}\Xi \rangle$, and the factors $\alpha_j = TN/n_{-j}^{\text{buf}}$ and $p_{j,NT}^{-1} = TN/|\mathcal{I}_j|$ are precisely what convert the purged-training and held-out-fold raw Grams into the same per-cell average: $\alpha_j \langle \mathcal{A}_{-j}^{\text{buf,pred}} \Delta, \mathcal{A}_{-j}^{\text{buf,pred}} \Xi \rangle = (n_{-j}^{\text{buf}})^{-1} \sum_{a \in \mathcal{I}_{-j}^{\text{buf}}} X_a(\Delta) X_a(\Xi)$ and $p_{j,NT}^{-1} \langle \Pi_j \mathcal{A}\Delta, \Pi_j \mathcal{A}\Xi \rangle = |\mathcal{I}_j|^{-1} \sum_{a \in \mathcal{I}_j} X_a(\Delta) X_a(\Xi)$ are both order-one averages comparable to $\langle \mathcal{A}\Delta, \mathcal{A}\Xi \rangle$. Thus the held-out fold, the buffered-training set, and the full sample are normalized identically. Proposition 4 derives their operator transport directly from the auxiliary randomization and the bounded buffer volume.

Scattering rather than held-out time blocks is essential: with scattered cells every period and unit retain training observations, so the latent time paths are estimable on every row, whereas a contiguous held-out time block has no training observations at its own time periods and makes the training information map singular by construction. Positive definiteness of the buffered map is then a statistical matter, supplied by Assumption 5.

Two opposing forces govern the window length. Stability makes the influence of u_{it} on $y_{i,t+h}$ decay like ρ_*^h for a modulus $\rho_* < 1$, so controlling the temporal leakage at the inferential scale requires $\rho_*^{q_T} \sqrt{TN} \rightarrow 0$, pushing q_T up to the order $\log(TN)/|\log \rho_*|$. The spatial radius separately has to make $\{\Omega_x(r_N) + \bar{\alpha}(r_N)\} \sqrt{NT} \rightarrow 0$. Training retention pushes the other way: under a balanced assignment the retained share is bounded below by $1 - CJ_{NT}^{-1}(q_T + 1)v_N(r_N)$ up to a fold-balance remainder, which tends to one when $(q_T + 1)v_N(r_N)/J_{NT} \rightarrow 0$. The data-driven rule of [Section 5.2](#) balances the two, estimating ρ_* from a preliminary fit, and the simulations of [Section 8](#) display the predicted interior optimum.

Every cell is held out once, but the resulting fold-specific score is covered by the limit theory only when the retained training geometry and the leakage remainder are uniformly stable. This is why the asymptotic analysis requires $(q_T + 1)v_N(r_N)/J_{NT} \rightarrow 0$. A fixed moderate J is a finite-sample approximation, not a theorem-covered replacement, and is shown together with retention, transport, and window-sensitivity diagnostics ([Remark 2](#), [Appendix A.2.7](#)).

Subsample-specific weights: the infeasible benchmark. For dynamic inference the natural benchmark weights are defined fold by fold, from information that excludes the primitive blocks attached to the held-out cells. Because predetermined covariates may themselves react to past idiosyncratic errors, and because their own unit-specific innovations may be contemporaneously

linked to those errors, purifying lagged outcomes alone is not enough. We instead use a fold-coupled recursive process. Let Π_j denote the projection that keeps fold- j cells. For each validation cell $(i, t) \in \mathcal{I}_j$, replace $\zeta_{it} = ((\varepsilon_{it}^{x,\text{id}})', u_{it})'$ by an independent copy ζ_{it}^* under the conditional law given $\mathcal{G}_0$, while leaving aggregate and common covariate innovation blocks fixed; outside $\mathcal{I}_j$ set $\zeta_{it}^{(-j)} = \zeta_{it}$. Write $u_{it}^{(-j)}$ and $\varepsilon_{it}^{x,\text{id},(-j)}$ for the corresponding components, and form $\varepsilon_t^{x,(-j)}$ by replacing only the unit-specific blocks in ε_t^x . Generate $\tilde{x}_{it,k}^{(-j)}$ from the covariate recursive map in Assumption 1 with the innovation sequence $\{\varepsilon_s^{x,(-j)}, \mathbf{u}_s^{(-j)}\}$, and generate $\tilde{y}_{it}^{(-j)}$ recursively from the original panel equation using those covariates and the coupled idiosyncratic errors. Thus every fold- j unit-specific primitive block is removed from the training design channels through which it could enter a future design value. The resulting paths are measurable with respect to $\mathcal{G}_0 \vee \sigma(\zeta_b : b \notin \mathcal{I}_j) \vee \sigma(\zeta_b^* : b \in \mathcal{I}_j)$. Stability and the single-input space-time coupling bound imply, uniformly over folds,

$$\max_{(s,i) \in \mathcal{I}_{-j}^{\text{buf}}} \left\{ \|y_{is} - \tilde{y}_{is}^{(-j)}\|_{p,0} + \max_k \|x_{is,k} - \tilde{x}_{is,k}^{(-j)}\|_{p,0} \right\} \leq C_p \{\rho_*^{q_T} + \Omega_x(r_N)\}, \quad 2 \leq p \leq q. \quad (\text{A.33})$$

Indeed, every deleted input that can affect a surviving buffered training cell is either more than q_T periods old or lies outside the radius- r_N spatial buffer. A telescoping replacement of the surviving temporal and spatial tails is therefore bounded by the geometric decay and the spatial-propagation envelope. The corresponding maximum over the realized training grid is $O_{p,0}(b_{NT}\{\rho_*^{q_T} + \Omega_x(r_N)\})$. Let $\mathcal{A}_{-j}^{\text{buf,pred}}$ denote the buffered training design built from both $\tilde{y}^{(-j)}$ and $\tilde{x}^{(-j)}$. This counterfactual replacement is an analytical device used only to define the benchmark, never computed; Proposition 7 controls its gap from the feasible design. The benchmark direction and weights for target ν and fold j are

$$\mathbf{q}_{\nu,j}^{\text{or}} = \left(\alpha_j \mathcal{P}_{\mathcal{T}_0} (\mathcal{A}_{-j}^{\text{buf,pred}})^* \mathcal{A}_{-j}^{\text{buf,pred}} \mathcal{P}_{\mathcal{T}_0} \right)^{-1} \mathcal{P}_{\mathcal{T}_0} \mathbf{D}_\nu, \quad \Psi_{\nu,j}^{\text{or}} = \mathcal{A} \mathbf{q}_{\nu,j}^{\text{or}}, \quad (\text{A.34})$$

where the inverse is the genuine inverse of the training information map on $\mathcal{T}_0$ —nonsingular under the restricted-eigenvalue condition of Assumption 5, so the defining property holds exactly and any numerical truncation used in computation is inactive regularization. The direction is indexed by the held-out fold j because the training complement depends on j ; the weights are then evaluated on $\mathcal{I}_j$. For each cell a , let $j(a)$ be its fold and define the effective benchmark weight

$$\Psi_{\nu,a}^{\text{or}} = [\Psi_{\nu,j(a)}^{\text{or}}]_a. \quad (\text{A.35})$$

The defining (debiasing) property holds on the training geometry: for every admissible perturbation $\Delta \in \mathcal{T}_0$,

$$\langle \mathbf{D}_\nu, \Delta \rangle_{\mathbb{H}} = \alpha_j \left\langle \mathcal{A}_{-j}^{\text{buf,pred}} \mathbf{q}_{\nu,j}^{\text{or}}, \mathcal{A}_{-j}^{\text{buf,pred}} \Delta \right\rangle. \quad (\text{A.36})$$

The one-step estimator, however, evaluates residuals on the *held-out* fold. The proof therefore establishes *fold-information transport*: the validation-fold operator (scaled by $p_{j,NT}^{-1}$) and the buffered-training operator (scaled by α_j) are asymptotically equivalent on the relevant perturbation spaces. Proposition 4 derives this property from auxiliary randomization, leverage, and the buffer-volume

restrictions in (5.2). Finally, the construction achieves its purpose through comparison rather than by declaring the entire empirical weight field predictable. The population-information weight is coupled to a pre-realization direction, and the remaining same-period dependence is controlled by the spatial buffer and Assumption 3. The fold-specific benchmark and feasible weights may depend on other observations from the same period; Lemma 3 removes that dependence with a leave-one-slice coupling, and Proposition 7 controls the surviving post-window descendants. Thus the proof uses martingale orthogonality across time periods, the spatial buffer, and the weak-mixing bound in Assumption 3; it does not invoke conditional independence of held-out cells.

A.3.6 Feasible Riesz replacement

Proposition 7 (Cross-fitted nuisance products, Riesz replacement, and dynamic leakage). *Under Assumptions 1–6, the tangent-normal rank-manifold decomposition, rowwise recovery, and fold transport imply*

$$\sum_j p_{j,NT} \frac{|\mathbf{r}_{\nu,j}^{\text{or}}(\hat{\Theta}_{-j}^0 - \Theta_0)|}{s_{\nu,NT}} + \sum_j \frac{|\langle \Pi_j(\hat{\Psi}_{\nu,-j}^0 - \Psi_{\nu,j}^{\text{or}}), \Pi_j \mathcal{A}(\hat{\Theta}_{-j}^0 - \Theta_0) \rangle|}{s_{\nu,NT}} = O_{p,0}(r_{\nu,NT}) = o_{p,0}(1). \quad (\text{A.37})$$

Here $r_{\nu,NT}$ is defined in (7.1).

Proof. Project the first-stage error into the true tangent space and its quadratic normal remainder. The tangent term is bounded by the fold-operator discrepancy of Proposition 4; the normal term is bounded by the rowwise singular-space expansion in Proposition 6 and incoherence. Replacing the empirical buffered weights by predictable weights contributes the temporal and spatial coupling terms of Lemma 3. After division by the target standard deviation, the four contributions are respectively the four terms in (7.1). $\square$

Auxiliary result (Feasible Riesz-weight replacement). *Under the conditions of Proposition 5, rank consistency, and Assumption 6,*

$$\left\{ \sum_j \|\Pi_j(\hat{\Psi}_{\nu,-j} - \Psi_{\nu,j}^{\text{or}})\|_F^2 \right\}^{1/2} = o_{p,0}(\|\Psi_{\nu}^{\text{or}}\|_F), \quad (\text{A.38})$$

$$\sum_j p_{j,NT} \left| \hat{\mathbf{r}}_{\nu,-j}(\hat{\Theta}_{-j}^0 - \Theta_0) \right| = o_{p,0}(s_{\nu,NT}), \quad (\text{A.39})$$

The replacement of the full spatial-kernel covariance form is established separately in Proposition 9.

A.3.7 Dynamic leakage control

Buffered local-replacement bound. Let $a = (i, t)$ be a validation cell and let $b = (\ell, s)$ be a candidate training cell. Construct the coupled design $Z_b^{[a]}$ by replacing the unit-specific primitive block $\zeta_{it} = ((\varepsilon_{it}^{x,\text{id}})', u_{it})'$ by an independent copy under the conditional law given $\mathcal{G}_0$, while holding

aggregate and common covariate innovation blocks fixed. Under [Assumptions 1 to 3](#) and the cross-fitting design in [Section 6.1](#), for every $2 \leq p \leq q$,

$$\|Z_b - Z_b^{[a]}\|_{p,0} \leq C\rho_*^{s-t}\omega_x\{d_N(\ell, i)\}\mathbf{1}\{s > t\} + C\omega_x\{d_N(\ell, i)\}\mathbf{1}\{s = t\},$$

with the convention that the right side is zero for $s < t$. Consequently, for every normalized finite-rank feature vector entering a buffered training information operator,

$$\left\| \frac{1}{NT} \sum_{b \in \mathcal{I}_{-j}^{\text{buf}}} \{Z_b Z_b' - Z_b^{[a]} (Z_b^{[a]})'\} \right\|_{\text{op}} = O_{p,0}(\rho_*^{qT} + \Omega_x(r_N)) + o_{p,0}(1), \quad (\text{A.40})$$

and the corresponding normalized target right-hand-side perturbation satisfies

$$\left\| \frac{1}{NT} \sum_{b \in \mathcal{I}_{-j}^{\text{buf}}} \{Z_b - Z_b^{[a]}\} \ell_b \right\|_{\mathbb{H}} = O_{p,0}(\rho_*^{qT} + \Omega_x(r_N)) + o_{p,0}(1), \quad (\text{A.41})$$

for every bounded target-loading array $\{\ell_b\}$ used in the Riesz right-hand side. After adding the remaining contemporaneous statistical dependence between spatially separated current-period terms, the full normalized fold perturbation is

$$O_{p,0}\{\rho_*^{qT} + \Omega_x(r_N) + \bar{\alpha}(r_N)\} \quad (\text{A.42})$$

in the operator norms used by the first-stage and Riesz equations.

Proof. A change in ζ_{it} cannot affect pre- t regressors, which gives the zero bound for $s < t$. For $s = t$, only the unit-specific covariate block can transmit the perturbation across units, and [Assumption 1\(ii\)](#) gives $\omega_x\{d_N(\ell, i)\}$. For $s > t$, the outcome companion recursion and the predetermined-covariate recursion multiply the same spatial envelope by the geometric temporal decay ρ_*^{s-t} . This proves the cellwise display.

We next sum the cellwise sensitivities over the surviving training cells. If a surviving cell lies inside the spatial radius, the buffer forces $s - t > q_T$, so the temporal contribution is bounded by a constant times ρ_*^{qT} after normalization. If a surviving cell lies outside the spatial radius, the normalized shell contribution is bounded by

$$C \sum_{r \geq r_N} (1+r)^{3d_s-1} \omega_x(r) = C\Omega_x(r_N),$$

where the factor $(1+r)^{3d_s-1}$ is the worst ordered-tuple shell count used for the finite-rank information and target-right-hand-side products. The bounded low-rank feature and localization bounds then give [\(A.40\)](#) and [\(A.41\)](#). These two displays show explicitly where the normalization by NT enters.

The preceding terms are causal-propagation terms. The remaining current-period covariance

between spatially separated score components is controlled by the conditional Davydov inequality. Summing the remote covariance shells gives

$$C \sum_{r \geq r_N} (1+r)^{3d_s-1} \alpha(r)^{\chi_\eta} = C\bar{\alpha}(r_N).$$

Combining this statistical-dependence tail with the causal-propagation bound proves (A.42), uniformly over folds and over the fixed-rank feature classes used below. $\square$

Auxiliary result (Purged dynamic leakage is negligible). *Under Assumptions 1 to 3, 5 and 6, the cross-fitting design in Section 6.1, and the rate regime (5.2),*

$$\sum_{j=1}^{J_{NT}} \left\langle \Pi_j(\hat{\Psi}_{\nu,-j} - \hat{\Psi}_{\nu,-j}^{\text{pred}}), \Pi_j U \right\rangle = o_{p,0}(s_{\nu,NT}),$$

where $\hat{\Psi}_{\nu,-j}^{\text{pred}}$ recomputes the feasible weight after replacing both post-buffer future lagged outcomes in the affected spatial neighborhood and predetermined covariates by their fold-coupled predictable counterparts.

Proof. Put $\Delta\Psi_j = \Pi_j(\hat{\Psi}_{\nu,-j} - \hat{\Psi}_{\nu,-j}^{\text{pred}})$ and $X_j = \langle \Delta\Psi_j, \Pi_j U \rangle$. We control the effect of the held-out idiosyncratic error on the training design, transfer that perturbation through the first-stage and Riesz maps, and then use the mean-zero score structure. A crude Cauchy–Schwarz bound against $\|U\|_F$ is not used.

Step 1: recursive comparison of the training design. Fix a validation cell $a = (i, t)$. In the training design only, replace the unit-specific primitive block $\zeta_{it} = ((\varepsilon_{it}^{x,\text{id}})', u_{it}')'$ by an independent copy under the conditional law given $\mathcal{G}_0$, holding the aggregate and common covariate innovation blocks fixed. Propagate this copy through the recursive system and retain the observed validation-cell regressor when the final weight is evaluated. Let $Z_b^{[a]}$ denote the resulting regressor at a surviving training cell $b = (\ell, s)$. The companion recursion and the unit-specific physical-dependence representation of the predetermined covariates imply, for all $p \leq q = 8 + \eta$,

$$\|Z_b - Z_b^{[a]}\|_{p,0} \leq C\rho_*^{s-t}\omega_x(d_N(\ell, i)) \mathbf{1}\{s > t\} + C\omega_x(d_N(\ell, i)) \mathbf{1}\{s = t\}. \quad (\text{A.43})$$

The canonical buffer removes all terms with $0 \leq s - t \leq q_T$ and $d_N(\ell, i) \leq r_N$. For a surviving cell, the effect of the held-out primitive block is therefore controlled either by the temporal tail $\rho_*^{q_T}$ or by the spatial-propagation point bound $\omega_x(r_N)$. Once the surviving spatial shells are summed in the normalized operator norm, this point bound is replaced by the tail $\Omega_x(r_N)$. Applying the spatial-mixing shell bound to the remaining contemporaneous statistical dependence adds $\bar{\alpha}(r_N)$. The finite-rank feature bound gives uniformly over folds

$$\|\hat{\mathcal{G}}_{\nu,-j} - \hat{\mathcal{G}}_{\nu,-j}^{\text{pred}}\|_{\text{op}} + \|\widehat{W}_{\nu,-j} - \widehat{W}_{\nu,-j}^{\text{pred}}\|_{\mathbb{H}} = O_{p,0}(\vartheta_{NT}\|W_\nu\|_{\mathbb{H}}), \quad \vartheta_{NT} := \rho_*^{q_T} + \Omega_x(r_N) + \bar{\alpha}(r_N). \quad (\text{A.44})$$

The first-stage tangent projection is included in (A.44): its perturbation follows from the local inverse-Hessian argument used in Proposition 6, with the coupled-design score in place of the leave-one-slice score.

Step 2: transfer through the Riesz equation. On the event from Lemma 1, both realized and predictable buffered information operators have eigenvalues in a common compact subset of $(0, \infty)$. The resolvent identity therefore yields

$$\|\widehat{q}_{\nu,-j} - \widehat{q}_{\nu,-j}^{\text{pred}}\|_{\mathbb{H}} \leq C\vartheta_{NT}\|W_{\nu}\|_{\mathbb{H}}.$$

Validation norm transport, target variance comparability, and $\sum_j p_{j,NT} = 1$ imply

$$\sum_{j=1}^{J_{NT}} \mathbb{E}_0 \|\Delta\Psi_j\|_F^2 \leq C\vartheta_{NT}^2 s_{\nu,NT}^2. \quad (\text{A.45})$$

There is no $J_{NT}^{1/2}$ loss: the validation masks are disjoint and the fold-specific normalizations are already incorporated in the Riesz equations.

Step 3: predictable part of the score. For each validation cell $a = (i, t)$, take the coordinatewise predictable projection

$$\Delta\Psi_{j,it}^0 := \mathbb{E}_0(\Delta\Psi_{j,it} \mid \mathcal{P}_{it}), \quad \Delta\Psi_{j,it}^1 := \Delta\Psi_{j,it} - \Delta\Psi_{j,it}^0.$$

Jensen's inequality preserves (A.45). The first component is a measurable function of the own current predetermined regressor and the past, so coordinatewise weak exogeneity gives $E_0(\Delta\Psi_{j,it}^0 u_{it} \mid \mathcal{F}_{t-1}) = 0$. Because conditional expectation onto $\mathcal{P}_{it} = \mathcal{F}_{t-1} \vee \sigma(x_{it})$ is coordinatewise in the current field, the transformed array $\{\Delta\Psi_{j,it}^0 u_{it} : i \leq N\}$ has conditional strong-mixing coefficients no larger than those of $\{(x_{it}, u_{it}) : i \leq N\}$. The Davydov-type covariance inequality for strong-mixing variables in Doukhan (1994, Davydov-type covariance inequality for strong-mixing variables) therefore gives period by period

$$E_0 \left[\left(\sum_i \Delta\Psi_{j,it}^0 u_{it} \right)^2 \middle| \mathcal{F}_{t-1} \right] \leq C \sum_i E_0 [(\Delta\Psi_{j,it}^0)^2 \mid \mathcal{F}_{t-1}].$$

Temporal martingale orthogonality, disjoint validation masks, and (A.45) therefore give

$$\sum_j \langle \Delta\Psi_j^0, \Pi_j U \rangle = O_{p,0}(\vartheta_{NT} s_{\nu,NT}) = o_{p,0}(s_{\nu,NT}). \quad (\text{A.46})$$

Step 4: centered coupling remainder. The centered component is a measurable functional of the omitted temporal and spatial tails. Apply the conditional mixing covariance inequality to the products $\Delta\Psi_{j,a}^1 u_a$, using the $L^{4+\eta/2}$ bound implied by Assumption 3. The same shell summation as

in (A.43), now over the disjoint validation cells, gives

$$\left\| \sum_j \langle \Delta \Psi_j^1, \Pi_j U \rangle \right\|_{2,0} \leq C \vartheta_{NT} \sqrt{NT} s_{\nu,NT} + o(s_{\nu,NT}). \quad (\text{A.47})$$

The rate regime (5.2) requires $\{\rho_*^{qT} + \Omega_x(r_N) + \bar{\alpha}(r_N)\} \sqrt{NT} \rightarrow 0$, so the right side of (A.47) is $o(s_{\nu,NT})$. Combining (A.46) and (A.47) proves the proposition. $\square$

Proof of Proposition 7. By Proposition 6 and $\delta_{NT} \leq \delta_{\max,NT}$,

$$\left\| \mathcal{P}_{\hat{\mathcal{T}}_{-j}} \mathbf{D}_\nu - \mathcal{P}_{\mathcal{T}_0} \mathbf{D}_\nu \right\|_{\mathbb{H}} = O_{p,0} \{ \delta_{\max,NT} \|\mathbf{W}_\nu\|_{\mathbb{H}} \}. \quad (\text{A.48})$$

Let

$$\mathcal{G}_{\nu,j}^{\text{or}} = \alpha_j \mathcal{P}_{\mathcal{T}_0} (\mathcal{A}_{-j}^{\text{buf,pred}})^* \mathcal{A}_{-j}^{\text{buf,pred}} \mathcal{P}_{\mathcal{T}_0}, \quad \hat{\mathcal{G}}_{\nu,-j} = \alpha_j \mathcal{P}_{\hat{\mathcal{T}}_{-j}} \mathcal{A}^* \Pi_{-j}^{\text{buf}} \mathcal{A} \mathcal{P}_{\hat{\mathcal{T}}_{-j}}.$$

Purged-training Gram nonsingularity follows from the purged-training secant lower bound and the fold-transport condition, and the upper bound gives $\left\| \alpha_j^{1/2} \Pi_{-j}^{\text{buf}} \mathcal{A} \mathcal{P} \right\|_{\text{op}} \leq C$ for every product tangent projection $\mathcal{P}$ in a δ_{NT} -neighborhood of $\mathcal{P}_{\mathcal{T}_0}$. The realized purged-training operator differs from its predictable benchmark version only through the geometrically small post-purge tail in the buffered local-replacement bound (A.42). Hence

$$\begin{aligned} \left\| \hat{\mathcal{G}}_{\nu,-j} - \mathcal{G}_{\nu,j}^{\text{or}} \right\|_{\text{op}} &\leq C \left\| \mathcal{P}_{\hat{\mathcal{T}}_{-j}} - \mathcal{P}_{\mathcal{T}_0} \right\|_{\text{op}} + C \left\| \mathcal{P}_{\hat{\mathcal{T}}_{-j}} - \mathcal{P}_{\mathcal{T}_0} \right\|_{\text{op}}^2 \\ &\quad + o_{p,0}(\delta_{\max,NT}) = O_{p,0}(\delta_{\max,NT}). \end{aligned} \quad (\text{A.49})$$

Write $\widehat{\mathbf{W}}_{\nu,-j} = \mathcal{P}_{\hat{\mathcal{T}}_{-j}} \mathbf{D}_\nu$, $\mathbf{W}_\nu = \mathcal{P}_{\mathcal{T}_0} \mathbf{D}_\nu$. The resolvent decomposition

$$\hat{\mathbf{q}}_{\nu,-j} - \mathbf{q}_{\nu,j}^{\text{or}} = \hat{\mathcal{G}}_{\nu,-j}^{-1} (\widehat{\mathbf{W}}_{\nu,-j} - \mathbf{W}_\nu) + \hat{\mathcal{G}}_{\nu,-j}^{-1} (\mathcal{G}_{\nu,j}^{\text{or}} - \hat{\mathcal{G}}_{\nu,-j}) (\mathcal{G}_{\nu,j}^{\text{or}})^{-1} \mathbf{W}_\nu,$$

together with (A.48) and (A.49) and the uniformly bounded inverse norms, gives

$$\left\| \hat{\mathbf{q}}_{\nu,-j} - \mathbf{q}_{\nu,j}^{\text{or}} \right\|_{\mathbb{H}} = O_{p,0} \{ \delta_{\max,NT} \|\mathbf{W}_\nu\|_{\mathbb{H}} \}. \quad (\text{A.50})$$

Applying the validation operator and using the validation-fold restricted upper information bound $\|\Pi_j \mathcal{A}(\Delta)\|_F \leq C p_{j,NT}^{1/2} \|\Delta\|_{\mathbb{H}}$ yields

$$\left[\sum_{j=1}^{J_{NT}} \left\| \Pi_j (\hat{\Psi}_{\nu,-j} - \Psi_{\nu,j}^{\text{or}}) \right\|_F^2 \right]^{1/2} = O_{p,0} \{ \delta_{\max,NT} \|\mathbf{W}_\nu\|_{\mathbb{H}} \}.$$

Assumption 6, Assumption 5, and validation fold transport give $s_{\nu,NT} \asymp \|\Psi_\nu^{\text{or}}\|_F \asymp \|\mathbf{W}_\nu\|_{\mathbb{H}}$ in probability. Since $\delta_{\max,NT} = o(1)$, this proves (A.38). For the one-step remainder, work on the

event $\hat{\mathbf{r}} = \mathbf{r}_0$, whose probability tends to one. With $\hat{\Delta}_{-j}^0 = \hat{\Theta}_{-j}^0(\mathbf{r}_0) - \Theta_0$, the exact identity

$$\hat{\mathbf{r}}_{\nu,-j}(\hat{\Delta}_{-j}^0) = \mathbf{r}_{\nu,j}^{\text{or}}(\hat{\Delta}_{-j}^0) - p_{j,NT}^{-1} \left\langle \Pi_j(\hat{\Psi}_{\nu,-j}^0 - \Psi_{\nu,j}^{\text{or}}), \Pi_j \mathcal{A}(\hat{\Delta}_{-j}^0) \right\rangle$$

shows that the left side of (A.39) is bounded by the two localized terms in (A.37). It is therefore $o_{p,0}(s_{\nu,NT})$ by the cross-fitted product rate (A.37). This step uses the derived cross-fitted product rate; it does not multiply the marginal operator bound (A.50) by an uncontrolled global Frobenius error.

Studentization is not reduced to a diagonal sum of squared residuals. The full-covariance spatial-kernel replacement is proved in Proposition 9, using the score-vector bound delivered by the same rowwise, fold-transport, and coupling decomposition. $\square$

A.4 CLT, studentization, and target-specific verification

Let

$$\bar{\mathcal{G}}_0 = \mathbb{E}_0[\mathcal{P}_{\mathcal{T}_0} \mathcal{A}^* \mathcal{A} \mathcal{P}_{\mathcal{T}_0}], \quad \bar{\mathbf{q}}_\nu = \bar{\mathcal{G}}_0^{-1} \mathbf{W}_\nu, \quad \tilde{\Psi}_{\nu,it} = [\mathcal{A} \bar{\mathbf{q}}_\nu]_{it}.$$

The population operator integrates over the stochastic predetermined design; $\bar{\mathbf{q}}_\nu$ is measurable with respect to the primitive frame $\mathcal{G}_0$. Because the design operator is cellwise, $\tilde{\Psi}_{\nu,it}$ depends on the current observation only through $(x_{it}, y_{i,t-1}, \dots, y_{i,t-P})$ and is therefore measurable with respect to $\mathcal{P}_{it} := \sigma(\mathcal{G}_0, x_{it}, \mathcal{F}_{t-1})$. Define

$$\xi_{\nu,t} = \sum_{i=1}^N \tilde{\Psi}_{\nu,it} u_{it}, \quad \tilde{L}_\nu = \sum_{t=1}^T \xi_{\nu,t},$$

$$V_{\nu,NT} = \sum_{t=1}^T E_0[\xi_{\nu,t}^2 \mid \mathcal{F}_{t-1}], \quad s_{\nu,NT}^2 = E_0[V_{\nu,NT}].$$

The conditional variance retains every contemporaneous covariance term inside $E_0[\xi_{\nu,t}^2 \mid \mathcal{F}_{t-1}]$.

Lemma 3 (Predictability, buffer decoupling, and benchmark CLT). *Under Assumptions 1 and 2,*

$$E_0[\xi_{\nu,t} \mid \mathcal{F}_{t-1}] = 0.$$

Consequently, $\{\sum_{s \leq t} \xi_{\nu,s}, \mathcal{F}_t\}$ is a martingale. No stronger vector conditional-mean restriction and no diagonal within-period covariance formula are used.

Proof. Write the product-space Riesz direction as $\bar{\mathbf{q}}_\nu = (\bar{q}_\nu^{(1)}, \dots, \bar{q}_\nu^{(M)}, \bar{q}_\nu^H)$. Because $\mathcal{G}_0$ is the conditional expectation, given $\mathcal{G}_0$, of the random information map, both $\mathcal{G}_0$ and its inverse on $\mathcal{T}$ are $\mathcal{G}_0$ -measurable. Hence $\bar{\mathbf{q}}_\nu$ is $\mathcal{G}_0$ -measurable. Cellwise evaluation of the design map gives

$$\tilde{\Psi}_{\nu,it} = \sum_{m=1}^P y_{i,t-m} [\bar{q}_\nu^{(m)}]_{it} + \sum_{k=1}^K x_{it,k} [\bar{q}_\nu^{(P+k)}]_{it} + [\bar{q}_\nu^H]_{it}.$$

All lagged outcomes in this display are $\mathcal{F}_{t-1}$ -measurable and the current predetermined regressor x_{it} is observed before u_{it} . Thus $\tilde{\Psi}_{\nu,it}$ is measurable with respect to $\mathcal{P}_{it} = \mathcal{F}_{t-1} \vee \sigma(x_{it})$. Assumptions 3 and 5 imply $E_0|\tilde{\Psi}_{\nu,it}u_{it}| < \infty$, so conditional iterated expectations are legitimate. Coordinatewise weak exogeneity then gives

$$\begin{aligned} E_0[\tilde{\Psi}_{\nu,it}u_{it} \mid \mathcal{F}_{t-1}] &= E_0\left[\tilde{\Psi}_{\nu,it}E_0(u_{it} \mid x_{it}, \mathcal{F}_{t-1}) \mid \mathcal{F}_{t-1}\right] \\ &= 0. \end{aligned}$$

Summing the finite number N of integrable terms yields $E_0(\xi_{\nu,t} \mid \mathcal{F}_{t-1}) = 0$. Notice that the argument does not condition u_{it} on x_{jt} , $j \neq i$, and therefore does not invoke vector sequential exogeneity. It also does not eliminate contemporaneous covariance. Because the weights may depend on current own regressors, the correct predictable variance is kept in its unfactored form,

$$E_0(\xi_{\nu,t}^2 \mid \mathcal{F}_{t-1}) = \sum_{i,j} E_0\left[\tilde{\Psi}_{\nu,it}\tilde{\Psi}_{\nu,jt}u_{it}u_{jt} \mid \mathcal{F}_{t-1}\right],$$

which retains every off-diagonal term and does not pull random current-regressor weights through the conditional expectation. The partial sums are therefore a square-integrable martingale with the stated filtration. $\square$

Auxiliary result (Space–time buffer decoupling). *Let L_ν^{or} be the score formed with the true-tangent empirical buffered Riesz weights and let $\tilde{L}_\nu$ be the population predictable score. Under Assumptions 1, 3, 5, 6, together with the cross-fitting design in Section 6.1 and the rate regime (5.2),*

$$L_\nu^{\text{or}} - \tilde{L}_\nu = O_{p,0}\left([\rho_*^{qT} + \Omega_x(r_N) + \bar{\alpha}(r_N)]\sqrt{NT} s_{\nu,NT} + \varepsilon_{G,NT}s_{\nu,NT}\right) = o_{p,0}(s_{\nu,NT}).$$

Proof. Couple one held-out idiosyncratic error slice to an independent copy and propagate the replacement through the stable outcome recursion and recursive covariate map. The temporal buffer leaves an L_2 discrepancy bounded by $C\rho_*^{qT}$, while summing the local propagation envelope over surviving spatial shells leaves the causal tail $C\Omega_x(r_N)$. The spatial buffer and conditional Davydov inequality leave a remaining contemporaneous statistical tail bounded by $C\bar{\alpha}(r_N)$. Apply the resolvent identity to the empirical and population information operators; Assumption 5 bounds both inverses, and Proposition 4 contributes $\varepsilon_{G,NT}$. Proposition 10 converts the cellwise discrepancies to the target-score scale. Summing over folds uses their validation shares, whose weights sum to one; no $J_{NT}^{1/2}$ loss occurs. $\square$

Auxiliary result (Conditional weighted spatial-slice Gaussian approximation). *Let $\{Y_{n,i} : i \in I_n\}$ be a triangular array that, conditional on a sigma-field $\mathcal{C}_n$, is centered and strongly mixing on a finite metric space with the envelope and volume growth in Assumption 3. Put $S_n = \sum_i Y_{n,i}$ and*

$\sigma_n^2 = E(S_n^2 \mid \mathcal{C}_n) > 0$. Assume also that, with probability tending to one,

$$c \sum_i E(Y_{n,i}^2 \mid \mathcal{C}_n) \leq \sigma_n^2 \leq C \sum_i E(Y_{n,i}^2 \mid \mathcal{C}_n).$$

For some $\delta \in (0, \eta/4]$, define

$$L_n = \frac{\sum_i E(|Y_{n,i}|^{2+\delta} \mid \mathcal{C}_n)}{\sigma_n^{2+\delta}}, \quad A_n(r) = \frac{\max_k E\{(\sum_{i \in B_N(k,r)} Y_{n,i})^2 \mid \mathcal{C}_n\}}{\sigma_n^2}.$$

If, for some $r_n \rightarrow \infty$,

$$L_n + A_n(r_n) + |I_n|^{1/2} \bar{\alpha}(r_n) \rightarrow_p 0,$$

then, for every fixed $U < \infty$,

$$\sup_{|u| \leq U} \left| E\left(e^{iuS_n/\sigma_n} \mid \mathcal{C}_n\right) - e^{-u^2/2} \right| \rightarrow_p 0. \quad (\text{A.51})$$

The conclusion is uniform over a fixed-dimensional family of weights whenever the three displayed quantities converge uniformly over the standardized unit sphere.

Proof. This is a conditional triangular-array version of the random-field central limit arguments in [Bolthausen \(1982, random-field CLT for stationary mixing fields\)](#) and [Jenish and Prucha \(2009, random-field CLT and uniform law for arrays\)](#). The mapping to those blocking theorems is as follows. After conditioning on $\mathcal{C}_n$, the weights, metric balls, conditional second moments, and mixing envelope are fixed, so the triangular array can be viewed pathwise as a nonidentically distributed strong-mixing random field. The quantity L_n is the Lindeberg/truncation term; $A_n(r_n)$ is the maximum normalized variance of a local ball and rules out a non-negligible small block; and $|I_n|^{1/2} \bar{\alpha}(r_n)$ is the large-block separation error obtained by summing all remote covariance shells. Thus the displayed condition is the conditional analogue of the usual requirements that the largest local block and the remote-covariance remainder vanish relative to the variance. Random predictable weights require no additional probabilistic step after conditioning; they enter only through these three normalized quantities. The use of $\bar{\alpha}(r_n)$, rather than the point coefficient $\alpha(r_n)^{\chi_\eta}$, is essential because the blocking proof sums all shells outside the separation radius. We give the characteristic-function proof to make these modifications explicit and because the same argument is used again in the hybrid time-space proof. Work on a realization of $\mathcal{C}_n$, suppress it from the notation, and normalize $\sigma_n^2 = 1$. Truncate each summand at a level $c_n \uparrow \infty$ chosen so slowly that $c_n^{-\delta} L_n = o(1)$ and $c_n^2 A_n(r_n) = o(1)$; recenter the truncated summands. Conditional Hölder and Markov inequalities show that the discarded sum is $o_p(1)$, uniformly for bounded characteristic-function arguments. It is therefore enough to treat bounded, centered summands, still denoted Y_i .

Let $N_i = B_N(i, r_n)$, $S_i = \sum_{j \in N_i} Y_j$, and $S_i^c = S_n - S_i$. Denote the conditional characteristic

function by $\varphi_n(u)$. Since $EY_i = 0$,

$$\varphi'_n(u) = i \sum_i E\{Y_i(e^{iuS_n} - e^{iuS_i^c})\} + i \sum_i \text{Cov}(Y_i, e^{iuS_i^c}).$$

Davydov's inequality, followed by the metric-shell summation, bounds the second term uniformly on $|u| \leq U$. Indeed, for each i , the function $e^{iuS_i^c}$ is measurable with respect to variables at distance larger than r_n from i , and the shell-counting bound gives

$$\sum_i |\text{Cov}(Y_i, e^{iuS_i^c})| \leq C_U \sum_i (EY_i^2)^{1/2} \sum_{\ell \geq r_n} (1 + \ell)^{3d_s - 1} \alpha(\ell)^{\chi_\eta}.$$

By Cauchy–Schwarz and $\sum_i EY_i^2 \asymp 1$, this is bounded by

$$C_U |I_n|^{1/2} \bar{\alpha}(r_n) \left(\sum_i EY_i^2 \right)^{1/2} = o(1).$$

For the first term, use $e^{iuS_n} - e^{iuS_i^c} = iuS_i e^{iuS_i^c} + R_i(u)$, where $|R_i(u)| \leq C_U |S_i|^{1+\delta/2} \min\{1, |S_i|^{1-\delta/2}\}$. Conditional Rosenthal, the definition of $A_n(r_n)$, and the truncated $(2 + \delta)$ th moments imply

$$\sum_i E|Y_i R_i(u)| \leq C_U \{L_n^{\delta/(2+\delta)} + A_n(r_n)^{\delta/(4+2\delta)}\} = o(1).$$

Next replace $e^{iuS_i^c}$ by e^{iuS_n} . The resulting error is bounded by $C_U \sum_i E|Y_i| |S_i|^2$, which has the same upper bound. Thus

$$\varphi'_n(u) = -u E \left[e^{iuS_n} \sum_i Y_i S_i \right] + o(1).$$

The omitted remote covariances satisfy

$$|1 - \sum_i E(Y_i S_i)| \leq \sum_i \sum_{j: d_N(i,j) > r_n} |\text{Cov}(Y_i, Y_j)| \leq C |I_n|^{1/2} \bar{\alpha}(r_n) = o(1),$$

where the final inequality again uses Davydov's bound, Cauchy–Schwarz over i , and the same shell-tail envelope. A second application of the local-moment bound shows that $\text{Cov}(e^{iuS_n}, \sum_i Y_i S_i) = o(1)$, uniformly on bounded u . Consequently

$$\sup_{|u| \leq U} |\varphi'_n(u) + u\varphi_n(u)| = o(1).$$

Since $\varphi_n(0) = 1$, variation of constants for this differential equation gives (A.51). Restoring the conditioning and the original scale proves the scalar statement. A compactness argument on the standardized unit sphere gives the fixed-dimensional uniform conclusion. $\square$

Auxiliary result (Conditional variance regularity and global Lindeberg bounds). *Under Assump-*

tions 1–6, the deterministic variance is positive and of the target scale, and the predictable quadratic variation satisfies

$$V_{\nu,NT}/s_{\nu,NT}^2 \rightarrow_p 1.$$

For some $\delta \in (0, \eta/4]$,

$$\frac{1}{s_{\nu,NT}^{2+\delta}} \sum_{i,t} E_0 \left(|\tilde{\Psi}_{\nu,it} u_{it}|^{2+\delta} \mid \mathcal{F}_{t-1} \right) = o_{p,0}(1), \quad (\text{A.52})$$

and the corresponding radius- r_N neighborhood ratio tends to zero. These conclusions hold uniformly over variance-normalized linear combinations of a fixed-dimensional target vector.

Proof. Variance noncancellation and predictable-quadratic-variation stabilization are imposed in Assumption 6. For the moment statement, note that $\tilde{\Psi}_{\nu,it}$ is generally measurable with respect to $\sigma(x_{it}, \mathcal{F}_{t-1}, \mathcal{G}_0)$, not merely $\mathcal{F}_{t-1}$. Therefore the conditional moment is evaluated by first conditioning on the own current predetermined regressor:

$$\begin{aligned} & E_0 \left(|\tilde{\Psi}_{\nu,it} u_{it}|^{2+\delta} \mid \mathcal{F}_{t-1} \right) \\ &= E_0 \left[|\tilde{\Psi}_{\nu,it}|^{2+\delta} E_0 \{ |u_{it}|^{2+\delta} \mid x_{it}, \mathcal{F}_{t-1} \} \mid \mathcal{F}_{t-1} \right] \\ &\leq C E_0 \left(|\tilde{\Psi}_{\nu,it}|^{2+\delta} \mid \mathcal{F}_{t-1} \right), \end{aligned}$$

where the last inequality uses Assumption 3. Summing over cells, taking expectations, and applying Markov's inequality reduce the desired conditional Lindeberg display to the unconditional bound below. Interpolating between the maximal weight and its squared Frobenius norm, using variance comparability, and applying the single-cell part of (4.5) gives

$$\frac{\sum_{i,t} E_0 |\tilde{\Psi}_{\nu,it}|^{2+\delta}}{s_{\nu,NT}^{2+\delta}} \leq C \left(\frac{\|\tilde{\Psi}_{\nu}\|_{\max}^2}{E_0 \|\tilde{\Psi}_{\nu}\|_F^2} \right)^{\delta/2} = o(1).$$

The local-block ratio required by the conditional spatial-slice CLT is exactly the second term in (4.5) under the maintained spatial design; under the known-cluster alternative the same display is read with the cluster block $\mathcal{C}_N(k)$. Uniformity over a fixed target vector follows from the uniform target clause and compactness of the standardized unit sphere. $\square$

Primitive variance-field sufficient condition. Writing $v_t = E_0(\xi_{\nu,t}^2 \mid \mathcal{F}_{t-1})$, there exist coupled variables $v_t^{(H)}$, measurable with respect to the primitive slices $e_{t-H}, \dots, e_{t-1}$ and independent auxiliary copies, such that

$$\|v_t - v_t^{(H)}\|_{2,0} \leq C \rho_*^H E_0 v_t, \quad E_0(v_t^2) \leq C(E_0 v_t)^2$$

uniformly in t . The first inequality is a primitive physical-dependence restriction on the predictable second-moment field; it is automatic, for example, when the conditional covariance kernel is a uniformly Lipschitz functional of the same stable recursive state as the regressors.

Predictable-variance stabilization fact. Suppose the hypotheses of Lemma 3 and the primitive variance-field sufficient condition above hold and

$$d_{\nu,NT} := \frac{\max_t E_0 v_t}{s_{\nu,NT}^2} \rightarrow 0.$$

Then the predictable-variance stabilization clause of Assumption 6 holds. In particular, the conclusion applies to time-averaged standard targets for which Appendix A.4.2 verifies $d_{\nu,NT} \rightarrow 0$. For a time-local target, predictable-variance stabilization remains a genuine target requirement unless a cross-sectional weighted law of large numbers for its conditional covariance kernel is verified separately.

Proof. For each t , construct $v_t^{(H)}$ with independent auxiliary copies for primitive slices older than $t - H$. Conditional on $\mathcal{G}_0$, the sequence $\{v_t^{(H)}\}$ is $(2H + 1)$ -dependent. Therefore

$$\text{Var}_0\left(\sum_t v_t^{(H)}\right) \leq CH \sum_t E_0(v_t^2) \leq CH(\max_t E_0 v_t) \sum_t E_0 v_t = CH d_{\nu,NT} s_{\nu,NT}^4.$$

Minkowski's inequality and the coupling bound give

$$\left\| \sum_t (v_t - v_t^{(H)}) \right\|_{2,0} \leq C \rho_*^H \sum_t E_0 v_t = C \rho_*^H s_{\nu,NT}^2.$$

Choose $H = H_{NT} \rightarrow \infty$ so slowly that $H_{NT} d_{\nu,NT} \rightarrow 0$ and $\rho_*^{H_{NT}} \rightarrow 0$. The two displays imply $\text{Var}_0(V_{\nu,NT}) = o(s_{\nu,NT}^4)$; Chebyshev's inequality proves $V_{\nu,NT}/s_{\nu,NT}^2 \rightarrow_p 1$. $\square$

Auxiliary result (Hybrid spatial-temporal benchmark central limit theorem). *Under the hypotheses of Lemma 3 and the predictable-variance clause of Assumption 6,*

$$\frac{\tilde{L}_\nu}{s_{\nu,NT}} \xrightarrow{d} N(0, 1).$$

The conclusion holds jointly for a fixed finite target vector after normalization by its deterministic covariance matrix.

Proof. Write

$$X_t = \xi_{\nu,t}/s_{\nu,NT}, \quad a_t = E_0(X_t^2 \mid \mathcal{F}_{t-1}), \quad A_t = \sum_{r \leq t} a_r.$$

By the predictable-variance clause of Assumption 6, $A_T \rightarrow_p 1$. The proof permits a finite or slowly growing number of time periods to carry a nonnegligible share of the variance; those time periods are Gaussianized cross-sectionally rather than being forced into a no-dominant-time period martingale theorem.

Step 1: conditional characteristic-function errors. Fix $U < \infty$ and choose $\tau_{NT} \downarrow 0$ sufficiently

slowly. Let

$$\mathcal{T}_H = \{t : a_t > \tau_{NT}\}, \quad \mathcal{T}_L = \{1, \dots, T\} \setminus \mathcal{T}_H.$$

On $\{A_T \leq C\}$, $|\mathcal{T}_H| \leq C/\tau_{NT}$. For $t \in \mathcal{T}_H$, apply Lemma 3, conditionally on $\mathcal{F}_{t-1}$, to $Y_{it} = \tilde{\Psi}_{\nu,it} u_{it}/s_{\nu,NT}$. The slice Lyapunov and radius- r_N ratios are bounded by their global counterparts divided by $\tau_{NT}^{1+\delta/2}$ and τ_{NT} , respectively. The remote mixing term is bounded by Assumption 3 and the spatial-radius restriction in (5.2). A diagonal choice of τ_{NT} therefore gives predictable errors $\epsilon_{t,NT}(u)$ such that, uniformly on $|u| \leq U$,

$$E_0(e^{iuX_t} \mid \mathcal{F}_{t-1}) = e^{-u^2 a_t/2} + \epsilon_{t,NT}(u), \quad E_0 \sum_{t \in \mathcal{T}_H} \sup_{|u| \leq U} |\epsilon_{t,NT}(u)| = o(1). \quad (\text{A.53})$$

The expectation in the last display follows from the explicit bounded truncation used in Lemma 3; it is not obtained by multiplying a pointwise $o_p(1)$ bound by the number of high variance time periods.

For $t \in \mathcal{T}_L$, conditional centering and Taylor's formula yield, for every $\varepsilon > 0$,

$$\begin{aligned} & \left| E_0(e^{iuX_t} \mid \mathcal{F}_{t-1}) - e^{-u^2 a_t/2} \right| \\ & \leq C_U E_0[X_t^2 \mathbf{1}\{|X_t| > \varepsilon\} \mid \mathcal{F}_{t-1}] + C_U \varepsilon a_t + C_U a_t^2. \end{aligned} \quad (\text{A.54})$$

Summing (A.54) over $\mathcal{T}_L$, first letting $NT \rightarrow \infty$ and then $\varepsilon \downarrow 0$, gives $o_p(1)$ by (A.52), conditional spatial Rosenthal, and $\sum_{t \in \mathcal{T}_L} a_t^2 \leq \tau_{NT} A_T$. Uniform integrability from the $(2+\delta)$ th-moment bound upgrades this to L^1 . Combining the high- and low-time bounds gives

$$E_0 \sum_{t=1}^T \sup_{|u| \leq U} \left| e^{u^2 a_t/2} E_0(e^{iuX_t} \mid \mathcal{F}_{t-1}) - 1 \right| = o(1) \quad (\text{A.55})$$

when the sum is stopped before A_t exceeds any fixed constant.

Step 2: the martingale characteristic-function recursion. The usual product-of-conditional-characteristic-functions identity is not valid because future conditional variances depend on past idiosyncratic errors. We instead use the predictable exponential compensator. For $C > 2$, let $\tau_C = \inf\{t : A_t > C\}$ and replace X_t, a_t by zero for $t \geq \tau_C$, so the stopped cumulative variance never exceeds C . Define

$$Z_t(u) = \exp\left(iu \sum_{r \leq t} X_r\right) \exp\left[-\frac{u^2}{2} \{1 - A_t\}\right], \quad Z_0(u) = e^{-u^2/2}.$$

Since a_t is $\mathcal{F}_{t-1}$ -measurable,

$$E_0\{Z_t(u) \mid \mathcal{F}_{t-1}\} - Z_{t-1}(u) = Z_{t-1}(u) \left[e^{u^2 a_t/2} E_0(e^{iuX_t} \mid \mathcal{F}_{t-1}) - 1 \right].$$

On the stopped array, $\sup_{t, |u| \leq U} |Z_t(u)| \leq \exp\{U^2(C+1)/2\}$. Summing conditional expectations

and applying (A.55) therefore gives

$$\sup_{|u| \leq U} |E_0 Z_T(u) - e^{-u^2/2}| \rightarrow 0.$$

the predictable-variance clause of Assumption 6 implies $P_0(\tau_C \leq T) \rightarrow 0$ and $\exp[-u^2\{1 - A_T\}/2] \rightarrow_p 1$, uniformly on bounded u . Boundedness on the stopped event then yields

$$E_0 \exp\left(iu \sum_{t=1}^T X_t\right) \rightarrow e^{-u^2/2}.$$

Lévy's theorem proves the scalar limit. The conditional slice bounds and the predictable-variance stabilization assumption hold uniformly over the variance-normalized unit sphere for a fixed target vector; Cramér–Wold gives the joint conclusion. $\square$

Proposition 8 (One-step expansion). *Under Assumptions 1–6, the rate regime (5.2), and (7.1), and, when a finite numerical iterate is used, the implementation requirement (6.5),*

$$\check{\phi}_\nu - \phi_\nu(\Theta_0) = \tilde{L}_\nu + o_{p,0}(s_{\nu,NT}).$$

Proof. Work on the event on which the rank selector recovers $\mathbf{r}_0$ and the first-stage localization bounds in Theorem 2 hold for all folds; its probability tends to one. For fold j , put $\hat{\Delta}_{-j} = \hat{\Theta}_{-j}^0 - \Theta_0$ and insert $\hat{\mathbf{R}}_{-j}^0 = \mathbf{U} - \mathcal{A}\hat{\Delta}_{-j}$ into the one-step formula. The plug-in part is decomposed into its true-tangent component and the quadratic normal remainder associated with the low-rank manifold. The fold-specific Riesz identity in Proposition 3 cancels the true-tangent component against the correction term based on the benchmark fold weight. Therefore the fold contribution equals the benchmark weighted error term plus three remainder blocks: the quadratic normal remainder, the feasible-to-benchmark Riesz-weight replacement, and the buffer/coupling error created by replacing the purged training law with the predictable population law.

The quadratic normal remainder is controlled by the rowwise and tangent-space rates in Theorem 2; the replacement of estimated by true tangent spaces contributes $O_{p,0}(b_{NT}\delta_{\max,NT}s_{\nu,NT})$; Proposition 7 controls both the feasible Riesz-weight and nuisance-product terms; and Lemma 3 controls the temporal and spatial leakage terms. After summing over folds with weights $p_{j,NT}$, these bounds are exactly the components of $r_{\nu,NT}s_{\nu,NT}$ in (7.1). Since $r_{\nu,NT} \rightarrow 0$, the feasible one-step estimator is $\tilde{L}_\nu + o_{p,0}(s_{\nu,NT})$. $\square$

Additional proof details for Theorem 4. Proposition 8 gives an $o_p(s_{\nu,NT})$ remainder. Lemma 3, proved under predictable-variance stabilization, gives $\tilde{L}_\nu/s_{\nu,NT} \Rightarrow N(0, 1)$. Slutsky's theorem proves the scalar result. The joint result follows from the uniform fixed-dimensional version of Lemma 3 and Cramér–Wold. $\square$

A.4.1 Studentization

Proposition 9 (Spatial-kernel covariance consistency with generated weights). *Under the dependence, moment, bandwidth, and generated-weight replacement rates in (7.4), the feasible spatial-kernel variance estimator satisfies*

$$\hat{s}_{\nu,\text{sp}}^2/s_{\nu,NT}^2 \rightarrow_p 1.$$

The same statement holds for any fixed finite vector of targets after applying the estimator to fixed linear combinations of the corresponding scores. For full cross-sectional means, the sufficient bandwidth condition in the main text is obtained by inserting the full-mean leverage scale into (7.4).

Proof. Let $s_{\nu,\text{sp},0}^2$ denote the infeasible kernel variance computed with benchmark weights and true errors. Spatial-kernel consistency for this infeasible object follows from Bolthausen’s random-field CLT argument (Bolthausen, 1982, random-field CLT for stationary mixing fields), the random-field CLT and uniform-law framework of Jenish and Prucha (Jenish and Prucha, 2009, random-field CLT and uniform law for arrays), and Conley’s spatial HAC kernel-covariance construction (Conley, 1999, spatial HAC covariance estimator), applied with the conditional covariance-tail and local fourth-moment restrictions in Assumption 3. The metric volume factor $v_N(h_N)$ and the kernel bandwidth appear because only neighbors inside the kernel window enter each local covariance sum. The tail condition in (7.4) makes the truncation bias and remote covariance contribution negligible, so $s_{\nu,\text{sp},0}^2/s_{\nu,NT}^2 \rightarrow_p 1$.

It remains to replace benchmark weights and true errors by the feasible cross-fitted weights and residuals. The Riesz replacement theorem gives L^2 -small weight error on validation cells, while the first-stage theorem and the one-step expansion give residual replacement errors of order $r_{\nu,NT}$. Cauchy–Schwarz over each kernel neighborhood and the metric volume bound give the total replacement rate $v_N(h_N)r_{\nu,NT}$, which is $o_p(1)$ by (7.4). Hence $\hat{s}_{\nu,\text{sp}}^2 - s_{\nu,\text{sp},0}^2 = o_p(s_{\nu,NT}^2)$, proving ratio consistency. Fixed-vector covariance consistency follows by the Cramér–Wold device. $\square$

A.4.2 Target verification

The main theorem is stated for generic linear targets $\phi_\nu(\Theta_0) = \langle \mathbf{D}_\nu, \Theta_0 \rangle_{\mathbb{H}}$. The empirical and simulation targets are entries, cross-sectional means, group means, between-group contrasts, and cumulative-persistence summaries. For these standard targets, the tangent projection and leverage calculations are straightforward consequences of low-rank projection algebra, incoherence, and the Riesz normal equations. We record the verification once to make clear that no new theorem is needed for each target type.

Proposition 10 (Standard linear targets). *Let $\mathbf{D}_\nu$ be an entry, group mean, full mean, between-group contrast, or finite cumulative-persistence direction formed from a fixed number of lag coefficient matrices. For these targets, the tangent-projection and leverage calculations entering Assumption 6 reduce to the standard low-rank projection bounds displayed below. Whenever the corresponding relevance, variance, and dependence-block dispersion requirements in Assumption 6 hold, the five*

main theorems apply. Consequently the one-step estimator obeys the feasible central limit theorem and, when the relevant studentization conditions are imposed, the spatial-kernel limit under the main framework or the cluster-score limit under Appendix A.2.4.

Proof. For an entry target the direction is a coordinate matrix. Incoherence of the low-rank singular spaces bounds its tangent projection and prevents a single cell from dominating the Riesz score. For a group or full mean with group weights π_i , the tangent projection is the sum of the row-space and column-space projections of the matrix $\pi e'_t$. Orthogonal-projection algebra gives the usual decomposition into row, column, and intersection terms; under incoherence,

$$\|\mathcal{P}_{\mathcal{T}_0}(\pi e'_t)\|_{\mathbb{H}}^2 \lesssim T^{-1}\|\pi\|_2^2 + N^{-1}\|\pi\|_1^2, \quad (\text{A.56})$$

with the corresponding group-size specialization. A between-group contrast is the same calculation with signed weights, and its variance keeps the covariance between the two group influence scores. A cumulative-persistence target is a fixed linear combination of lag coefficient targets, so its relevance and variance follow by finite-dimensional linear combination. Thus the only remaining target-specific requirements are the variance and dependence-block dispersion conditions stated in Assumption 6. When those hold, applying the five main theorems to these directions gives the stated inference results. $\square$

B Additional Simulation Results

B.1 Infeasible benchmark with true tangent spaces

Table 17 reports the infeasible benchmark estimator at $(T, N) = (100, 100)$, beside the feasible estimator. With the true tangent spaces the estimator is essentially unbiased for all eight targets, the empirical coverage of the nominal 95% interval lies in $[0.948, 0.969]$ (mean 0.960), and the mean estimated standard error matches the Monte Carlo standard deviation, supporting the conclusion that the spatial-kernel studentizer (6.9) is correctly calibrated under the weak-dependence variance scale of Proposition 9. Consistent with Proposition 10, the mean targets are far more precisely estimated than the entry targets: in this design the infeasible benchmark standard error of the autoregression-coefficient full mean is several times smaller than the autoregression-coefficient entry. This reflects the loading-projection term in (A.56); by Proposition 10 the size of the gain is governed by that projection $\pi'_G P_{\Lambda, m} \pi_G$ and is not a generic $g_N^{-1/2}$ law.

B.2 Feasible convergence study

Table 14 reports the feasible estimator over the geometric grid $T = N \in \{50, 100, 200, 400\}$ at the data-driven exclusion window, separately for the autoregression coefficient a_{it} and the slope coefficient b_{it} and for the two leading targets, the cross-sectional full mean and the between-group contrast. Two patterns stand out. First, the root-mean-square error contracts at the $\sqrt{TN}$ rate: each doubling of the panel more than halves it—for the lag full mean, $0.70 \rightarrow 0.088 \rightarrow 0.034 \rightarrow 0.018$;

for the covariate full mean, $0.134 \rightarrow 0.008 \rightarrow 0.003 \rightarrow 0.001$ —and the mean estimated standard error tracks the Monte Carlo standard deviation at every size, the steeper mean contraction reflecting the extra $\sqrt{g_N}$ factor of [Proposition 10](#). Second, the empirical coverage of the nominal 95% interval rises to nominal as the panel grows and stays there: the lag full mean moves $0.88 \rightarrow 0.93 \rightarrow 0.95 \rightarrow 0.94$ and the lag contrast $0.80 \rightarrow 0.92 \rightarrow 0.95 \rightarrow 0.96$, while the covariate targets are already near nominal by $(100, 100)$ (0.96–0.97). The smallest cell $(50, 50)$ is the genuinely hard regime—the weakly identified lag block and the near-singular Riesz solve leave the entry and contrast targets noisy there, the small- g_N corner anticipated by [Proposition 10](#)—but by $(100, 100)$ every target is within a few Monte Carlo standard errors of nominal, and by $(200, 200)$ the lag *mean*, the hardest object because its variance is the smallest, sits at 0.95. The capped exclusion window of [Remark 3](#) is what holds the lag means at nominal at the largest panels, and [Appendix B.3](#) traces that window’s role. Monte Carlo standard errors for the $R = 1000$ coverage entries are ≈ 0.007 , so the across-size movement toward nominal exceeds simulation noise. Both theory objects of the model—the autoregressive coefficient and the slope coefficient—are thus recovered with RMSE contracting at the predicted rate and inference at the nominal level. At the largest panel the procedure is at nominal across all four target *types* for both coefficients ([Table 15](#)): entry, group-mean, full-mean, and contrast targets all cover at 0.94–0.96, with RMSE ordered as the leverage theory predicts — the pooled full and group means most precise, the single-cell entry least, the contrast intermediate. The debiasing step and the cost of feasibility are isolated in the next two tables. [Table 16](#) contrasts the first-stage plug-in with the debiased one-step: at this large panel the plug-in is already nearly unbiased, so the debiasing’s contribution is the *valid studentized inference* it carries, not a large bias correction. [Table 17](#) contrasts the feasible estimator with the infeasible oracle handed the true tangent spaces: the feasible estimator pays only a small price for estimating the low-rank structure — its coverage is within a couple of points of the oracle’s and its RMSE and standard errors essentially match it — which is the empirical content of the feasible $\sqrt{TN}$ central limit theorem.

B.3 Forward-exclusion-window sensitivity

[Table 18](#) and [Figure 1](#) study sensitivity to the space–time exclusion buffer q at $T = N = 100$. The mechanism is most visible for the cross-sectional *mean* targets, the paper’s primary empirical object: pooled own-error leakage is coherent across units and therefore does not average out, whereas in an entry target it is a single-cell effect. A moderate purge at $q \in \{0, 1, 2, 3\}$ holds the lag full mean at the nominal level (coverage 0.950, 0.960, 0.965, 0.960) and the covariate full mean above it (0.97–0.98). An over-long purge is counterproductive: as q rises it deletes more training information — the retained training share falls from 0.92 at $q = 0$ to 0.56 at $q = 6$ — so the RMSE inflates (the lag full-mean RMSE rises from 0.071 at $q = 1$ to 0.104 at $q = 6$) and coverage erodes (lag full mean 0.940 at $q = 4$, 0.918 at $q = 6$). The forward purge is thus a bias–variance trade-off with an interior optimum at the recommended short radius, the finite-sample counterpart of the post-purge geometric-tail condition $\rho_*^{qT} \sqrt{TN} \rightarrow 0$ balanced against the training-retention restriction in [\(5.2\)](#); the capped window of [Remark 3](#) keeps the degradation at large q mild rather than the sharp collapse

Table 14: Feasible convergence.

Notes: Bias, RMSE, mean estimated standard error, and empirical coverage of the nominal 95% interval over $T = N \in \{50, 100, 200, 400\}$, with $R = 1000$ replications except $R = 500$ at $T = N = 400$.

$T=N$	bias	RMSE	mean s.e.	cov.	pc cov.	pc len.
<i>Autoregressive coefficient a_{it}</i>						
full mean						
50	-0.035	0.698	0.256	0.876	0.880	1.014
100	0.014	0.088	0.063	0.929	0.925	0.247
200	0.003	0.034	0.030	0.953	0.948	0.118
400	0.000	0.018	0.018	0.940	0.940	0.071
contrast						
50	0.028	1.048	0.199	0.801	0.800	0.785
100	0.000	0.028	0.023	0.919	0.923	0.091
200	-0.000	0.009	0.009	0.948	0.947	0.035
400	0.000	0.004	0.004	0.956	0.954	0.016
<i>Slope coefficient b_{it}</i>						
full mean						
50	0.001	0.134	0.043	0.890	0.898	0.168
100	-0.000	0.008	0.007	0.964	0.966	0.028
200	0.000	0.003	0.003	0.956	0.955	0.011
400	0.000	0.001	0.001	0.964	0.960	0.005
contrast						
50	0.004	0.268	0.084	0.879	0.878	0.331
100	-0.000	0.015	0.015	0.963	0.963	0.058
200	0.000	0.005	0.005	0.968	0.965	0.021
400	0.000	0.002	0.002	0.948	0.948	0.010

Table 15: Coverage and RMSE by target type.

Target type	bias	RMSE	cov.	pc cov.
<i>Autoregressive coefficient a_{it}</i>				
entry	-0.000	0.045	0.950	0.946
group mean	0.000	0.018	0.948	0.944
full mean	0.000	0.018	0.940	0.940
contrast	0.000	0.004	0.956	0.954
<i>Slope coefficient b_{it}</i>				
entry	0.003	0.024	0.950	0.944
group mean	0.000	0.002	0.962	0.958
full mean	0.000	0.001	0.964	0.960
contrast	0.000	0.002	0.948	0.948

Table 16: Plug-in versus debiased estimates at $T = N = 400$.

Target	plug-in bias	plug-in RMSE	debiased bias	debiased RMSE	cov.
<i>Autoregressive coefficient a_{it}</i>					
full mean	−0.004	0.016	0.000	0.018	0.940
contrast	0.000	0.004	0.000	0.004	0.956
<i>Slope coefficient b_{it}</i>					
full mean	0.000	0.001	0.000	0.001	0.964
contrast	0.000	0.002	0.000	0.002	0.948

Table 17: Infeasible oracle (true tangent spaces) versus the feasible estimator at $T = N = 100$.

Object	estimator	bias	RMSE	cov.	mean s.e.
Autoregression a_{it} (full mean)	oracle	0.011	0.097	0.948	0.070
	feasible	0.014	0.088	0.929	0.063
Autoregression a_{it} (contrast)	oracle	−0.000	0.024	0.956	0.023
	feasible	0.000	0.028	0.919	0.023
Slope b_{it} (full mean)	oracle	−0.000	0.007	0.968	0.007
	feasible	−0.000	0.008	0.964	0.007
Slope b_{it} (contrast)	oracle	0.000	0.014	0.961	0.015
	feasible	−0.000	0.015	0.963	0.015

an uncapped over-purge would produce. Under the scattered cell-level folds the own-cell exclusion already controls the single-cell leakage, so the incremental value of the forward purge is concentrated in the pooled mean targets, which is precisely where it is reported to matter. We emphasize that the near-nominal coverage already at $q = 0$ for some targets is design-specific finite-sample behavior — a property of this DGP’s dependence range and panel size — and not evidence that the buffer is unnecessary in general: under stronger or longer-range temporal dependence the $q = 0$ leakage would bite, and the theorem requires a growing exclusion window regardless.

Table 18: Purge-radius sensitivity.

q	lag cov.	lag RMSE	cov. cov.	cov. RMSE	retained
0	0.950	0.081	0.976	0.005	0.917
1	0.960	0.071	0.973	0.005	0.841
2	0.965	0.073	0.979	0.006	0.772
3	0.960	0.078	0.972	0.006	0.710
4	0.940	0.103	0.978	0.007	0.654
6	0.918	0.104	0.951	0.010	0.556

Remark 3 (Forward exclusion window for the minimal-variance targets). [Table 18](#) exposes a tension specific to the minimal-variance mean targets. The post-purge geometric-tail condition $\rho_*^{q_T} \sqrt{TN} \rightarrow 0$ pushes the radius up, yet an over-long purge inflates the standard error of the autoregression-coefficient means and costs coverage, while a moderate radius $q \in \{1, 2, 3\}$ leaves all targets near nominal even at the smallest panel. For finite-sample sensitivity we therefore also report

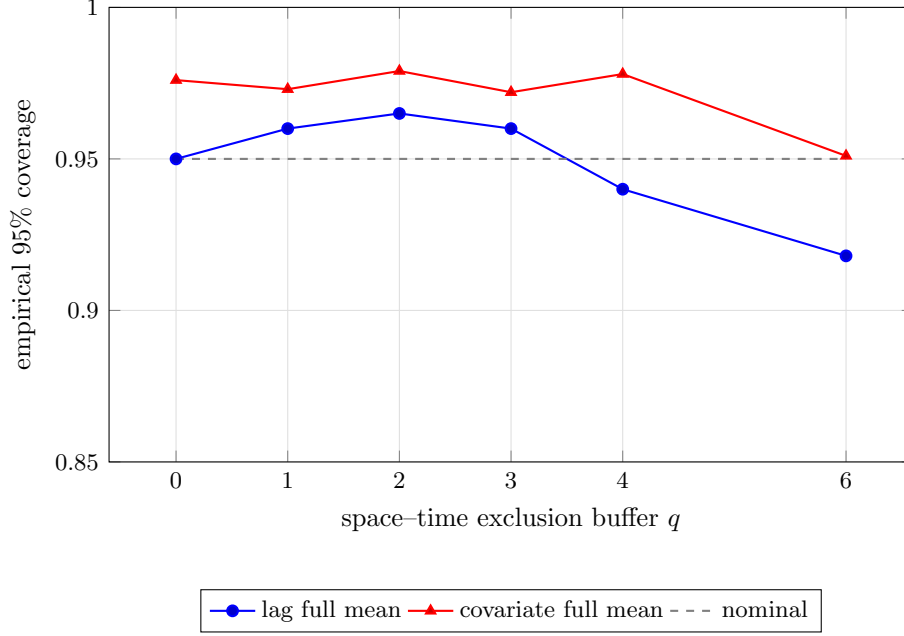


Figure 1: Purge-radius sensitivity.

capped radii, $q = \min\{\lceil c_q \log(TN)/|\log \bar{\rho}_{NT}| \rceil, q_{\max}\}$. A fixed small cap (here $q_{\max} = 3$) is a tuning experiment, not a substitute for the decay restrictions in (5.2); it is theorem-covered only along sequences for which the cap itself grows sufficiently fast and retention is preserved. Target-specific caps can be diagnostically useful, especially for minimal-variance means, but any interval based on a shorter-than-required window must be labelled finite-sample/sensitivity rather than asymptotically validated by the purge theorem. Table 18 reports the corresponding coverage trade-off.

B.4 Theorem-justification experiments

Four further experiments target specific results of the paper directly.

Rank consistency. The cross-fitted rank criterion recovers the true rank $(1, 1, 1)$ with probability 0.95 at $(50, 50)$ and 1.00 at both $(100, 100)$ and $(200, 200)$: the finite-sample image of the selection-consistency guarantee.

Why the scattered, purged folds are necessary. On a contiguous (time-block) fold the buffered training information map is numerically singular — minimum eigenvalue -9×10^{-16} , condition number 4.5×10^{12} — whereas the scattered cell-level design is well conditioned (minimum eigenvalue 0.039, condition number 123). The scattered geometry is not a stylistic choice; it is what makes the debiasing-weight solve well posed.

Cross-sectional dependence. Under a decaying spatial-AR(1) error structure (compliant with Assumption 3, with no pervasive common factor), the diagonal interval for the lag full mean

under-covers (0.87) while the period-cluster diagnostic interval recovers more of the nominal level in this finite design (0.90); for the covariate full mean, which the lag-driven dependence does not load on, both are 0.94. In this finite DGP, the period-cluster diagnostic standard error moves coverage toward nominal where contemporaneous dependence bites, with the residual gap closing as N grows. This is simulation evidence for this design, not a general theorem: the formal theorem-backed object remains the spatial-kernel standard error with a metric and bandwidth satisfying (7.4).

B.5 Stress tests

All sweeps below fix the panel at $T = N = 100$ with $R = 400$ replications under the decaying cross-sectional-dependence DGP, and report coverage and RMSE of the lag full mean — the hardest, dependence-dominated target. Because (100, 100) is a moderate finite sample under strong dependence, the lag full mean already under-covers there (around 0.88 diagonal, 0.92 spatial-kernel); what each stress test isolates is the *sensitivity* to the design choice, not the absolute level, which improves with N (Table 21).

Fold-floor sensitivity. The theory requires a slowly growing J_{NT} . The finite-sample implementation uses the floor rule (5.3), and Table 19 reports sensitivity to nearby realized fold counts. The retained share increases with J , as smaller validation folds create smaller buffers around held-out cells. The table should be read as a robustness check for the finite-sample floor, not as evidence for a separate fixed- J asymptotic theorem.

Table 19: Fold-floor sensitivity and retained training share (lag full mean, $T = N = 100$).

J	cov. (diag.)	cov. (spatial-kernel)	RMSE	retained share
6	0.887	0.920	0.098	0.582
8	0.885	0.922	0.081	0.673
10	0.885	0.922	0.068	0.732
12	0.880	0.907	0.079	0.772

Notes: “Retained share” is $\tau_{\text{nv}} = J^{-1} \sum_{j=1}^J |\mathcal{I}_{-j}^{\text{buf}}| / (NT - |\mathcal{I}_j|)$, the average fraction of candidate non-validation cells retained after buffering. The table evaluates finite-sample fold floors; it is not a fixed- J asymptotic theorem.

No-buffer and insufficient-buffer failure. A companion robustness calculation re-runs the same panel under four fold designs. The scattered, space–time-buffered design is well conditioned (buffered training-Gram minimum eigenvalue $+0.04$, condition number 2.3×10^2), whereas the contiguous time-block control is singular (minimum eigenvalue -9.7×10^{-16} , condition number 9.4×10^{12} , with units that receive zero training support) and its intervals collapse (coverage 0.03). Among scattered designs the temporal purge is the workhorse; the spatial radius is costly at this size (it cuts the retained share from 0.70 to 0.34 and inflates RMSE roughly fivefold), which is why the headline runs set it to zero. This is the empirical demonstration of the paper’s central design contribution.

Rank misspecification. Table 20 sets the true lag rank to two and estimates with lag rank one (under), two (correct), and three (mild over). Under-ranking is the dangerous direction: it biases the lag mean and gives the lowest coverage (0.75). The correct rank improves coverage (0.80), and mild over-ranking improves it further (0.85) at a substantial RMSE cost. The asymmetry — under-ranking biased, over-ranking merely noisy — is why the empirical specification errs toward retaining, rather than dropping, interactive factors.

Table 20: Rank misspecification (true lag rank 2, lag full mean, $T = N = 100$).

estimated lag rank	cov. (diag.)	cov. (spatial-kernel)	RMSE
1 (under)	0.748	0.772	0.108
2 (correct)	0.802	0.812	0.150
3 (over)	0.845	0.858	0.292

Cross-sectional dependence and the variance estimators. Table 21 compares the diagonal and spatial-kernel intervals as N grows under the spatial-AR(1) within-date error DGP. The spatial-kernel interval improves on the diagonal one at every size and moves toward nominal as N grows, while the covariate full mean — which does not load on the lag-driven dependence — is at nominal throughout.

Table 21: Cross-sectional-dependence stress (lag full mean coverage).

$T = N$	cov. (diagonal)	cov. (spatial-kernel)
120	0.870	0.897
200	0.880	0.923
300	0.837	0.873

Notes: $R = 300$ replications; the $N = 300$ row is within Monte Carlo error of the $N = 200$ row (a coverage estimate at $R = 300$ carries a standard error of about 0.017). The spatial-kernel estimator is the theorem-backed object here because the DGP supplies a metric and the bandwidth matches it.

C Empirical Appendix

This section provides additional details on empirical data construction, robustness checks, and diagnostics.

C.1 Data construction

We apply the estimator to two real, publicly available panels that exercise different regimes of the theory. The first is a panel of monthly regional unemployment rates in which the object of interest is the *idiosyncratic* persistence that survives after a pervasive common business-cycle factor is removed; the data-driven rank criterion selects a single dynamic order and a small number of interactive factors. The second is a very large regional housing panel in which the cross-sectional and time

dimensions are both large enough to deliver high finite-sample precision. Each panel is estimated in three specifications — a full-sample baseline, a baseline restricted to the shorter window on which local covariates are available, and a covariate-augmented fit on that same restricted window — so that the effect of *restricting the sample* is cleanly separated from the effect of *adding covariates*. In both cases the procedure is exactly the cross-fitted debiased one-step of [Sections 5 and 6](#) applied to the heterogeneous low-rank model in its AR(2) form, $\tilde{y}_{it} = a_{1,0,it}\tilde{y}_{i,t-1} + a_{2,0,it}\tilde{y}_{i,t-2} + h_{0,it} + u_{it}$, where the second lag is the predetermined generated regressor, $a_{1,0,it}$ and $a_{2,0,it}$ are the autoregressive coefficients, and $\mathbf{H}_0$ is the low-rank interactive component. The notation b and β is reserved for observed-covariate slopes. Every number in [Tables 11, 22, 23, 25 and 26](#) is produced by the companion package from the public data and written directly to the table files; none is entered by hand. Each target is accompanied by two reported standard errors: a *diagonal cell-heteroskedastic sandwich* standard error — the heteroskedasticity-robust estimator that weights the squared cross-fitted scores and treats cells as conditionally independent (the analog of a diagonal-type sandwich for the debiased moment, not the classical OLS diagonal estimator), which we call the “diagonal” standard error for short — and a by-period clustered sensitivity standard error. The latter is a by-period clustered sensitivity interval, not theorem-backed in the present framework and not guaranteed conservative. In empirical tables without a credible spatial metric, the theorem-backed interval is the diagonal cell-heteroskedastic interval only under the conditional-independence special case. For the metropolitan applications we also supply a geographic metric — metro centroids from the Census CBSA gazetteer — and admissible Bartlett bandwidths, and compute the geographic spatial-kernel interval directly for both applications ([Table 11](#)). The geographic spatial-kernel standard errors are close to the diagonal standard errors, which suggests that residual spatial dependence is limited at the reported bandwidths. The by-period clustered interval remains a wider sensitivity diagnostic; it is not theorem-backed in this framework and is not generally guaranteed conservative. The table headers abbreviate it as “period-cluster sens. s.e.” and “period-cluster sens. CI”. Thus the recommended hierarchy for empirical inference is: diagonal sandwich as the conditional-independence benchmark; geographic spatial-kernel standard errors as the theorem-aligned dependence-robust implementation once metro coordinates and a bandwidth are fixed; and period-cluster intervals only as sensitivity diagnostics. The recovered coefficient heterogeneity, its evolution over time, and its geographic distribution appear in [Figures 2 to 5](#).

This subsection records, for replication, exactly which series enter each panel, how they are made model-ready, and how the two tuning constants — the number of folds and the length of the forward-exclusion window — are set. Every step is performed by the companion package’s data scripts on the raw public downloads; no series is edited by hand.

Sources, frequency, and coverage. The housing panel is a monthly metropolitan house-value index for the top and bottom price tiers of each metro, which we stack as two units per metro. Its raw series runs from January 2000 through May 2026; with two lags, the usable post-lag estimation sample has $T = 315$ monthly observations. The labor panel is a monthly, not-seasonally-adjusted

metropolitan unemployment-rate panel. The full raw labor panel also runs through May 2026, and the headline unemployment specification below uses the covariate-complete January 2005–December 2024 window. The exact raw-file vintages, download dates, and source locations are recorded in the replication package’s data manifest. The covariates used in specification C are metropolitan *building-permit growth* (Census Building Permits Survey, monthly), *population growth* (Census annual estimates), and *real-GDP growth* (Bureau of Economic Analysis annual metropolitan GDP). The three covariates are jointly complete only over 2005–2024, which is exactly why the restricted specifications B and C use that window: B holds the covariate sample fixed while dropping the covariates, so that the move A→B isolates the effect of shortening the sample and B→C the effect of adding the covariates.

Stationarity transforms. House-price *levels* carry a unit root, so each tier series is converted to monthly log growth (a one-step log-difference); the per-series augmented Dickey–Fuller screen of [Appendix C.2](#) supports this as the lightest transform that delivers stationarity, and the rare already-stationary series is left in levels. Unemployment *rates* are bounded and economically mean-reverting, so we model them in levels after metro-specific deseasonalization and removal of common-cycle movements through the interactive block. Because the metropolitan unemployment series is not seasonally adjusted, we remove its within-year seasonality metro-by-metro by subtracting each area’s twelve month-of-year means and adding back its grand mean. This deseasonalization is *level-preserving*: it strips the calendar pattern without altering the persistence dynamics or the cross-metro level differences that define the groups, and it avoids importing the model-based revisions that an official seasonal-adjustment filter would inject. A small number of short gaps (at most six consecutive months) are filled by linear interpolation; metros with longer gaps are dropped.

Covariate construction and frequency matching. Permits, population, and GDP are published by county or already by CBSA; we aggregate counties to metropolitan areas with the 2023 OMB delineation file and align them to each panel by CBSA code. The two annual series (population, GDP) are interpolated linearly to a monthly grid before differencing, and all three covariates are expressed as twelve-month log growth rates, so that they share the units and the seasonal treatment of the outcome. Each covariate is winsorized at its first and ninety-ninth percentiles — the permit-growth series in particular has extreme local spikes that would otherwise ill-condition the first-stage decomposition — and standardized, and each enters the model *predetermined*, lagged one month relative to the outcome it helps predict. Payroll employment is deliberately *excluded* from the labor-panel covariates: employment and the unemployment rate are linked by a near-accounting identity, so employment growth would be collinear with the very own-dynamics the estimator is measuring and would mechanically distort them.

Group definitions. The housing contrast compares the two *price tiers*: top-tier (more expensive) versus bottom-tier segments of the same metros, a split that is given by the data rather than estimated. The labor contrast splits metros at the *median of their sample-average unemployment*

rate, separating structurally tight from structurally slack labor markets. In both cases the grouping variable is a time-invariant characteristic of the unit, so it does not use the autoregressive dynamics being estimated, and it yields two roughly equal groups for the between-group target.

Folds J and buffer windows. The implementation uses scattered cell-level folds together with a forward time buffer and, when relevant, a spatial/unit buffer. For a validation cell (i, t) , the time buffer removes training cells whose time index falls within the forward window q_T , while the spatial buffer removes nearby units within the unit-distance window r_N . In the Monte Carlo lattice design, DGPs 2–3 use the spatial metric $d_N(i, j) = |i - j|$; in the empirical applications, geographic spatial-kernel calculations use great-circle CBSA centroid distances when coordinates are available.

The asymptotic theory uses a slowly growing number of scattered folds $J = J_{NT}$. The finite-sample implementation uses the fold-floor rule (5.3), with $J_{\min} = 10$ as the Monte Carlo baseline. For each run, we report the realized fold count and the retained training shares after validation folds and buffers are removed. These diagnostics document the practical implementation of the growing- J_{NT} rule; they are not evidence for a separate fixed- J theorem.

Cross-sectional dependence and the two standard errors. Metropolitan areas do not move independently: national shocks — interest-rate changes, recessions, the pandemic — hit them together within the same month. The diagonal standard error treats the units as independent and therefore understates uncertainty whenever such comovement is present. The theory-covered standard error is the spatial-kernel estimator of Theorem 5 when the score field is indexed by a credible metric and satisfies the conditional spatial-mixing envelope. When such a metric is not imposed, we report by-period clustered standard errors only as sensitivity intervals. These clustered standard errors are not the primary object of Theorem 5, which is stated for spatially weak dependent arrays indexed by a metric. They are also not treated as a validated saturated-kernel theorem in this paper, because the feasible replacement rate would have to hold at the most demanding bandwidth. The gap between diagonal and by-period clustered standard errors is therefore reported as an empirical diagnostic of contemporaneous comovement; the by-period clustered interval is not theorem-backed in the present framework and is not guaranteed conservative. With a credible geographic metric and a bandwidth satisfying (7.4), the geographic spatial-kernel interval is the theorem-aligned dependence-robust interval; we supply metro centroids and report it for both applications in Table 11. Its closeness to the diagonal suggests limited residual spatial dependence at the reported bandwidths, while the by-period clustered interval remains a non-theorem-backed sensitivity diagnostic.

C.2 Empirical robustness and diagnostics

C.2.1 Idiosyncratic persistence in a regional unemployment panel

We use a monthly metropolitan unemployment-rate panel, taking the *monthly*, not-seasonally-adjusted unemployment rate of metropolitan statistical areas. Working at the monthly rather than

the annual frequency lengthens the time dimension from a few dozen to several hundred periods, which is what the dependence-robust theory needs; the cost is a pronounced within-year seasonal pattern, which we remove metro-by-metro by subtracting each area’s month-of-year means and adding back its grand mean. This is a level-preserving deseasonalization: it strips the calendar seasonality without touching the persistence dynamics or the cross-metro level differences that define the groups. A handful of short gaps are filled by linear interpolation. The headline sample, specification B, is 2005–2024 ($T = 240$ months) — the window on which the metropolitan covariates introduced below are complete — and comprises the $N = 315$ metros that match those covariates; the longer 2000–2026 sample ($N = 388$, $T = 317$) is reported as specification A. The reported T is the usable post-lag estimation sample; the raw series includes the required initial lag observation.

Unemployment rates are bounded and economically mean-reverting, so we work in *levels* after metro-specific deseasonalization and let the interactive block $\mathbf{H}_0$ absorb the strong common business-cycle component, which alone accounts for roughly three-quarters of the panel variance. The autoregressive coefficients then measure the persistence of the *idiosyncratic* unemployment dynamics that remain after the common comovement is removed. At the monthly frequency the two own-lags are nearly collinear — consecutive monthly rates move almost in lockstep — so the second lag is only weakly identified; consistent with this, the cross-fitted criterion of [Section 5](#) sets the second-lag rank r_2 to *zero*. In the headline specification B no observed covariates enter, so $r_2 = 0$ should not be read as a zero covariate-slope rank; covariates enter separately in specification C. The idiosyncratic dynamics are therefore a heterogeneous AR(1) — the monthly counterpart of the AR(1) the same criterion selects from the annual data — so we report the second lag as structurally absent, omit its row, and read cumulative persistence as the lag-1 mean a_1 , with a_2 structurally zero. For the interactive block, the cross-fitted criterion selects a *single* factor, $(r_1, r_2, r_H) = (1, 0, 1)$ ([Table 24](#)), which we adopt as the headline specification — the data-driven, theorem-aligned choice, so the reported intervals are exactly the output of the rank-selection theorem. A single factor leaves more of the strong common business cycle — which accounts for roughly three-quarters of the panel variance — in the residual comovement, which is precisely why the period-cluster standard errors below are sizeable. As a robustness check we also report the more heavily factor-absorbing $(1, 0, 3)$: as [Table 24](#) shows, the average persistence is essentially invariant to the interactive rank, staying within $[0.91, 0.94]$ as r_H ranges over $\{1, 2, 3, 4\}$, so the conclusions below do not depend on this choice. The two groups split the metros at the median of their sample-average unemployment rate, separating structurally tight from structurally slack labor markets.

On the headline sample the average idiosyncratic persistence is $+0.91$ ([Table 22](#)): metro unemployment, net of the common cycle, is highly persistent month to month. Under the diagonal cell-heteroskedastic sandwich standard error this is precisely estimated ($[0.85, 0.97]$) under the conditional-independence special case. Under the period-cluster sensitivity diagnostic, which is not the spatial-kernel theorem-backed object and is not guaranteed conservative, the interval widens to roughly $[0.78, 1.03]$ and no longer excludes a unit root — a direct read-out of how much common-cycle comovement the parsimonious single-factor specification leaves in the residuals. The high-

and low-unemployment groups carry similar persistence (0.87 versus 0.91), a difference that is not statistically distinguishable from zero under both the diagonal interval and the period-cluster sensitivity diagnostic. At the monthly frequency, then, the heterogeneity in own-persistence *across* the unemployment-level groups is modest; the economically salient finding is the high overall persistence of the idiosyncratic component and, as the robustness panel shows, its stability across sample and covariates.

Table 22: Idiosyncratic persistence in metro unemployment.

Target	Estimate	s.e.	period-cluster sens.	s.e.	95% CI (diag.)	95% period-cluster sens. CI
Lag-1 persistence, global mean	+0.909	0.033		0.063	[+0.845, +0.972]	[+0.784, +1.033]
Lag-1 persistence, high-unemployment metros	+0.867	0.053		0.103	[+0.764, +0.970]	[+0.665, +1.069]
Lag-1 persistence, low-unemployment metros	+0.915	0.029		0.057	[+0.857, +0.972]	[+0.803, +1.026]
Lag-1 persistence, high–low contrast	−0.012	0.010		0.013	[−0.031, +0.007]	[−0.038, +0.014]
Long-run multiplier $1/(1-a_1)$	+11.0	3.9		—	[+3.3, +18.6]	—

Notes: Headline specification B: $N = 315$ metropolitan areas matched to the covariate window, $T = 240$ monthly observations, 2005–2024. Outcome is the not-seasonally-adjusted unemployment rate in levels, deseasonalized metro-by-metro by subtracting each area’s month-of-year means and adding back its grand mean. Unemployment rates are bounded and economically mean-reverting, so we model them in levels after metro-specific deseasonalization and removal of common-cycle movements through the interactive block. Heterogeneous low-rank model with a *single*-factor interactive block, ranks $(1, 0, 1)$ — the cross-fitted criterion’s selection (Table 24) and our headline specification. The second-lag rank is zero — whose row is therefore omitted — so the idiosyncratic dynamics are a heterogeneous AR(1) and cumulative persistence equals the lag-1 mean. The more heavily factor-absorbing $(1, 0, 3)$ is reported only as a robustness row in Table 24; the persistence is essentially unchanged. The high- and low-unemployment groups split the metros at the median of their sample-average unemployment rate. The high–low contrast is a *separately* debiased target with its own Riesz representer, so it need not equal the arithmetic difference of the two debiased group means. To reconcile the three relevant quantities: the plug-in group difference is +0.001 (the first-stage cell means barely differ); the difference of the two separately debiased group means is −0.048, but with standard error ≈ 0.060 (it differences two imprecise pooled means); and the separately debiased contrast is −0.012 with standard error 0.010. These are statistically consistent — the noisy group difference −0.048 (0.060) comfortably covers −0.012 — and the contrast is far more precise because the common business-cycle component cancels in the difference direction, so its Riesz representer is better conditioned (the faster contrast rate of Proposition 10). We therefore report the separately debiased contrast as the preferred estimate. “s.e.” is the diagonal cell-heteroskedastic sandwich standard error; “period-cluster” is the within-period by-period clustered sensitivity standard error. It is reported as a diagnostic for contemporaneous metro comovement; it is not the primary spatial-kernel object of Theorem 5, is not theorem-backed in the present framework, and is not guaranteed conservative unless additional saturated-bandwidth replacement conditions are verified. Because a_1 is close to one, the long-run multiplier $1/(1-a_1)$ is large and imprecisely estimated; it is a nonlinear transform reported only as a descriptive summary, with its diagonal interval only, and is not one of the theorem-backed linear targets.

Table 23 traces the persistence across the three specifications and is the panel’s main robustness evidence. Restricting the full 2000–2026 sample to the covariate window (A→B) leaves the estimate essentially unchanged (0.912 to 0.909). Adding the two predetermined local covariates, metropolitan population growth and real-GDP growth (B→C), moves it only modestly (to 0.937), and both slope coefficients are economically small and statistically insignificant (population +0.008, GDP −0.003). This is broadly the pattern the structure anticipates: because the interactive block already absorbs the common local-economy movements, *observed* local fundamentals carry almost no additional predictive content for the idiosyncratic dynamics, and the persistence estimate is robust to their inclusion. The sample-average companion radius stays below one in every specification, so the reported linear target remains sub-unit in the headline and robustness windows. This sample-average radius should not be read as a certificate of the pointwise uniform AR(1) stability bound

in Assumption 1. The period-by-period coefficient-path diagnostic in Figure 4 exceeds one in a minority of months: most excursions are small and occur outside 2020, while the large March–April 2020 spike is the clearest pandemic break. Because that path is a cross-sectional average of noisy first-stage coefficient estimates around a near-integrated true persistence, excursions above one are finite-sample stability stress diagnostics rather than direct evidence that the true pointwise envelope violates Assumption 1. We therefore add a pre-pandemic robustness specification (spec D, 2005–2019, the same matched metros). Excluding the COVID period *raises* the sample-average idiosyncratic persistence to 0.988 (s.e. 0.008) — near-integrated, with the upper confidence limit marginally touching one — and the cross-fitted rank criterion drops the interactive factor entirely, selecting $(1, 0, 0)$ rather than the full sample’s $(1, 0, 1)$. The common pandemic movement is therefore absorbed by the interactive block once the COVID window is included, lowering the *idiosyncratic* sample-average persistence from its near-integrated pre-COVID level to 0.909. The sample and group-average targets — the objects of inference — stay below one in all four specifications.

Table 23: Unemployment persistence across specifications.

Spec.	Sample	N	T	Lag-1 mean (s.e.)	Lag-2	Cum. a_1+a_2	Comp. radius
A. Full sample	2000–2026	388	317	+0.912 (0.029)	—	+0.912	0.912
B. Covariate window	2005–2024	315	240	+0.909 (0.033)	—	+0.909	0.909
C. + covariates	2005–2024	315	240	+0.937 (0.027)	—	+0.937	0.937
D. Pre-COVID	2005–2019	315	180	+0.988 (0.008)	—	+0.988	0.988

Notes: Heterogeneous AR(1) (headline ranks $(1, 0, 1)$; second lag structurally absent) of the deseasonalized monthly unemployment rate. Specification A is the full monthly sample; the reported T is the usable post-lag estimation sample, with the raw series including the required initial lag observation. B restricts to the 2005–2024 window and the $N = 315$ metros on which the covariates are complete; C adds two predetermined covariates — metropolitan population growth and real-GDP growth (winsorized at the 1st/99th percentiles and standardized) — to that same sample, so that $A \rightarrow B$ isolates the sample-restriction effect and $B \rightarrow C$ the covariate effect. Specification D is the pre-COVID 2005–2019 sub-sample on the same $N = 315$ metros, estimated at the headline ranks $(1, 0, 1)$ for comparability with B; its own cross-fitted criterion selects $(1, 0, 0)$, dropping the interactive factor that the pandemic common movement induces in the full sample. In C the slope coefficients are population growth +0.008 (0.009) and real-GDP growth -0.003 (0.009), both statistically insignificant. Standard errors in parentheses are the diagonal sandwich; the reported radius is the companion radius of the sample-average autoregressive coefficients and should not be interpreted as a pointwise uniform stability certificate. Employment growth is deliberately *not* used as a covariate: it is a labor-market identity with the unemployment rate and would mechanically confound the own-dynamics.

C.2.2 House-price momentum and a price-tier contrast

We use a metropolitan house-value index, taking the top and bottom price-tier series for each metro and stacking them as separate units in a single panel, so that $N = 610$ units (305 metros $\times$ two tiers) enter estimation. The raw housing series runs from January 2000 through May 2026. With two lags, the usable post-lag estimation sample has $T = 315$ monthly observations. The outcome is monthly log price growth, a stationary transform, and the tier indicator defines the two groups for the contrast. This is the large- N , large- T regime in which the $\sqrt{TN}$ theory predicts high finite-sample precision. The cross-fitted criterion selects $(r_1, r_2, r_H) = (1, 1, 1)$.

Table 25 reports the estimates. House-price growth exhibits strong, very precisely estimated

Table 24: Rank selection and interactive-rank robustness, unemployment.

(a) cross-fitted rank criterion (top 5)			(b) persistence vs. interactive rank r_H			
(r_1, r_2, r_H)	CV loss	criterion	r_H	lag-1	cum. a_1+a_2	radius
$(1, 0, 1)^*$	0.119	0.128	1*	+0.909	+0.909	0.909
$(1, 1, 0)$	0.159	0.168	2	+0.935	+0.935	0.935
$(0, 1, 1)$	0.174	0.182	3	+0.940	+0.940	0.940
$(0, 2, 0)$	0.232	0.240	4	+0.912	+0.912	0.912
$(1, 0, 0)$	0.236	0.240				

Notes: Panel (a): the cross-fitted rank criterion of [Section 5](#) over the candidate box, on the headline sample (B). “CV loss” is the cross-fitted prediction loss; “criterion” adds the rank penalty; the starred row $(1, 0, 1)$ is the selected, most parsimonious model — a single interactive factor — and is our headline specification. Panel (b): debiased full-mean lag-1 persistence, cumulative persistence, and companion radius as the interactive rank r_H is varied with the two lag ranks held at $(r_1, r_2) = (1, 0)$; the starred row $r_H = 1$ is the same headline target and sample as [Tables 22](#) and [23](#). The persistence is stable across r_H (within $[0.91, 0.94]$), so the more heavily factor-absorbing $(1, 0, 3)$, reported here only as a robustness check, yields the same conclusions as the headline single-factor $(1, 0, 1)$ specification. The notation r_1, r_2 refers to the first- and second-lag ranks; covariate-slope ranks, when present in specification C, are separate components of the general rank vector.

short-run momentum: the AR(2) lag-1 coefficient is +1.250 (period-cluster sensitivity interval $[1.239, 1.261]$) and the lag-2 coefficient is -0.419 (period-cluster sensitivity interval $[-0.431, -0.407]$). The lag-1 coefficient exceeds one, but the sample-average AR(2) summary has companion spectral radius $\hat{\rho}_{pt} \approx 0.65$ — the companion eigenvalues are a complex-conjugate pair with modulus $\sqrt{0.419} \approx 0.65$; equivalently, the roots of the lag polynomial $1 - 1.250z + 0.419z^2$ have modulus $1/\sqrt{0.419} \approx 1.55$, outside the unit circle. This is a statement about the average dynamic summary and does not certify the pointwise uniform stability bound for the heterogeneous AR(2) coefficient matrices. The cumulative persistence $a_1 + a_2 = 0.831$ (period-cluster sensitivity interval $[0.827, 0.834]$) is well below one. Prices thus display momentum at one month’s horizon that partially reverses at the next, the well-documented house-price momentum pattern. Cumulative persistence is a *linear* combination of the lag-coefficient means, hence an [Assumption 6](#)-regular target covered by [Theorem 5](#) when paired with the spatial-kernel studentizer or the diagonal conditional-independence special case. The period-cluster interval in [Table 25](#) is reported as the same sensitivity diagnostic used for the lag means, not as a separate theorem-backed studentizer. The nonlinear long-run multiplier and the full response-path intervals produced by the package are *not* linear targets; their dependence-robust inference would require a separate nonlinear-response bootstrap or Delta-method theory and is not part of the main empirical claims.

The price-tier contrast is small and only marginally significant under the by-period clustered sensitivity diagnostic: top-tier (more expensive) segments have a lag-1 coefficient about +0.018 larger than bottom-tier segments, with a diagonal interval $[0.008, 0.027]$ that excludes zero but a wider period-cluster interval $[-0.005, 0.040]$ that marginally includes it. We therefore read the tier difference as suggestive rather than decisive under this non-theorem diagnostic. The intervals here are two orders of magnitude tighter than in the unemployment panel, the empirical counterpart of the simulation’s large-panel cells in which the entry, slope, and contrast coverage is near nominal and the standard errors contract at the $\sqrt{TN}$ rate (the autoregression-coefficient *mean* coverage, as

in Table 14, requires the capped exclusion window of Remark 3).

Table 25: House-price dynamics in the metro house-price panel.

Target	Estimate	s.e.	period-cluster sens.	s.e.	95% CI (diag.)	95% period-cluster sens. CI
Lag-1 coefficient, all segments	+1.250	0.004		0.006	[+1.243, +1.257]	[+1.239, +1.261]
Lag-2 coefficient, all segments	-0.419	0.004		0.006	[-0.427, -0.412]	[-0.431, -0.407]
Lag-1 coefficient, top price tier	+1.259	0.005		0.010	[+1.250, +1.268]	[+1.240, +1.278]
Lag-1 coefficient, bottom price tier	+1.241	0.004		0.006	[+1.233, +1.249]	[+1.229, +1.254]
Lag-1 coefficient, top-bottom contrast	+0.018	0.005		0.012	[+0.008, +0.027]	[-0.005, +0.040]
Cumulative persistence ($a_1 + a_2$)	+0.831	0.002		0.002	[+0.827, +0.834]	[+0.827, +0.834]
Long-run multiplier $1/(1-a_1-a_2)$	+5.90	0.06		—	[+5.78, +6.02]	—

Notes: Headline specification A (full sample): $N = 610$ metro-tier units (305 metros $\times$ two price tiers), $T = 315$ months; outcome is monthly log price growth. Heterogeneous low-rank AR(2) with a rank-one interactive block; ranks $(1, 1, 1)$ selected by the cross-fitted criterion. The contrast is the top-minus-bottom price-tier difference, estimated within the single panel. “s.e.” is the diagonal standard error; “period-cluster” is the within-period by-period clustered sensitivity standard error, reported only as a diagnostic and not as a theorem-backed or guaranteed-conservative interval. Cumulative persistence is the linear combination $a_1 + a_2$, reported with both standard errors. The long-run multiplier $1/(1 - a_1 - a_2)$ is a nonlinear transform reported only as a descriptive summary and not a theorem-backed linear target, so only its diagonal interval is shown and it is not part of the theorem-backed linear-target claims.

Table 26 supports the conclusion that the momentum pattern is not an artifact of the sample period or of omitted local fundamentals. Restricting to the covariate window ($A \rightarrow B$) leaves cumulative persistence unchanged at 0.831 and the lag-coefficient pattern intact; adding building-permit growth, population growth, and real-GDP growth as predetermined covariates ($B \rightarrow C$) moves cumulative persistence only to 0.801, with all three slope coefficients statistically indistinguishable from zero (building permits +0.010, population +0.001, GDP -0.001). The lag-1 and lag-2 means and the sub-unit companion radius are stable throughout: the one-month momentum and its partial reversal are properties of the price dynamics themselves, not reflections of contemporaneous construction or income growth. The pre-pandemic robustness specification (spec D, 2000–2019) supports the conclusion that the COVID interval does not drive the result: cumulative persistence is essentially unchanged (0.832 versus 0.831) and the cross-fitted criterion selects the same ranks $(1, 1, 1)$. In contrast to the unemployment panel — where excluding COVID drops the interactive factor and raises idiosyncratic persistence — the house-price momentum structure is insensitive to the pandemic window, consistent with house prices being far less disrupted than labor markets in 2020.

C.2.3 Estimator diagnostics

Geographic spatial-kernel standard errors. The theorem-aligned geographic spatial-kernel standard errors are reported in the main text in Table 11. The table uses great-circle distances between CBSA centroids and Bartlett bandwidths of 300 and 600 km. The geographic standard errors are close to the diagonal standard errors, which suggests that residual spatial dependence is limited at the reported bandwidths. The by-period clustered interval remains a wider sensitivity diagnostic; it is not theorem-backed in this framework and is not generally guaranteed conservative. For housing, the geographic calculation is on the coordinate-matched subsample, so it is a theorem-aligned robustness check for spatial dependence rather than the exact standard error for the full

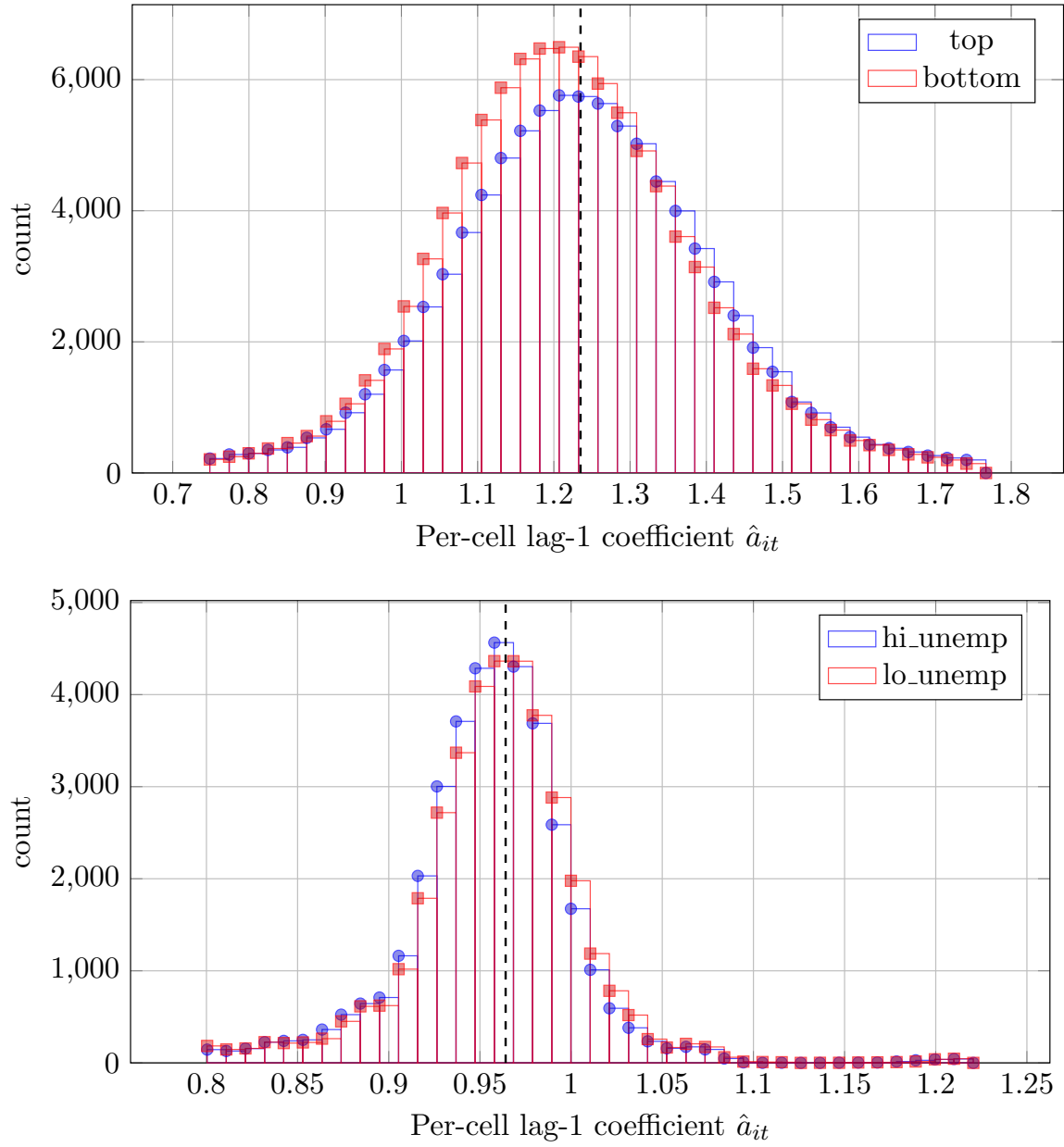


Figure 2: Heterogeneity of the estimated lag-1 coefficient.

Notes: Histogram of the per-cell estimated lag-1 coefficient $\hat{a}_{it}$ across all metro-time cells, split by group (top: house-price tiers; bottom: metro unemployment groups). The dashed line marks the global mean. The spread is the coefficient heterogeneity that the low-rank structure makes estimable.

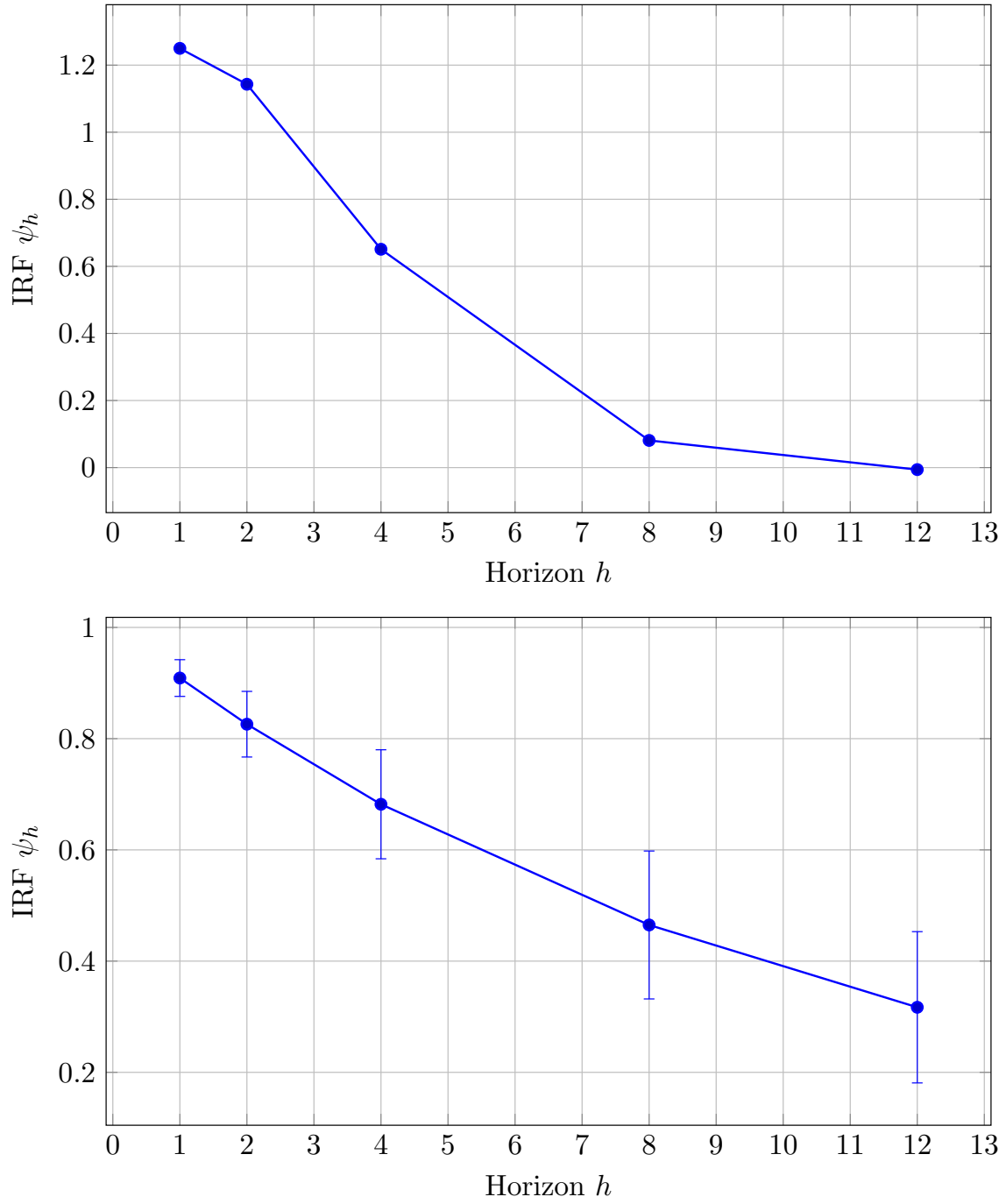


Figure 3: Estimated impulse-response paths.

Notes: Estimated impulse-response path ψ_h against horizon h (top: housing; bottom: metro unemployment), with pointwise 95% bands. The bands are the baseline diagonal delta-method intervals; dependence-robust response-path bands would require a separate nonlinear-response bootstrap or Delta-method analysis and are not part of the main claims.

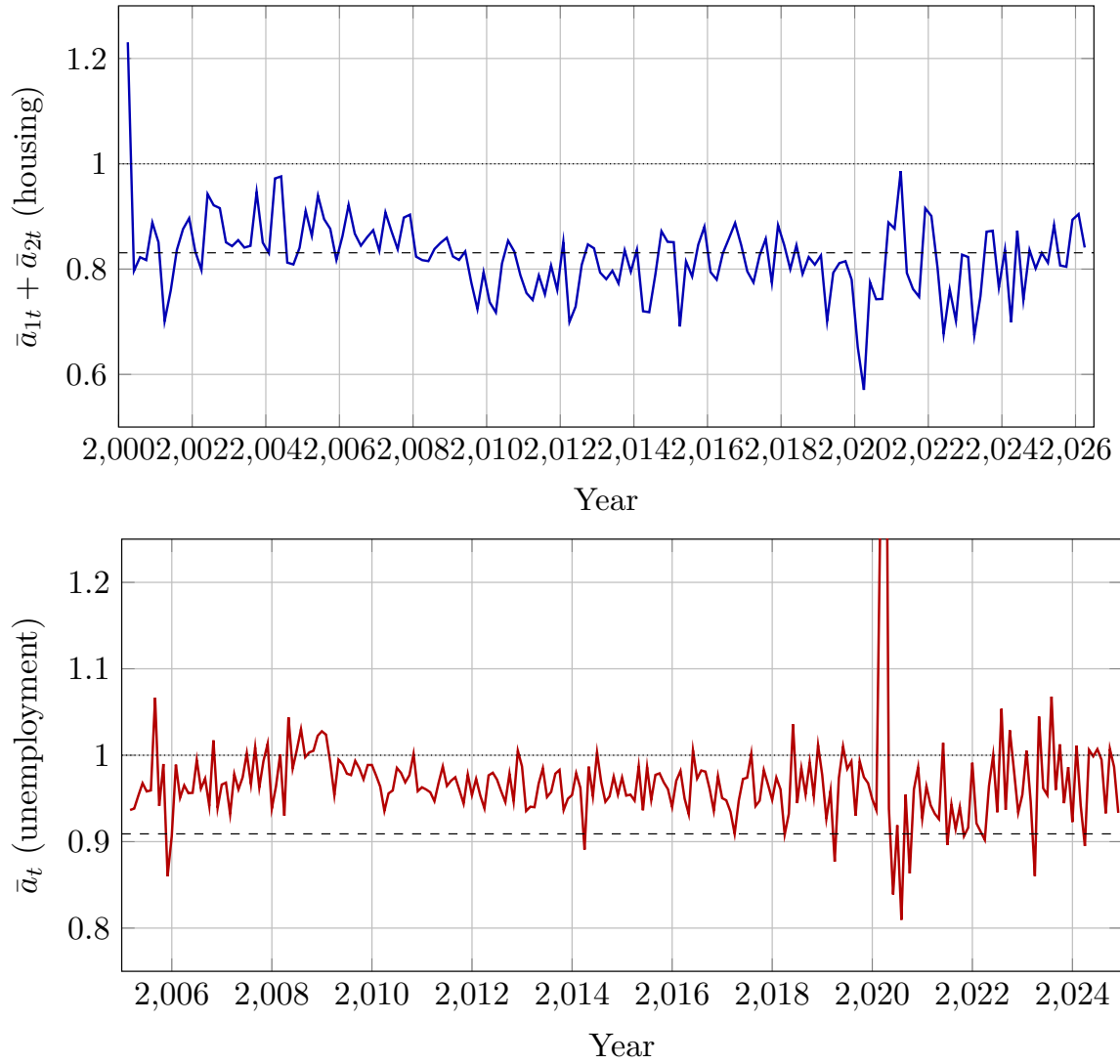


Figure 4: Time path of the average dynamic persistence.

Notes: Cross-sectional average of the estimated dynamic persistence in each period — the cumulative persistence $\bar{a}_{1t} + \bar{a}_{2t}$ of the period's lag means for housing (AR(2), top) and the lag-1 mean $\bar{a}_t$ for unemployment (AR(1), bottom); the plotted path is the first-stage coefficient matrix average, so its level gap from the dashed line is the debiasing correction. The dashed line is the full-sample debiased headline value (0.831 housing, 0.909 unemployment); the dotted line is the stationarity boundary at one. The time-varying coefficient matrix recovers an economically interpretable profile: housing momentum compresses sharply at the onset of the 2020 pandemic, and the unemployment coefficient-path diagnostic has several excursions above one, with the large March–April 2020 spike standing out from smaller pre- and post-pandemic excursions. The unemployment curve is a cross-sectional average of estimated lag-1 coefficients, not a pointwise stability certificate; Assumption 1 constrains the true companion products, while the reported inference targets are sample and group-average linear summaries. A homogeneous model would force each path to a single horizontal line.

Table 26: House-price dynamics across specifications.

Spec.	Sample	N	T	Lag-1 (s.e.)	Lag-2 (s.e.)	Cum. a_1+a_2	Comp. radius
A. Full sample	2000–2026	610	315	+1.250 (0.004)	−0.419 (0.004)	+0.831	0.648
B. Covariate window	2005–2024	392	239	+1.282 (0.005)	−0.451 (0.005)	+0.831	0.672
C. + covariates	2005–2024	392	239	+1.279 (0.017)	−0.478 (0.018)	+0.801	0.691
D. Pre-COVID	2000–2019	621	239	+1.198 (0.005)	−0.366 (0.005)	+0.832	0.605

Notes: Heterogeneous low-rank AR(2) (ranks (1, 1, 1) in A and B; (1, 1, 1, 1, 1, 1) in C, one rank-one coefficient matrix per covariate) of monthly log house-price growth. Specification A is the full sample; B restricts to the 2005–2024 window and the metro-tier units on which the covariates are complete; C adds three predetermined covariates — building-permit growth, metropolitan population growth, and real-GDP growth (each winsorized at the 1st/99th percentiles and standardized) — to that same sample, so that A→B isolates the sample-restriction effect and B→C the covariate effect. Specification D is the pre-COVID 2000–2019 sub-sample (the full set of metro-tier units passing the log-growth stationarity screen over that window); its cross-fitted criterion selects the same ranks (1, 1, 1) as the full sample. In C the covariate coefficients are building-permit growth +0.010 (0.020), population growth +0.001 (0.002), and real-GDP growth −0.001 (0.003), all statistically insignificant. Standard errors in parentheses are the diagonal sandwich; the reported companion radius is computed from the sample-average autoregressive coefficients. A radius below one describes the average AR(2) summary in each specification and should not be interpreted as a pointwise uniform stability certificate.

Housing heterogeneity map file not found.

Unemployment heterogeneity map file not found.

Figure 5: Geographic heterogeneity of the estimated dynamic persistence.

Notes: State-level averages, over the metros in each state, of the estimated dynamic persistence: cumulative persistence $a_1 + a_2$ for housing (top, headline spec A) and lag-1 persistence a_1 for unemployment (bottom, headline spec B); darker is more persistent. House-price momentum concentrates in the coastal and Sun Belt markets (Nevada, Florida, California, Arizona, the Mountain West) and is weakest in the interior and Rust Belt; idiosyncratic unemployment persistence is uniformly high but peaks in the Mountain West and Mid-Atlantic. The spread across states is the geographic face of the coefficient heterogeneity the low-rank structure makes estimable. The lower-48 states are shown (Alaska and Hawaii are in the underlying data but omitted from the map). If the rendered figures are absent, the source data are still sufficient to reproduce the maps in the replication package.

$N = 610$ headline estimate.

Several further diagnostics support the specification. First, the debiasing step is not cosmetic: [Table 27](#) contrasts the first-stage plug-in estimate with the debiased one-step. The corrections move every target in the expected direction — the housing lag means shift outward by about 0.02, and the unemployment lag-1 mean down by 0.067, a larger correction because the parsimonious single-factor specification leaves more first-stage regularization bias for the one-step to remove. The clearest case is the unemployment between-group contrast: the plug-in reports +0.001 and the debiased estimator corrects it to -0.012 , flipping the sign, so without debiasing the regularization bias would point the comparison the wrong way.

Table 27: Plug-in versus debiased estimates.

Target	plug-in	debiased	shift	diagonal s.e.
<i>Housing (spec A)</i>				
Lag-1 mean	+1.229	+1.250	+0.021	0.004
Lag-2 mean	−0.398	−0.419	−0.021	0.004
Lag-1 top–bottom contrast	+0.015	+0.018	+0.003	0.005
<i>Unemployment (spec B)</i>				
Lag-1 mean	+0.976	+0.909	−0.067	0.033
Lag-1 high–low contrast	+0.001	−0.012	−0.013	0.010

Notes: “plug-in” is the first-stage cell-mean of the estimated coefficient matrix; “debiased” is the cross-fitted one-step; “shift” is their difference. The shift is the bias the one-step removes; it is largest, relative to the estimate, for the unemployment contrast, where it changes the sign.

Second, the model fits well and the residuals are clean ([Table 28](#)). The dynamic structure explains 81–88% of the variation that a no-dynamics (interactive-block-only) baseline leaves unexplained. The residual first-order autocorrelation is essentially zero in housing and small in unemployment, so the selected lag order captures the own-dynamics; the average cross-sectional residual correlation is below 0.01 and the residual first-singular-value share is under 10%, so the interactive block has absorbed the common comovement — which is what the period-cluster sensitivity checks and the idiosyncratic-persistence interpretation require. A modest residual second-order autocorrelation remains in housing, consistent with the well-documented higher-order structure of monthly house-price growth, but it is small and leaves the lag means unaffected.

Third, the covariate-augmented results do not hinge on any single covariate ([Table 29](#)). Forcing each covariate’s coefficient to zero in turn leaves the cumulative persistence within about 0.02 of the full-covariate value in both panels and the companion radius below one throughout, so the autoregressive conclusions are a property of the dynamics, not of the controls.

Finally, comparing against a homogeneous benchmark shows what the heterogeneity buys ([Table 30](#)). The benchmark is a pooled two-way fixed-effects AR with a single common autoregressive coefficient — the textbook homogeneous dynamic panel. Its pooled coefficient is close to the heterogeneous *mean* (housing 1.22 versus 1.25; unemployment 0.92 versus 0.91), so pooling recovers the average dynamics. But it collapses to one number a coefficient that in fact varies widely across cells — the lag-1 coefficients spans roughly $[0.96, 1.53]$ in housing and $[0.90, 1.03]$ in unemployment

Table 28: Goodness of fit and residual adequacy.

Specification	RMSE	R_{out}^2	R_{dyn}^2	resid. ac(1)	resid. ac(2)	xs resid. corr	first-SV share
Housing A	0.296	0.913	0.812	+0.004	+0.145	+0.009	0.041
Housing B	0.252	0.936	0.828	+0.014	+0.138	+0.006	0.046
Housing C	0.239	0.943	0.845	+0.021	+0.127	+0.003	0.034
Unemployment A	0.258	0.992	0.860	−0.081	+0.053	+0.005	0.094
Unemployment B	0.239	0.993	0.846	−0.061	+0.070	+0.005	0.099
Unemployment C	0.215	0.994	0.876	−0.063	+0.053	+0.002	0.071

Notes: RMSE is the in-sample root-mean-square residual. R_{out}^2 is variance explained relative to the outcome; R_{dyn}^2 is the improvement over a no-dynamics baseline that keeps only the interactive block (all lag/covariate ranks set to zero). “resid. ac(k)” is the average over units of the lag- k residual autocorrelation; “xs resid. corr” is the average pairwise cross-sectional residual correlation; “first-SV share” is the share of the residual variance carried by its leading singular value (leftover common-factor strength). Near-zero residual autocorrelation and cross-sectional correlation, and a small first-SV share, indicate that the dynamics and the common factor are both captured.

Table 29: Covariate forced-rank robustness (spec C).

Specification	lag-1	cum. $a_1 + a_2$	comp. radius
<i>Housing</i>			
full (permits + population + GDP)	+1.279	+0.801	0.691
drop permits	+1.289	+0.823	0.683
drop population	+1.268	+0.819	0.670
drop GDP	+1.336	+0.826	0.714
<i>Unemployment</i>			
full (population + GDP)	+0.937	+0.937	0.937
drop population	+0.946	+0.946	0.946
drop GDP	+0.954	+0.954	0.954

Notes: Each row re-estimates the covariate-augmented specification C with one covariate’s block rank forced to zero — the covariate stays in the design but its coefficient is pinned at zero (effectively dropped). The lag dynamics and sub-unit companion radius are stable throughout, so no single covariate drives the autoregressive estimates.

— and it fits materially worse: letting the coefficients vary cuts the residual RMSE from 0.318 to 0.296 in housing and from 0.396 to 0.273 in unemployment, raising the R^2 from 0.79 to 0.91 and from 0.85 to 0.99. The heterogeneity is not noise to be averaged away; it is signal the homogeneous model discards.

Table 30: Homogeneous benchmark versus the heterogeneous estimator.

	lag-1	cum. a_1+a_2	RMSE	R^2
<i>Housing</i> (AR(2))				
Homogeneous (two-way FE)	+1.222	+0.814	0.318	0.793
Heterogeneous (mean)	+1.250	+0.831	0.296	0.913
Heterogeneous (5–95% range)	[0.96, 1.53]	—	—	—
<i>Unemployment</i> (AR(1))				
Homogeneous (two-way FE)	+0.921	+0.921	0.396	0.850
Heterogeneous (mean)	+0.909	+0.909	0.273	0.991
Heterogeneous (5–95% range)	[0.90, 1.03]	—	—	—

Notes: The homogeneous benchmark is a pooled two-way fixed-effects AR — one common autoregressive coefficient, with additive unit and time effects (the homogeneous analogue of the interactive block) — computed by `dlrhcs.empirical.homogeneous.benchmark`. “Heterogeneous (mean)” is the debiased lag-1 mean and cumulative persistence; the “5–95% range” is the 5th–95th percentile of the estimated per-cell lag-1 coefficients. RMSE and R^2 are in-sample, each application on its own headline sample (housing spec A; unemployment spec B). Both the homogeneous benchmark and the heterogeneous fit are computed on that sample.

The asymptotic theory is stated for the ideal rank-constrained minimizer (or an objective-gap-equivalent solution), whereas the implementation uses alternating rank-constrained least squares with a truncated-SVD warm start and random restarts. The diagnostics cannot certify global optimality and are not a proof of the objective-gap condition; they only document that the reported numerical solution is stable for the implemented nonconvex search (Table 31, headline unemployment specification; housing is comparable): the objective is monotonically non-increasing across sweeps and stabilizes; the buffered fold-Gram — the first-stage information map whose inverse is the debiasing direction — has minimum eigenvalue bounded well away from zero (0.12), so the debiasing weight is well posed; every Riesz-representer conjugate-gradient solve converged to tolerance 10^{-5} (mean 111 iterations); the retained training share stays high (0.69); and the cross-fitted rank criterion separates the selected $(1, 0, 1)$ from the next candidate by a clear margin (0.128 versus 0.168), so the selected rank is stable. The companion package logs these quantities for every fit: here the objective falls by only $\approx 10^{-4}$ at the final sweep and varies by only $\approx 5 \times 10^{-4}$ across restarts, so the reported solution is effectively restart-invariant.

C.2.4 What the two illustrations establish

The two applications are complementary and exercise different corners of the theory. The unemployment panel is the factor-dominated regime: a strong common cycle accounts for roughly three-quarters of the panel variance, and the data-driven criterion absorbs it into a *single*-factor interactive block while selecting a parsimonious heterogeneous AR(1) for the idiosyncratic dynamics

Table 31: Optimization diagnostics (headline unemployment specification).

Diagnostic	Value
Objective monotone, non-increasing across sweeps	yes
Final-sweep relative objective improvement	1×10^{-4}
Objective dispersion across restarts ($n=2$)	5×10^{-4}
Buffered fold-Gram minimum eigenvalue (training-map conditioning)	0.12
Riesz CG solves converged (tolerance 10^{-5})	100%
mean CG iterations	111
Retained training share (fraction of non-validation cells)	0.69
Rank-criterion value: selected (1, 0, 1) vs. next (1, 1, 0)	0.128 vs. 0.168

Notes: Computational diagnostics for the headline specification’s first stage and debiasing solves. A monotone objective, a well-conditioned fold-Gram (minimum eigenvalue bounded from zero), fully converged Riesz solves, a high retained training share, and a clear rank-criterion margin together indicate that the alternating-least-squares solution is numerically stable and well posed. They do not certify global optimality or the objective-gap condition required to substitute a finite nonconvex iterate for the ideal minimizer in the theory. The housing application is qualitatively the same.

that remain. The method isolates that idiosyncratic persistence — high, about 0.91, and, as the three specifications show, stable both across the sample window and to the inclusion of local population and GDP growth — once the common cycle is removed. The housing panel is the large- N , large- T regime: with both dimensions large the entry, mean, and contrast targets are estimated precisely and the standard errors contract at the $\sqrt{TN}$ rate, so that even an economically small price-tier difference is resolved to within a few thousandths. Under the by-period clustered sensitivity diagnostic the same contrast is only suggestive, but with the credible geographic metric in [Table 11](#) the spatial-kernel standard error is the operative theorem-aligned dependence-robust object and the coordinate-matched contrast is bounded away from zero. Together the two panels exercise the full set of targets — full means, group means, and contrasts — on real data, and corroborate the simulation evidence of [Section 8](#): pooled and large-panel targets are precise, while the informative content lies in the heterogeneity that the low-rank structure makes estimable. They also illustrate the rank selector adapting to the data, choosing an AR(2) for housing and the more parsimonious AR(1) for unemployment, with a single interactive factor sufficient for prediction in each (we retain additional factors in the labor panel only to absorb the common cycle more fully, with no effect on the conclusions).

These patterns line up with what the method is built to deliver. The housing standard errors are very tight—about 0.004–0.012, several times smaller than the unemployment panel’s—because that panel has the larger unit-by-time product, which is exactly where the estimator is sharpest; the two panels’ interval widths differ accordingly. The sample-average companion spectral radius of 0.65, below one despite a lag-1 coefficient above one, summarizes the average AR(2) dynamics, while the theorem’s stability assumption remains a pointwise condition on the true heterogeneous companion products. The two between-group contrasts show how sensitive empirical conclusions can be to contemporaneous comovement. The housing tier contrast is clearly nonzero under the diagonal interval; under the by-period clustered sensitivity diagnostic it is only marginal, but with a credible

geographic metric the spatial-kernel standard error in [Table 11](#) is the operative theorem-aligned dependence-robust object and the coordinate-matched contrast is bounded away from zero. In the labor panel the high-minus-low contrast is small and indistinguishable from zero under the diagonal, spatial-kernel, and period-cluster sensitivity intervals, so the finding is simply that, at the monthly frequency, idiosyncratic persistence does not differ materially between structurally tight and structurally slack labor markets. The data-driven ranks—a heterogeneous AR(2) with one interactive factor for housing, a heterogeneous AR(1) for unemployment—are the selection criterion adapting the model order to each panel rather than imposing it. Finally, the spread of the per-cell lag-1 coefficients in [Figure 2](#) is the heterogeneity itself: a homogeneous model collapses it to a single number, whereas the low-rank structure makes the entire coefficient matrix estimable once the common factor is absorbed—the paper’s central point: much of the informative content of a dynamic panel lies in the coefficient heterogeneity that becomes estimable once the low-rank structure and common factor are separated.